AMSE: all available evidence

Finding

Full AMSE improves spectral-power fidelity but sacrifices forecast accuracy. Lower AMSE weights retain more accuracy, yet no candidate in the completed joint campaign meets all three aggregate requirements plus the safety guard.

Coverage

September 19 full-AMSE/MSE pair; September 24 lambda=0.1 blend; October 4 trajectory re-evaluation, 500-update pilot, stopped/differentiable feedback, and three objective ablations. All 49 evaluated records across the four October phases are included, with repeated controls retained separately.

What is still pending

All six 2,000-update confirmation training runs finished. Seven evaluation tasks 15149394_0–6 and report 15149395 remain pending at the live queue check on October 7. Confirmation covers three seeds and 64 separate seasonal 2023 starts. No confirmation scores are available.

Related MGDA experiment

MGDA combines GraphCast and spectral-power gradients. Calibration completed; both 100-update smoke runs failed bitwise resume equality of residual parameters. The 2,000-update production comparison did not launch. These are engineering results, not evidence of forecast skill.

Reading this report

Positive reduction means better than Original GraphCast. Exact GraphCast loss is the headline metric; physical RMSE and relative spectral-power error are separate diagnostics. The companion complete PDF appends the existing 49-page field-level spectral report.

Protocols and interpretation

Historical accuracy versus spectral diagnostics

Historical exact-loss evaluation uses 32 matched 2022 starts and 40 six-hour leads. Spectral evaluation uses eight previously inspected 2023 starts, 30–60° latitude in both hemispheres, and zonal wavenumbers 1–180. Neither is the pending year-round 2023 evaluation.

Objective and measurements

Training AMSE interpolates spherical spectral MSE (lambda=0) and full AMSE (lambda=1), using a global T179 transform. Verification uses zonal Fourier spectra. Exact GraphCast loss retains its original normalization, latitude, pressure and variable weights; losses are pooled before the reduction ratio.

October joint qualification

On the same eight spectral starts: exact GC loss ratio <1, mean physical-RMSE ratio <1, and mean relative-power-error ratio <1; additionally no selected primitive field/day may exceed 2% RMSE harm versus GC. RMSE/power summaries use nine fields and days 1, 3, 7, 10. Power error is averaged over spectral bins then field/day cells.

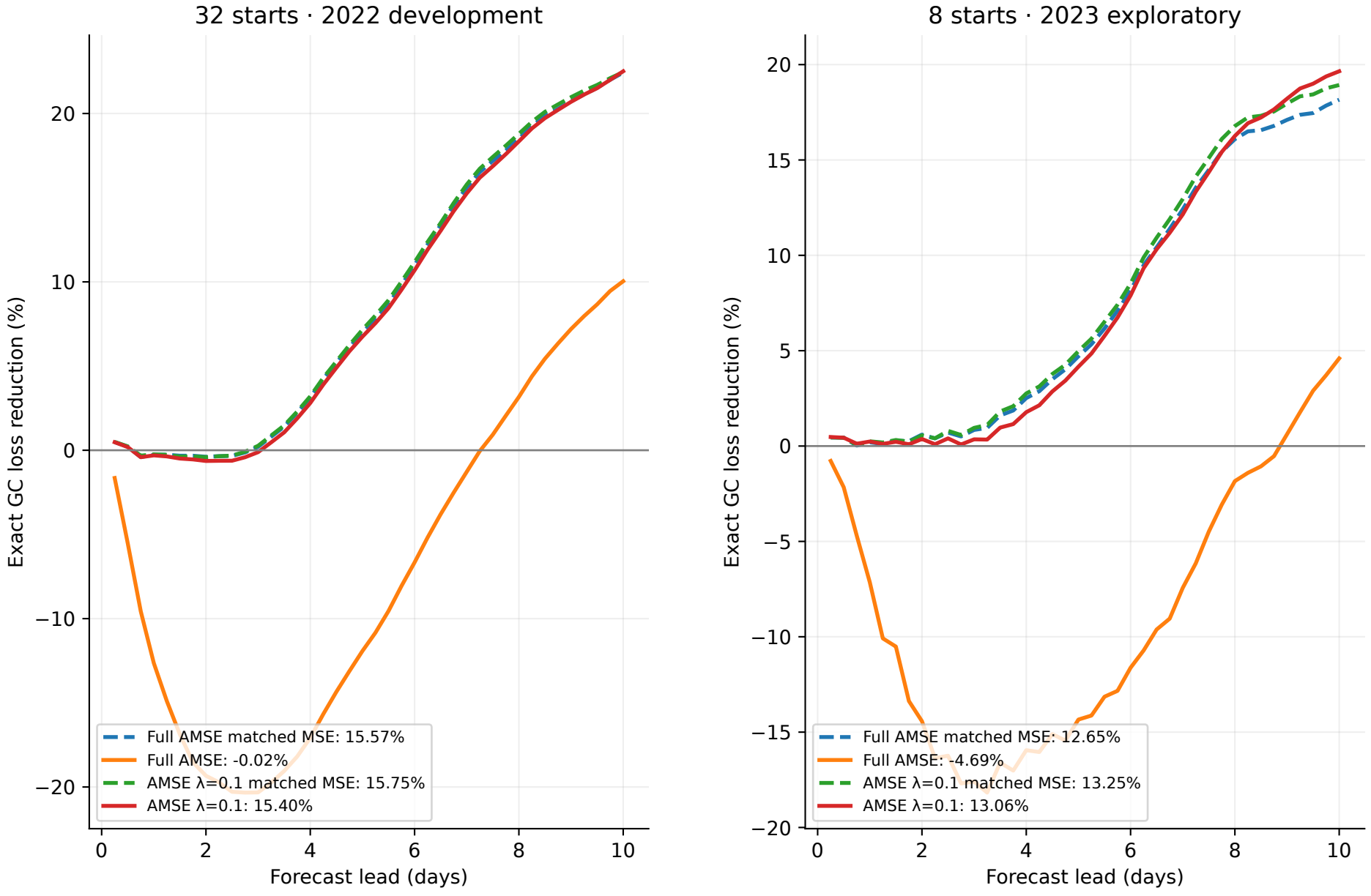
Baseline and uncertainty limits

Historical comparisons flag numerical baseline drift and use each model’s paired GC baseline. October scores use their phase’s canonical GC reference; controls vary between re-evaluations. Do not average repeated baselines or interpret tiny differences as established gains. Development selection has no confirmatory uncertainty claim.

Power is not accuracy

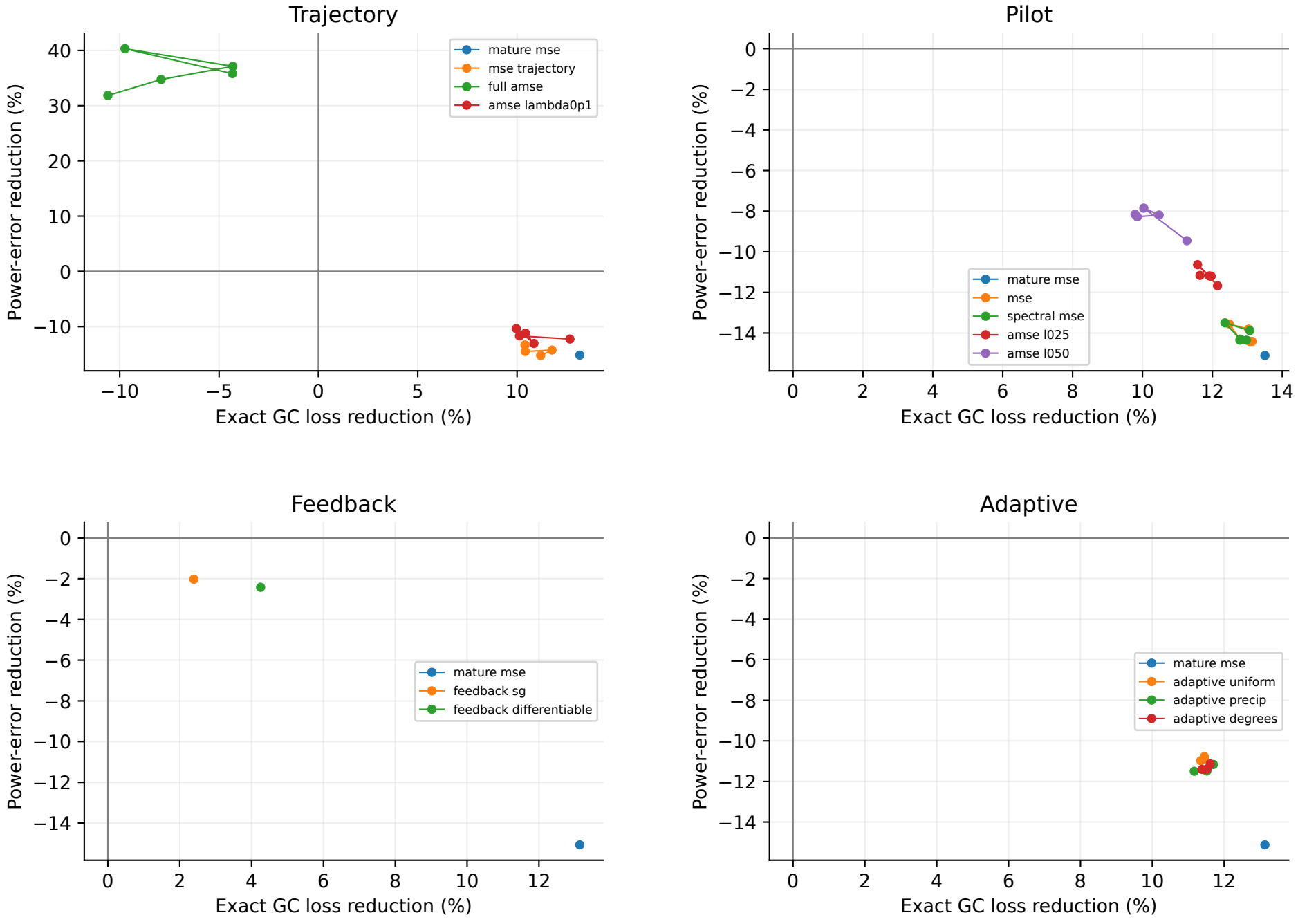
Matching power does not ensure the correct location or phase of weather features. Coherence and spectral-error decomposition in the appendix expose this distinction. No new forecast inference or training was run to prepare this report.

Historical AMSE: accuracy by forecast lead



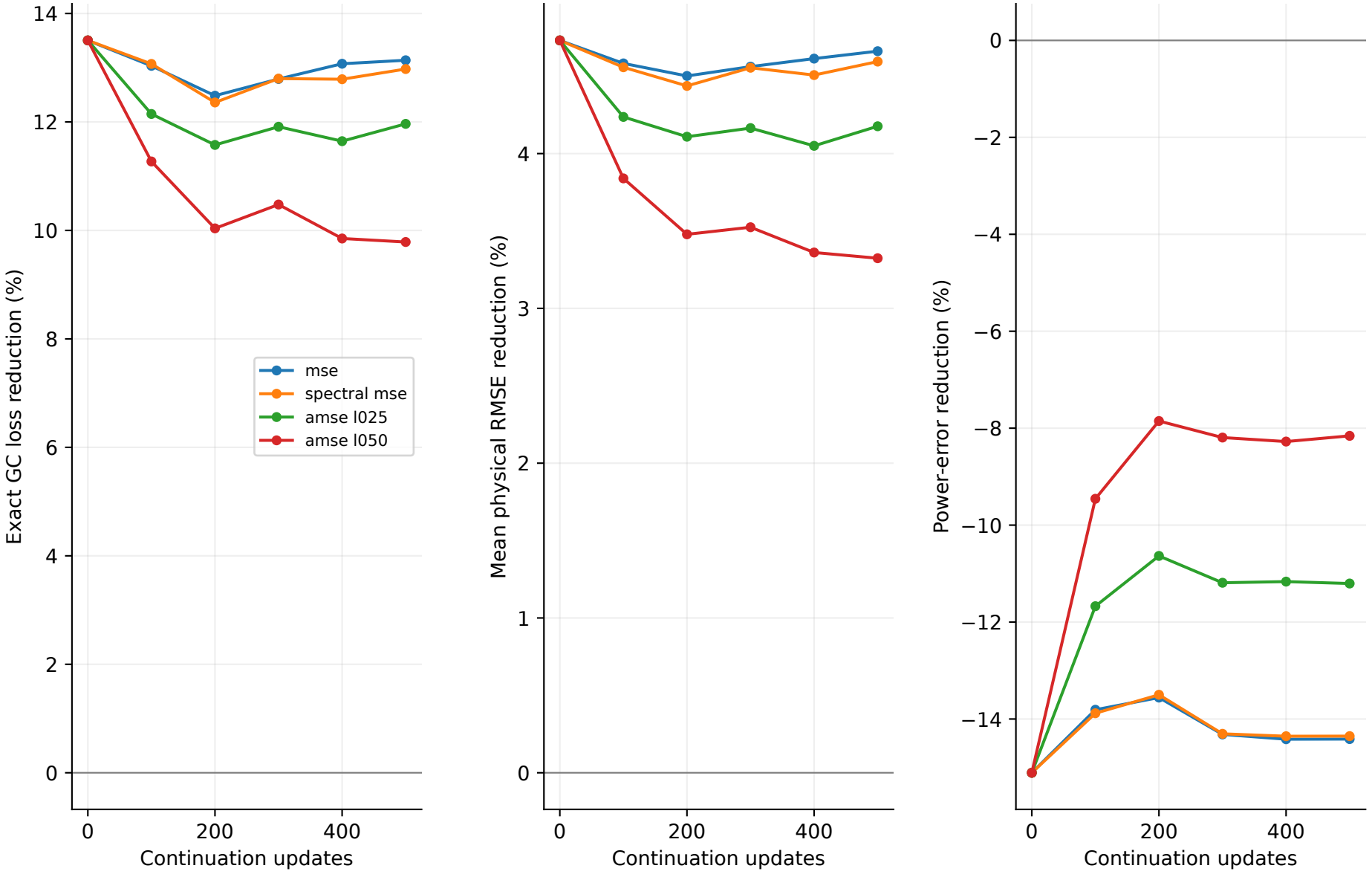
SWA 500–2000; paired baselines retained separately. Legend gives pooled rollout reduction, not mean lead percentage.

Accuracy and power: no joint winner yet



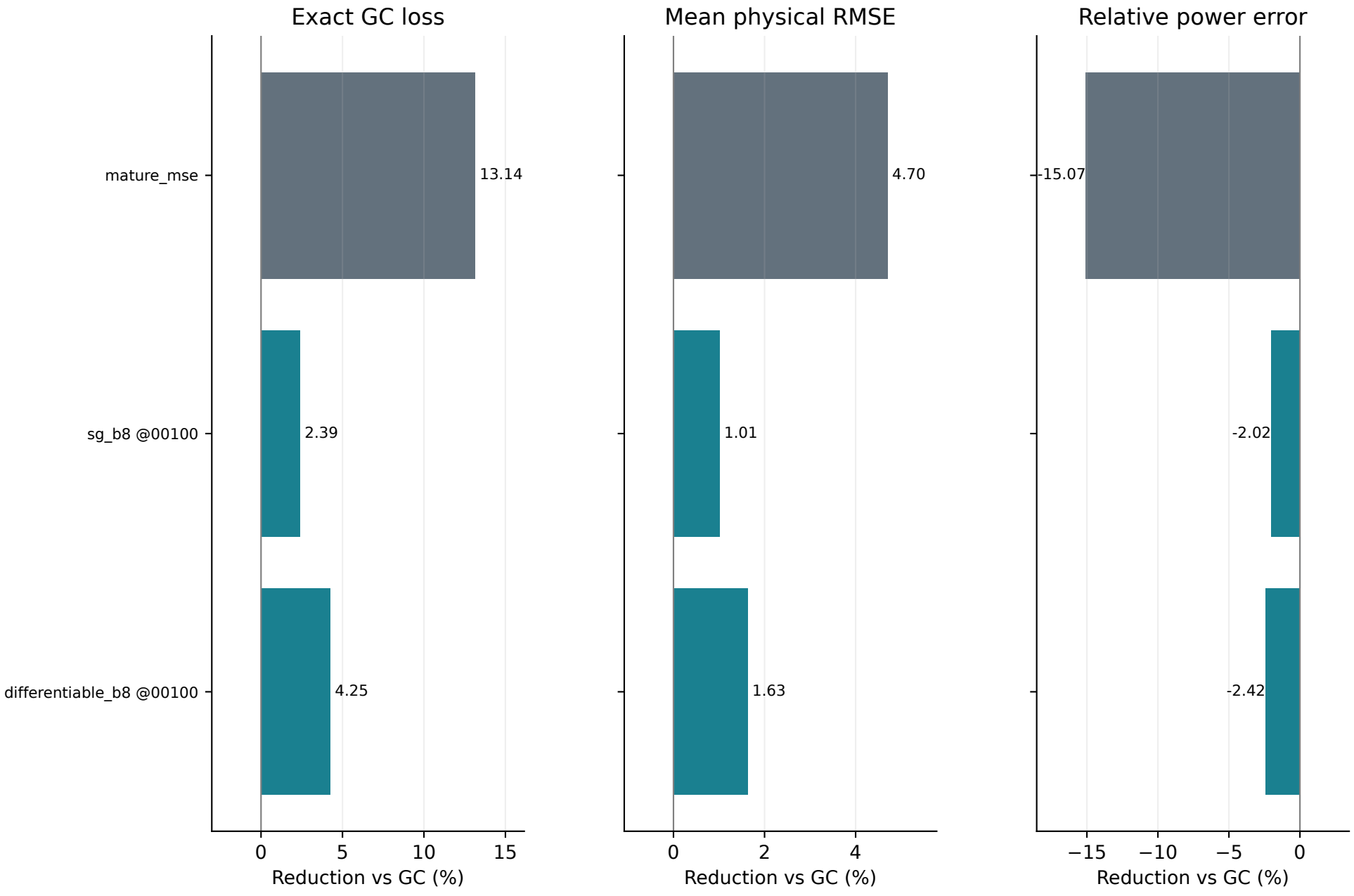
Same eight 2023 starts. Upper right improves both axes; qualification additionally requires physical RMSE and safety.

500-update pilot: increasing AMSE trades accuracy for power



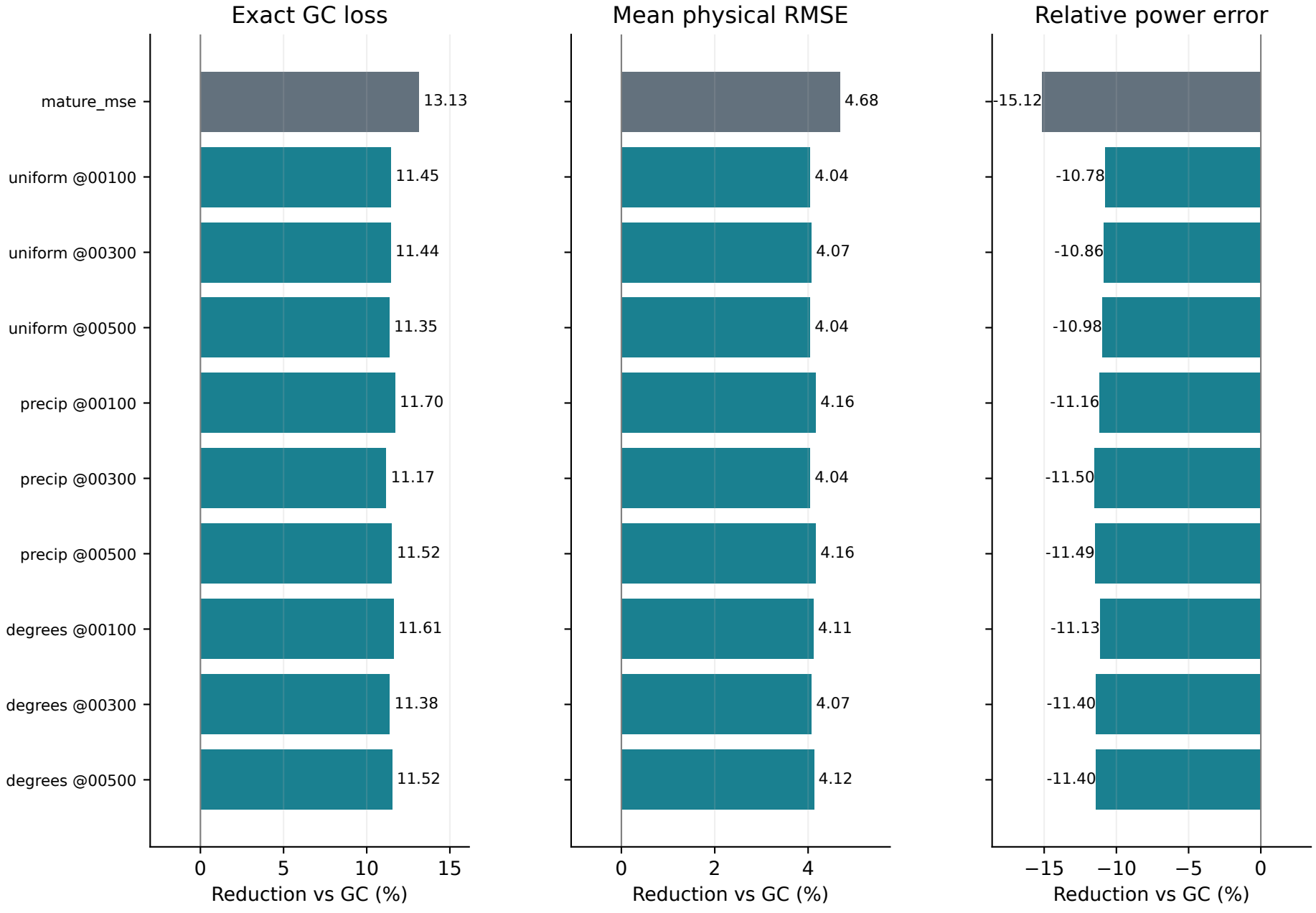
Common mature-MSE start; four matched 500-update continuations. All checkpoints shown; all fail joint qualification.

Feedback experiments: completed development results



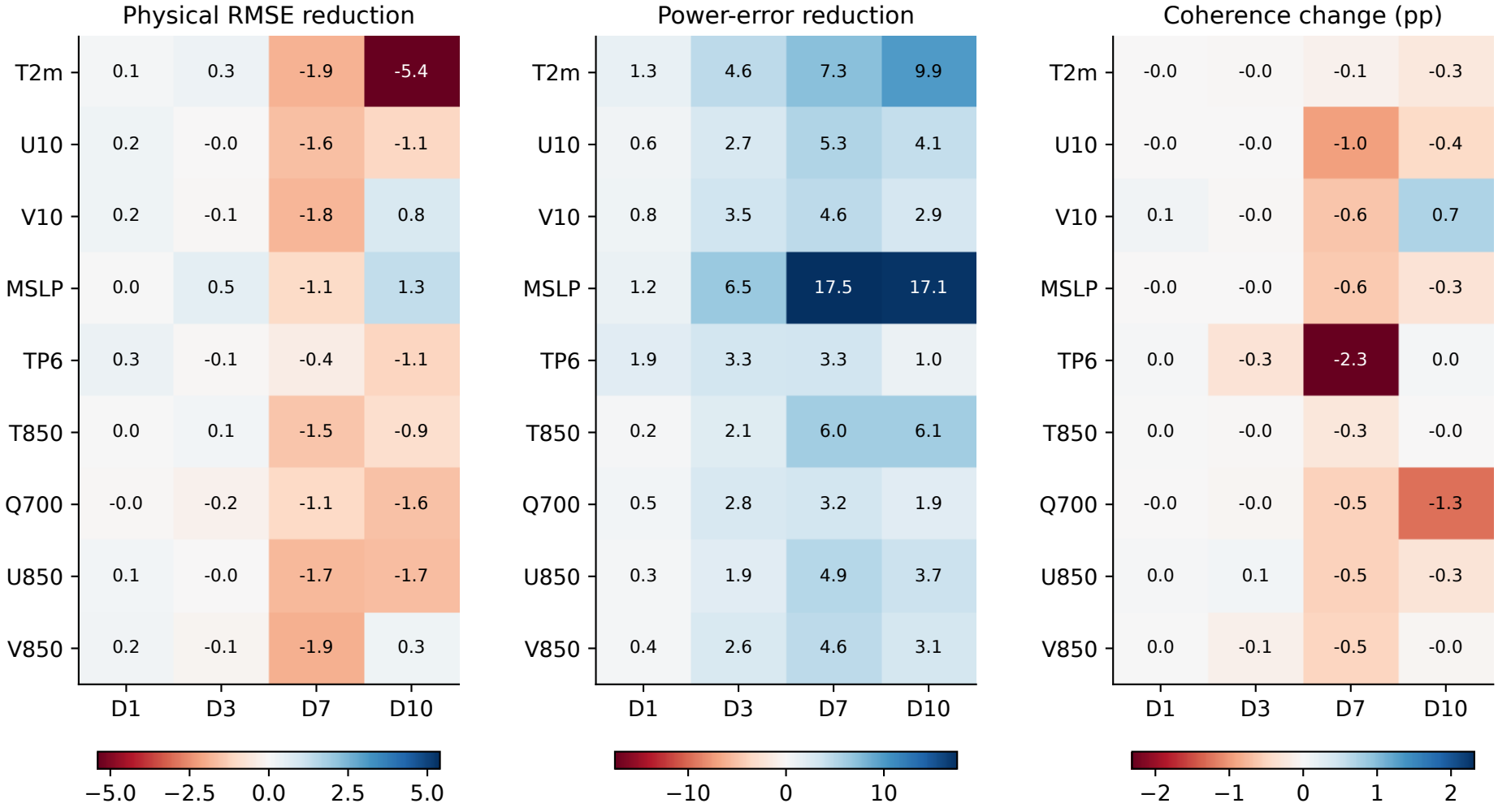
Exploratory eight-start 2023 diagnostics. Positive is better. No candidate passes the joint qualification gate.

Adaptive experiments: completed development results



Exploratory eight-start 2023 diagnostics. Positive is better. No candidate passes the joint qualification gate.

Locked recipe: where does it help or hurt?



adaptive_precip @100 versus phase MSE control. Positive is better; blue helps, red harms. This is development, not confirmation.

Complete checkpoint inventory: trajectory

Candidate	GC loss %	RMSE %	Power error %	Joint cells	Safety	Eligible
mature_mse	13.16	4.64	-15.15	8/36	pass	no
mse_trajectory_step00500	10.40	3.57	-13.31	5/36	pass	no
mse_trajectory_step01000	10.42	3.74	-14.49	5/36	pass	no
mse_trajectory_step01500	11.76	4.04	-14.23	9/36	pass	no
mse_trajectory_step02000	11.19	4.12	-15.22	9/36	pass	no
full_amse_step00500	-10.60	-5.28	31.84	2/36	FAIL	no
full_amse_step01000	-7.92	-3.90	34.74	4/36	FAIL	no
full_amse_step01500	-4.31	-2.63	37.12	8/36	FAIL	no
full_amse_step02000	-9.74	-4.46	40.31	1/36	FAIL	no
full_amse_swa	-4.33	-2.73	35.80	8/36	FAIL	no
amse_lambda0p1_step00500	9.96	3.30	-10.35	6/36	pass	no
amse_lambda0p1_step01000	10.85	3.71	-13.04	3/36	pass	no
amse_lambda0p1_step01500	10.42	3.56	-11.19	9/36	pass	no
amse_lambda0p1_step02000	10.12	3.69	-11.68	11/36	pass	no
amse_lambda0p1_swa	12.66	4.35	-12.24	9/36	pass	no

All percentage columns are reductions versus GC; positive is better. Joint cells improve both physical RMSE and power error.

Complete checkpoint inventory: pilot

Candidate	GC loss %	RMSE %	Power error %	Joint cells	Safety	Eligible
mature_mse	13.50	4.73	-15.11	9/36	pass	no
mse_step00100	13.03	4.58	-13.81	11/36	pass	no
mse_step00200	12.48	4.50	-13.56	11/36	pass	no
mse_step00300	12.79	4.56	-14.32	12/36	pass	no
mse_step00400	13.07	4.61	-14.42	11/36	pass	no
mse_step00500	13.14	4.66	-14.41	11/36	pass	no
spectral_mse_step00100	13.07	4.56	-13.88	11/36	pass	no
spectral_mse_step00200	12.36	4.44	-13.50	11/36	pass	no
spectral_mse_step00300	12.80	4.55	-14.31	11/36	pass	no
spectral_mse_step00400	12.79	4.51	-14.35	10/36	pass	no
spectral_mse_step00500	12.98	4.59	-14.35	11/36	pass	no
amse_l025_step00100	12.15	4.24	-11.67	13/36	pass	no
amse_l025_step00200	11.58	4.11	-10.64	13/36	pass	no
amse_l025_step00300	11.91	4.16	-11.19	13/36	pass	no
amse_l025_step00400	11.65	4.05	-11.17	12/36	pass	no
amse_l025_step00500	11.96	4.18	-11.20	12/36	pass	no
amse_l050_step00100	11.27	3.84	-9.46	11/36	pass	no
amse_l050_step00200	10.04	3.48	-7.85	10/36	pass	no
amse_l050_step00300	10.48	3.52	-8.19	8/36	pass	no
amse_l050_step00400	9.85	3.36	-8.28	7/36	pass	no
amse_l050_step00500	9.79	3.32	-8.16	8/36	pass	no

All percentage columns are reductions versus GC; positive is better. Joint cells improve both physical RMSE and power error.

Complete checkpoint inventory: feedback

Candidate	GC loss %	RMSE %	Power error %	Joint cells	Safety	Eligible
mature_mse	13.14	4.70	-15.07	10/36	pass	no
feedback_sg_b8_step00100	2.39	1.01	-2.02	8/36	pass	no
feedback_differentiable_b8_step00100	4.25	1.63	-2.42	13/36	pass	no

All percentage columns are reductions versus GC; positive is better. Joint cells improve both physical RMSE and power error.

Complete checkpoint inventory: adaptive

Candidate	GC loss %	RMSE %	Power error %	Joint cells	Safety	Eligible
mature_mse	13.13	4.68	-15.12	9/36	pass	no
adaptive_uniform_step00100	11.45	4.04	-10.78	13/36	pass	no
adaptive_uniform_step00300	11.44	4.07	-10.86	12/36	pass	no
adaptive_uniform_step00500	11.35	4.04	-10.98	12/36	pass	no
adaptive_precip_step00100	11.70	4.16	-11.16	15/36	pass	no
adaptive_precip_step00300	11.17	4.04	-11.50	12/36	pass	no
adaptive_precip_step00500	11.52	4.16	-11.49	11/36	pass	no
adaptive_degrees_step00100	11.61	4.11	-11.13	12/36	pass	no
adaptive_degrees_step00300	11.38	4.07	-11.40	12/36	pass	no
adaptive_degrees_step00500	11.52	4.12	-11.40	12/36	pass	no

All percentage columns are reductions versus GC; positive is better. Joint cells improve both physical RMSE and power error.

Pending work and what the evidence supports

Confirmation: training complete, evaluation pending

Six final step2000 checkpoints exist: MSE and locked AMSE recipe at seeds 22, 23 and 24. Evaluation uses 64 seasonal 2023 starts and averages separate-model metrics, never forecast fields. The recipe is lambda=0.25 with precipitation lambda=0; it was not qualified on development. Selection is locked before confirmation.

MGDA: blocked at engineering validation

Calibration job 15069232 completed. Smoke jobs 15069233_0 and _1 reached 100 updates but failed uninterrupted-versus-resumed bitwise residual-parameter equality. A failure shared by MGDA and MSE does not demonstrate failure of the scientific objective. No production forecast comparison is available.

2023 scope

The eight-case spectral diagnostics have completed and were inspected during development. The separate 122-start year-round memory/accuracy campaign remains pending numerical qualification; the earlier daily campaign failed baseline agreement. The 64-start AMSE confirmation is a third, separate evaluation.

Supported conclusion

Full AMSE produces materially better power fidelity with worse exact forecast loss. Weak AMSE produces an intermediate trade-off. The 500-update continuations, feedback tests and adaptive variants have not established a joint improvement. Small differences are particularly sensitive to observed numerical drift.

Recommended next evidence

Finish the already queued confirmation evaluations and inspect all fields, all seeds and paired intervals.
Repair and qualify resume/numerical behavior before running MGDA production or releasing year-round evaluation.
Preserve negative outcomes and keep selection fixed.

AMSE forecast quality across weather features

Original GC · MSE · AMSE λ=0.1 · Full AMSE

This report compares spectral power, forecast–truth coherence and deterministic forecast error using completed evaluations of the same 1° residual-Mamba architecture. The three trained models use stochastic weight averaging over updates 500, 1000, 1500 and 2000.

Full AMSE restores substantially more missing spectral power, but it also loses pattern alignment and much of the MSE-trained model's deterministic accuracy advantage. AMSE λ=0.1 stays much closer to MSE: its spectral changes are smaller and its forecast-loss performance is similar.

Model	Mean relative power error	Mean spectral coherence
Original GC	48.94%	0.5115
MSE	57.16%	0.5244
AMSE λ=0.1	55.58%	0.5242
Full AMSE	31.96%	0.4902

These spectral summaries give equal weight to 11 fields, zonal wavenumbers 1–180 and days 1, 3, 7 and 10. Raw spectra are pooled across eight 2023 starts before ratios are formed. The summaries include wind speed alongside its components and are secondary diagnostics, not the original GraphCast training loss.

The report contains a methods guide, spectral summary tables, exact GraphCast loss, global RMSE by variable, and three spectral plots for each field. The downloadable tables include every saved lead and all 83 global-RMSE channels.

Evidence scope: 32 historical 2022 development starts and eight exploratory 2023 starts, reported separately. No uncertainty intervals or new forecasts are included.

How to read the comparison

Definitions and fixed model references

Original GC is the baseline evaluation paired with the MSE SWA checkpoint from the original full-AMSE campaign. MSE is that same campaign's trained control. AMSE λ=0.1 comes from the later λ=0.1 campaign; Full AMSE comes from the original campaign. These four references are fixed throughout. Repeated GC evaluations are not averaged or silently substituted.

The separate λ=0.1 campaign has another matched MSE control. It is excluded from the four-model figures as requested, but retained in the source audit. Small cross-campaign differences may reflect run-to-run variability; the repeatability appendix records those differences.

Spectral diagnostics use zonal Fourier transforms of scalar fields within 30–60° latitude in both hemispheres. The zonal mean is removed at each latitude. Latitude weights are cosine weights, and raw powers and cospectra are averaged over starts before forming ratios. Curves are unsmoothed. Larger zonal wavenumber means finer scale, but wavelength depends on latitude.

Power error (%) = 100 × |forecast power − truth power| / truth power; lower is better. Coherence = real cospectrum / sqrt(forecast power × truth power); higher is better. ERA5 is a verification reference, not an additional forecast model.

Global RMSE is sqrt(mean squared physical error over starts and the global cosine-weighted grid). Upper-air variable-group RMSE additionally pools squared errors equally across 13 levels. Specific-level RMSE uses the stated pressure level. Neither metric is identical to pressure-weighted, difference-standard-deviation-normalized original GraphCast loss.

Exact GraphCast loss reduction is 100 × (1 − aggregate model loss / aggregate reference loss). Aggregation precedes the ratio. Displayed-reference and own-paired-reference comparisons are labeled separately. Wind-speed global RMSE was not saved; component RMSEs cannot reconstruct it. WS10 and WS850 have spectral diagnostics only.

Relative power error by field

Mean over days 1, 3, 7, 10 and k = 1–180 · percent · lower is better

Field	Original GC	MSE	AMSE λ=0.1	Full AMSE
T2m 2 m temperature	18.53	19.79	19.06	12.00
U10 10 m zonal wind	53.19	61.23	59.78	41.34
V10 10 m meridional wind	52.26	62.38	60.86	38.19
MSLP Mean sea-level pressure	17.26	22.76	21.08	14.70
TP6 6 h precipitation	67.69	81.09	78.22	26.40
T850 850 hPa temperature	47.39	52.64	51.02	25.83
Q700 700 hPa specific humidity	69.06	76.69	75.25	35.39
U850 850 hPa zonal wind	59.49	67.00	65.72	44.05
V850 850 hPa meridional wind	59.46	68.78	67.60	46.51
WS10 10 m wind speed	40.57	52.48	50.36	25.20
WS850 850 hPa wind speed	53.40	63.93	62.46	42.00

Each entry gives equal weight to the retained spectral bins, rather than weighting by their physical energy. Do not interpret these means as global RMSE or GraphCast loss. The individual field plots show scale and lead dependence that these averages conceal.

Spectral coherence by field

Mean over days 1, 3, 7, 10 and k = 1–180 · higher is better

Field	Original GC	MSE	AMSE λ=0.1	Full AMSE
T2m 2 m temperature	0.8606	0.8658	0.8652	0.8462
U10 10 m zonal wind	0.5044	0.5223	0.5217	0.4882
V10 10 m meridional wind	0.4901	0.5104	0.5092	0.4817
MSLP Mean sea-level pressure	0.7607	0.7660	0.7661	0.7226
TP6 6 h precipitation	0.2895	0.3016	0.3055	0.2683
T850 850 hPa temperature	0.5769	0.5876	0.5867	0.5533
Q700 700 hPa specific humidity	0.3161	0.3331	0.3322	0.2964
U850 850 hPa zonal wind	0.4336	0.4441	0.4437	0.4063
V850 850 hPa meridional wind	0.4287	0.4439	0.4426	0.4116
WS10 10 m wind speed	0.5281	0.5435	0.5429	0.5103
WS850 850 hPa wind speed	0.4382	0.4503	0.4499	0.4073

Each entry gives equal weight to the retained spectral bins, rather than weighting by their physical energy. Do not interpret these means as global RMSE or GraphCast loss. The individual field plots show scale and lead dependence that these averages conceal.

Rollout RMSE by variable in 2022

32 starts · all 40 leads pooled in squared-error space · lower is better

Variable group	Original GC	MSE	AMSE λ=0.1	Full AMSE
2 m temperature (K)	1.62	1.519	1.522	1.618
10 m zonal wind (m/s)	2.699	2.395	2.403	2.587
10 m meridional wind (m/s)	2.844	2.535	2.547	2.746
Sea-level pressure (hPa)	4.012	3.672	3.632	3.79
6 h precipitation (mm)	1.872	1.74	1.752	2.203
Temperature, all levels (K)	1.896	1.766	1.767	1.898
Specific humidity, all levels (g/kg)	0.7363	0.6784	0.6829	0.7712
Zonal wind, all levels (m/s)	5.312	4.868	4.886	5.294
Meridional wind, all levels (m/s)	5.52	5.03	5.05	5.457
Geopotential, all levels (m^2/s^2)	422.2	393.6	394.2	413
Vertical velocity, all levels (Pa/s)	0.127	0.1197	0.1205	0.1505

Largest Full AMSE rollout-RMSE increases versus MSE: 6 h precipitation +26.6%; vertical velocity, all levels +25.7%; specific humidity, all levels +13.7%.

These are physical-unit RMSEs, not the exact GraphCast loss. Upper-air groups pool all 13 levels equally. Small AMSE λ=0.1 differences are inconclusive without uncertainty estimates.

Rollout RMSE by variable in 2023

8 starts · all 40 leads pooled in squared-error space · lower is better

Variable group	Original GC	MSE	AMSE λ=0.1	Full AMSE
2 m temperature (K)	1.679	1.574	1.577	1.707
10 m zonal wind (m/s)	2.75	2.505	2.518	2.711
10 m meridional wind (m/s)	2.854	2.596	2.607	2.818
Sea-level pressure (hPa)	3.928	3.672	3.628	3.858
6 h precipitation (mm)	2.17	2.073	2.078	2.468
Temperature, all levels (K)	1.877	1.78	1.776	1.906
Specific humidity, all levels (g/kg)	0.7695	0.716	0.7198	0.8054
Zonal wind, all levels (m/s)	5.293	4.941	4.959	5.377
Meridional wind, all levels (m/s)	5.461	5.112	5.11	5.57
Geopotential, all levels (m^2/s^2)	422.4	394.8	394.1	421.9
Vertical velocity, all levels (Pa/s)	0.154	0.1492	0.1496	0.1748

Largest Full AMSE rollout-RMSE increases versus MSE: 6 h precipitation +19.0%; vertical velocity, all levels +17.1%; specific humidity, all levels +12.5%.

These are physical-unit RMSEs, not the exact GraphCast loss. Upper-air groups pool all 13 levels equally. Small AMSE λ=0.1 differences are inconclusive without uncertainty estimates.

Why sharper power can coexist with larger error

Measured mechanism and comparison with the AMSE paper

Power matching does not ensure that a feature is correctly located or timed. In each spectral mode, ordinary squared error equals an amplitude-mismatch term plus a decorrelation term. Restoring the amplitude of uncertain patterns can increase error even if coherence stays fixed.

$$\mathrm{MSE}_k = (\sqrt{P_f} - \sqrt{P_t})^2 + 2\sqrt{P_f P_t}(1 - \rho_k)$$

Day 10 T850	Amplitude error (K²)	Decorrelation (K²)	Total anomaly MSE (K²)
Original GC	0.301	17.203	17.504
MSE	1.336	12.388	13.724
AMSE λ=0.1	1.137	12.522	13.659
Full AMSE	0.480	15.704	16.184

These are midlatitude zonal-anomaly errors, not global T850 RMSE. The decorrelation term increases with both larger power and reduced coherence. AMSE includes a coherence penalty; finite network capacity, optimization and autoregressive dynamics still permit worse alignment on evaluation cases.

The original paper's main forecast-skill comparison uses nine-member lagged ensembles: CRPS improves while ensemble-mean RMSE changes little. This is a different target from individual-forecast RMSE. Its training also differs: 0.25° GraphCast, 32,500 updates, and a single-step-to-12-step curriculum. Here a frozen 1° GraphCast supports a residual branch trained for 2,000 updates with stopped physical-feedback gradients. These are plausible contributors, not isolated causal findings.

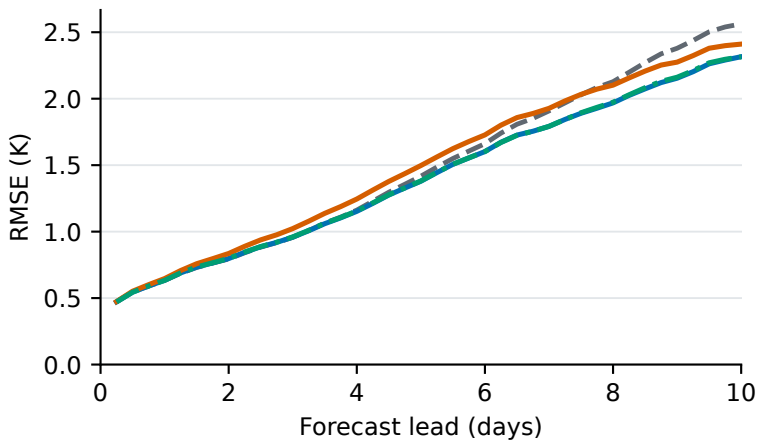
Source: Subich et al., sections 2.4 and 3.2 · arxiv.org/html/2501.19374v2

Forecast RMSE by variable

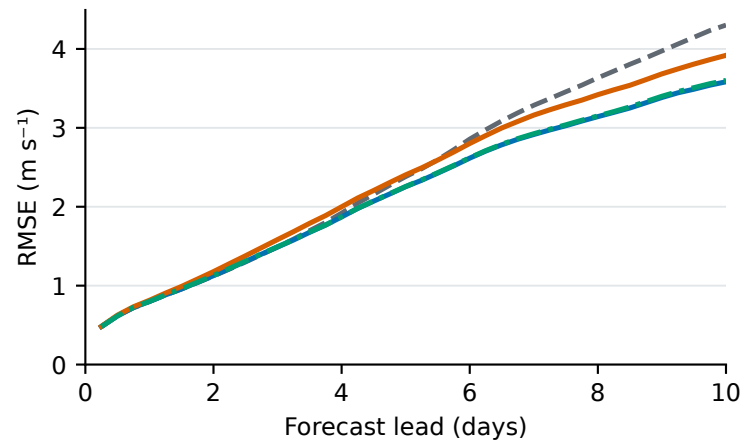
32 matched 2022 starts · 6-hourly leads through day 10 · residual models use SWA checkpoints

Original GC · MSE · AMSE λ=0.1 · Full AMSE

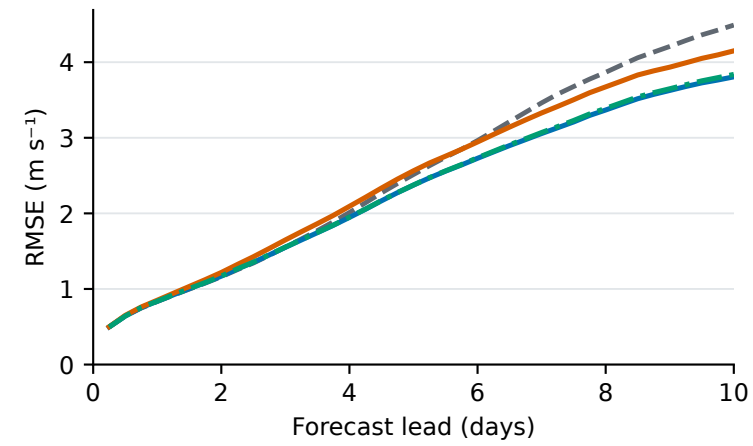
2 m temperature



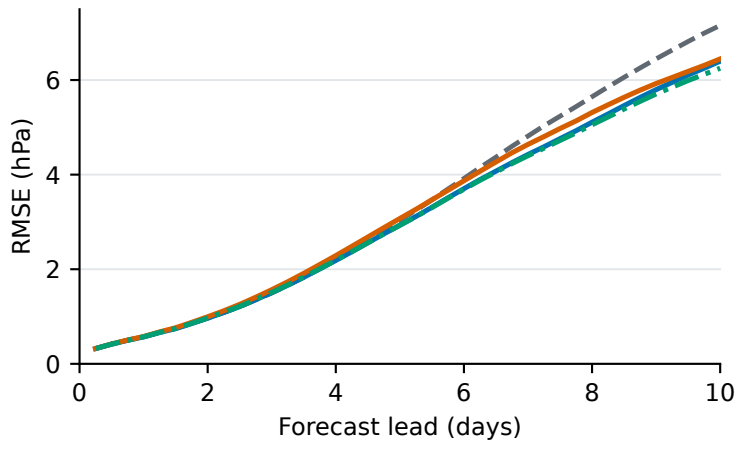
10 m zonal wind



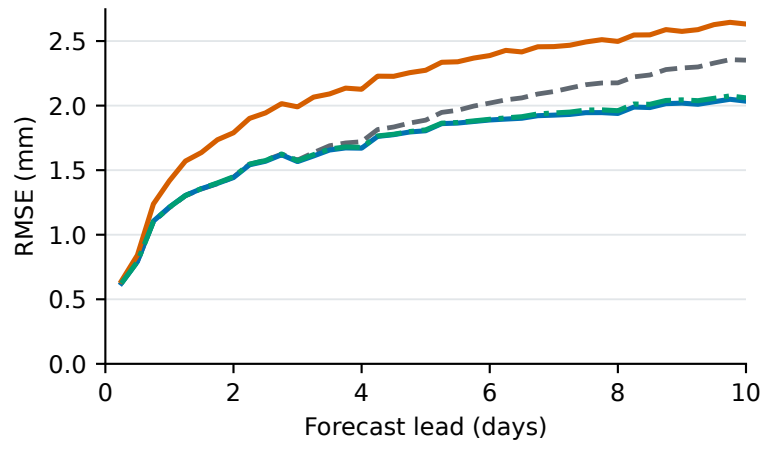
10 m meridional wind



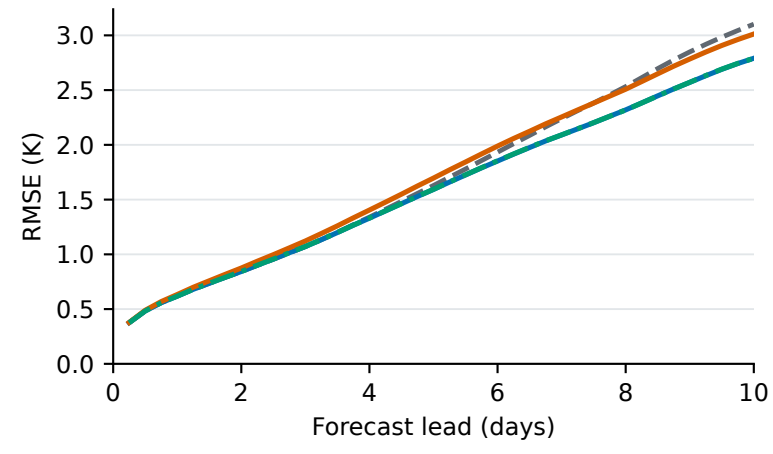
Mean sea-level pressure



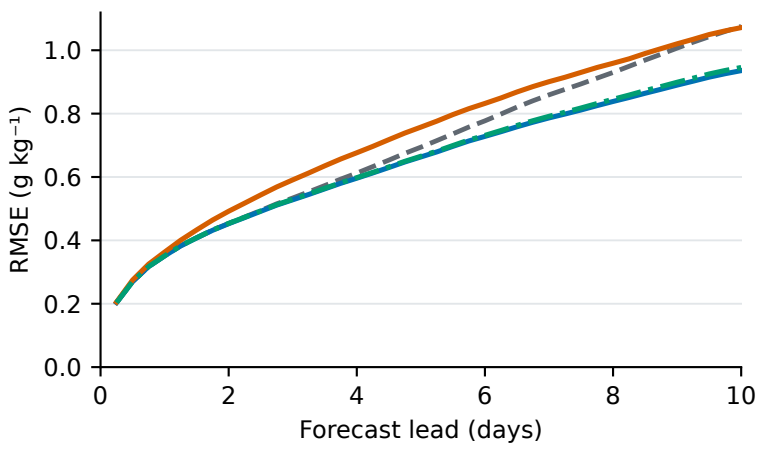
6 h precipitation



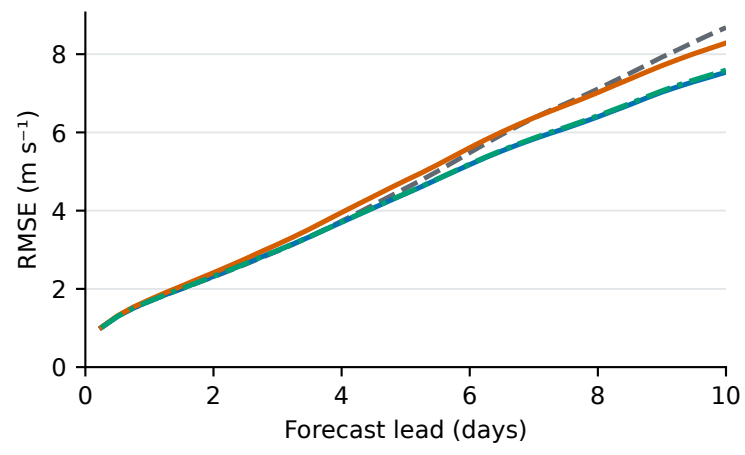
Temperature · all pressure levels



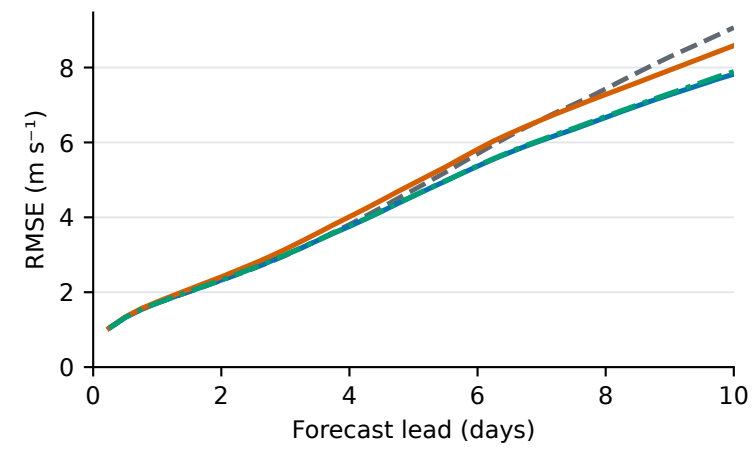
Specific humidity · all levels



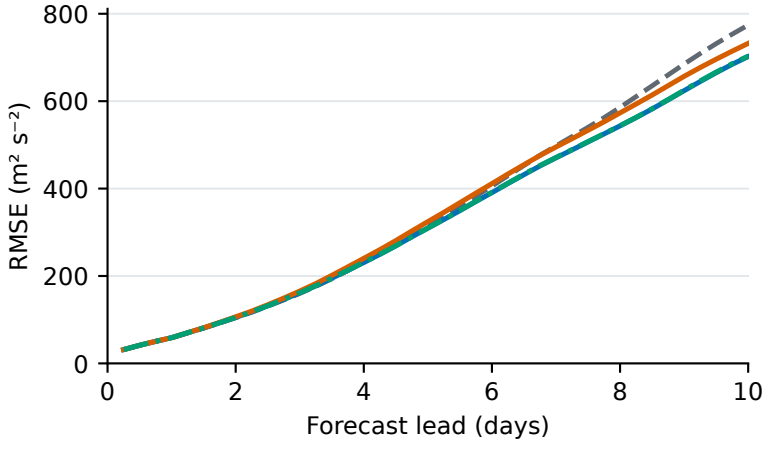
Zonal wind · all pressure levels



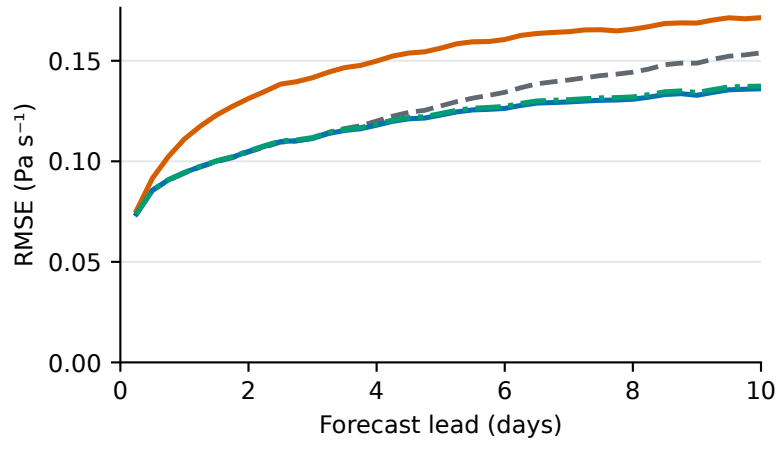
Meridional wind · all levels



Geopotential · all pressure levels



Vertical velocity · all levels



How to read these panels

Lower RMSE is better.
Each panel has its own physical units.

Upper-air groups pool squared error equally across all 13 pressure levels before taking the square root.

Global latitude-weighted errors pooled over the saved starts. These physical-variable RMSEs differ from the exact weighted GraphCast loss.

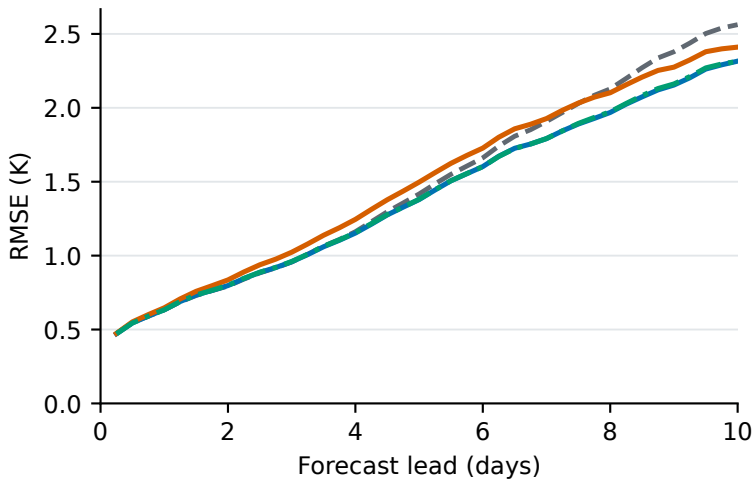
One common Original GC reference is shown; repeated baseline evaluations have small differences. No uncertainty intervals are available.

Global RMSE for the spectral diagnostic fields

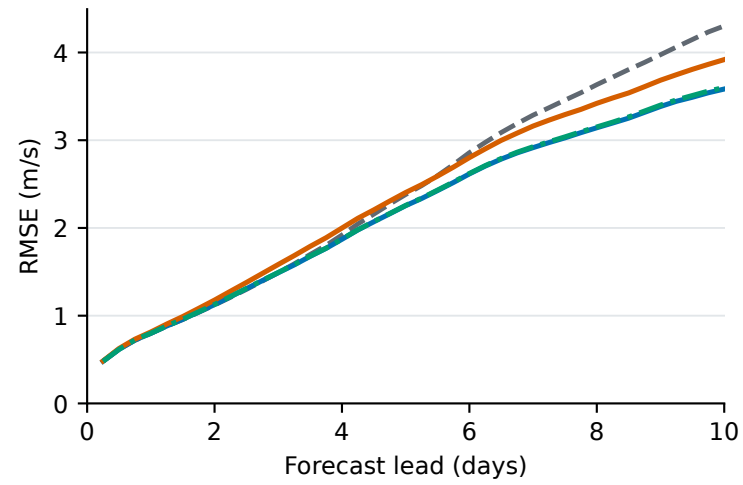
32 matched 2022 starts · 6-hourly leads through day 10 · residual models use SWA checkpoints

Original GC · MSE · AMSE λ=0.1 · Full AMSE

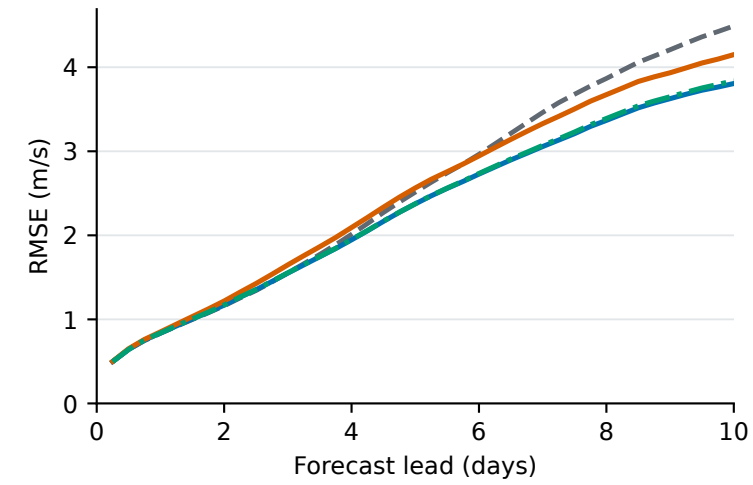
T2m · 2 m temperature



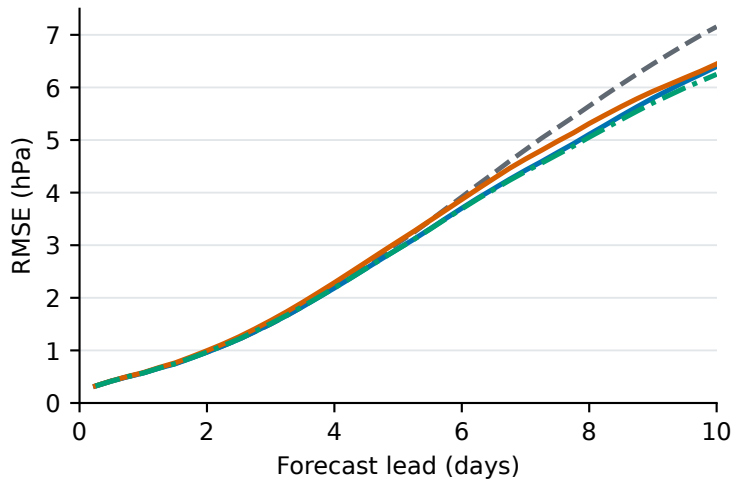
U10 · 10 m zonal wind



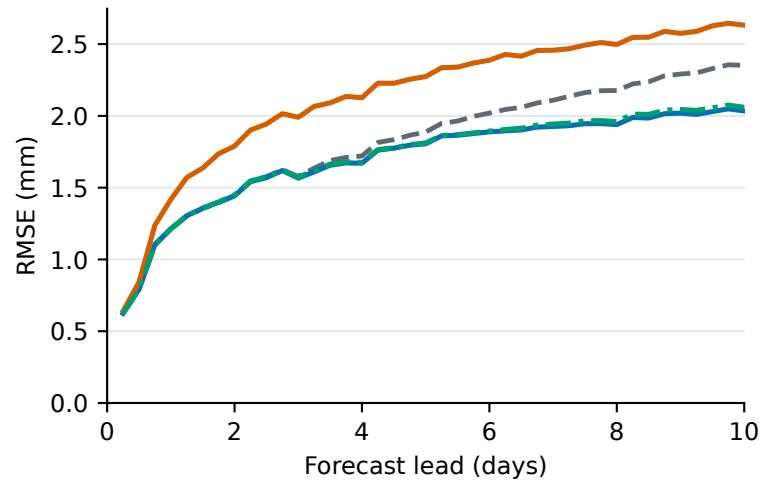
V10 · 10 m meridional wind



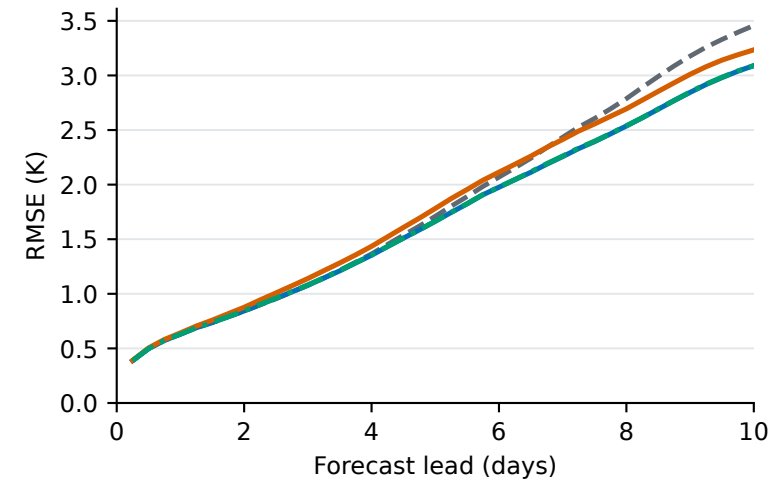
MSLP · Mean sea-level pressure



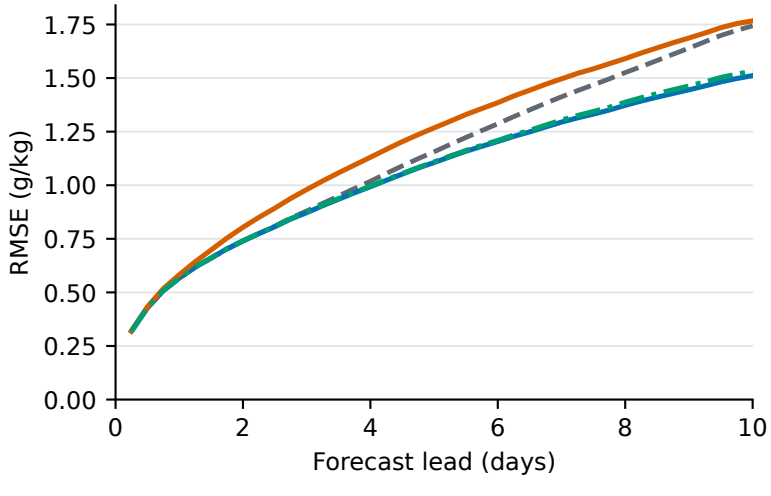
TP6 · 6 h precipitation



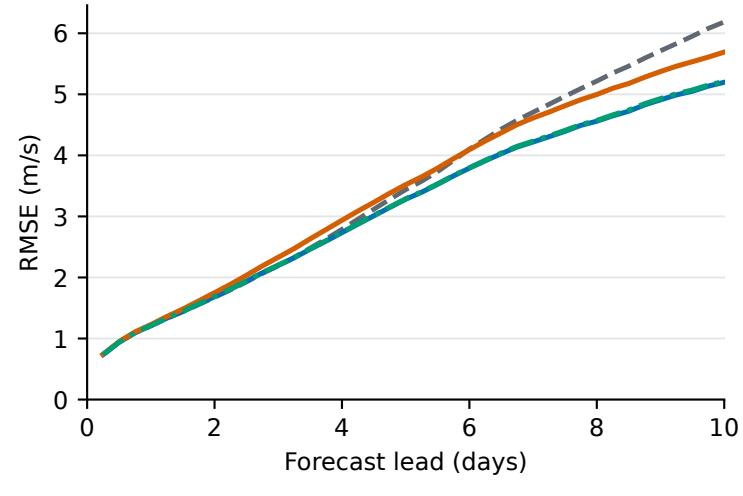
T850 · 850 hPa temperature



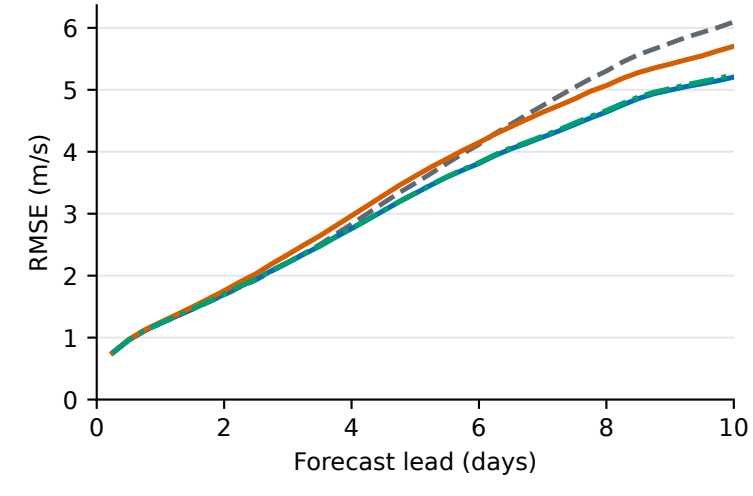
Q700 · 700 hPa specific humidity



U850 · 850 hPa zonal wind



V850 · 850 hPa meridional wind



Global latitude-weighted physical RMSE at the specified level; the companion spectral curves use only 30–60° latitude in each hemisphere.

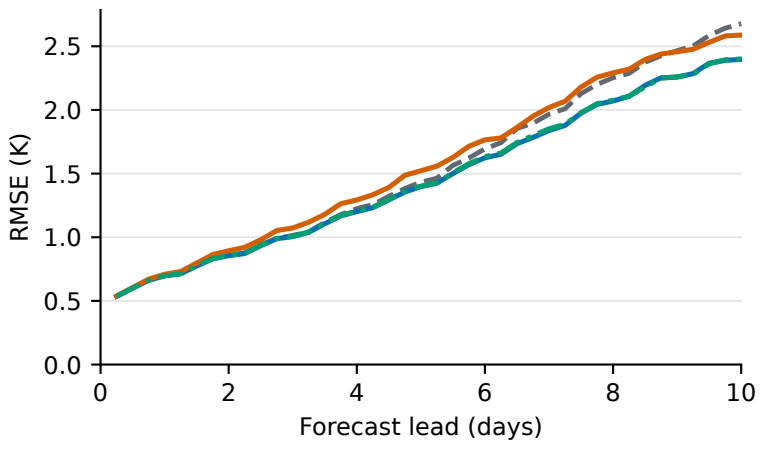
Global wind-speed RMSE (WS10 / WS850) was not saved; it cannot be reconstructed from component RMSEs. No uncertainty intervals.

Forecast RMSE by variable

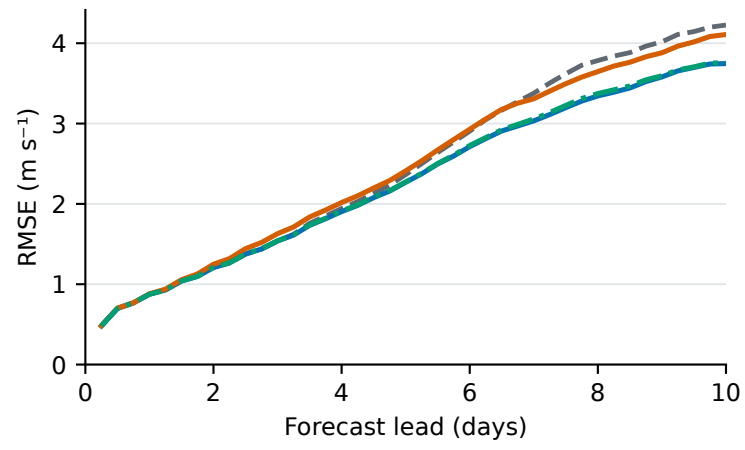
8 matched 2023 starts · 6-hourly leads through day 10 · residual models use SWA checkpoints

Original GC · MSE · AMSE λ=0.1 · Full AMSE

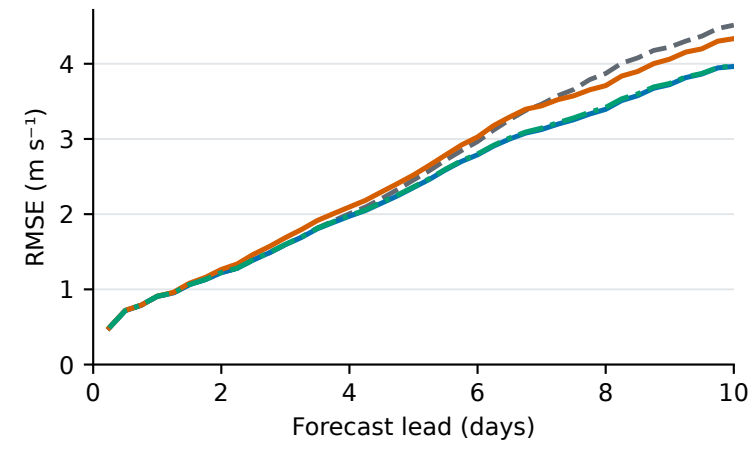
2 m temperature



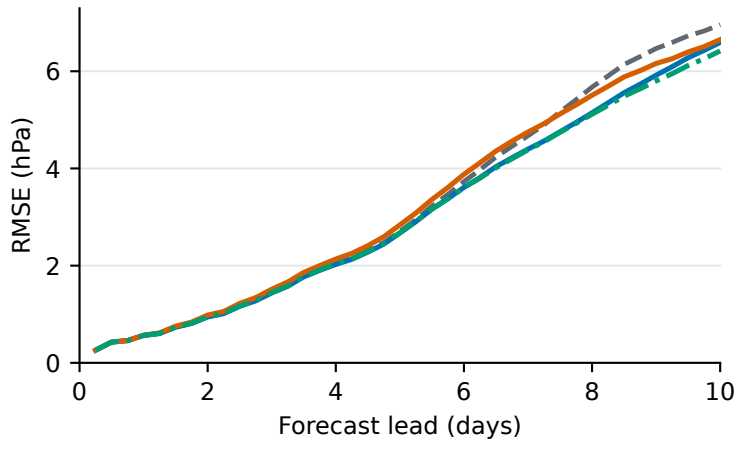
10 m zonal wind



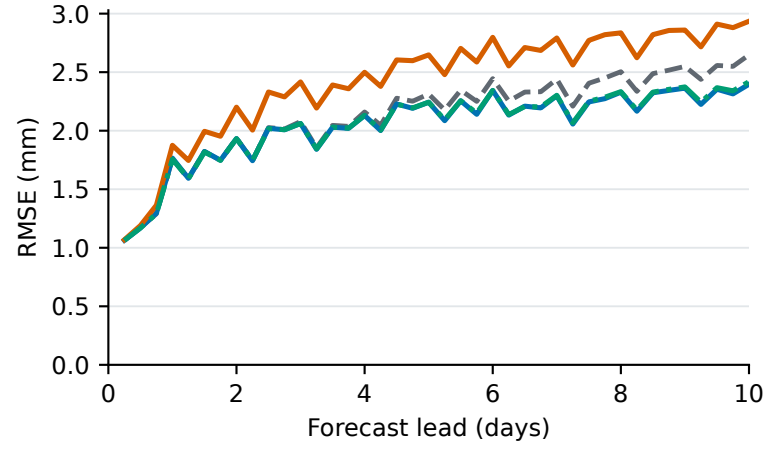
10 m meridional wind



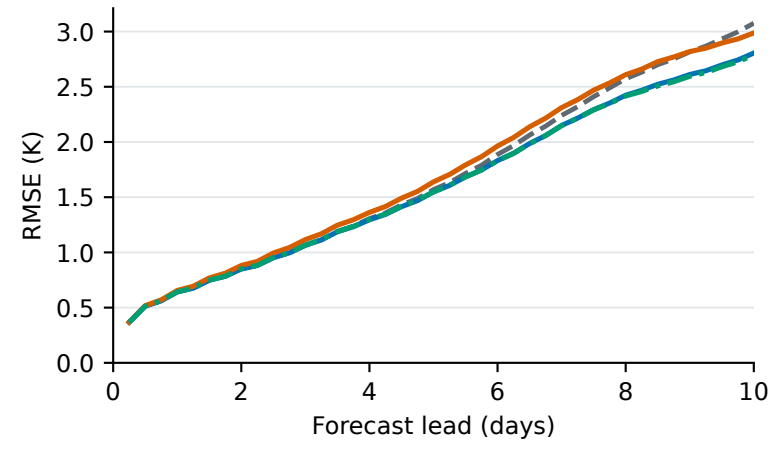
Mean sea-level pressure



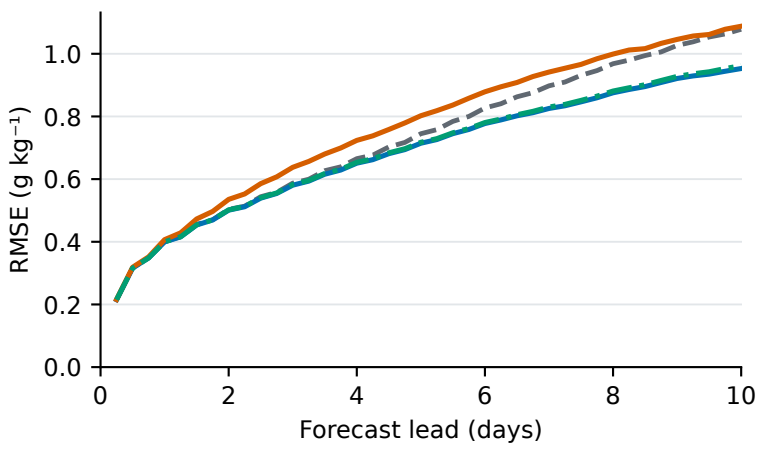
6 h precipitation



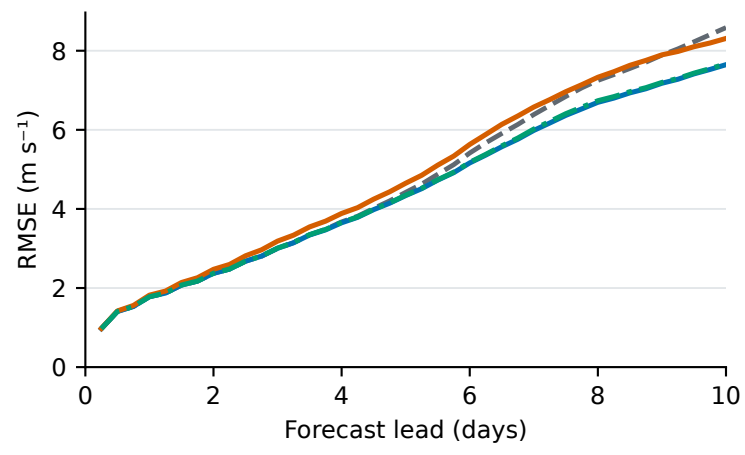
Temperature · all pressure levels



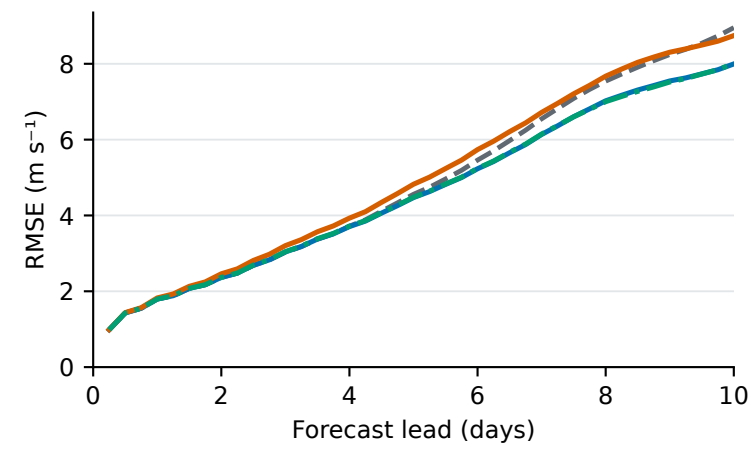
Specific humidity · all levels



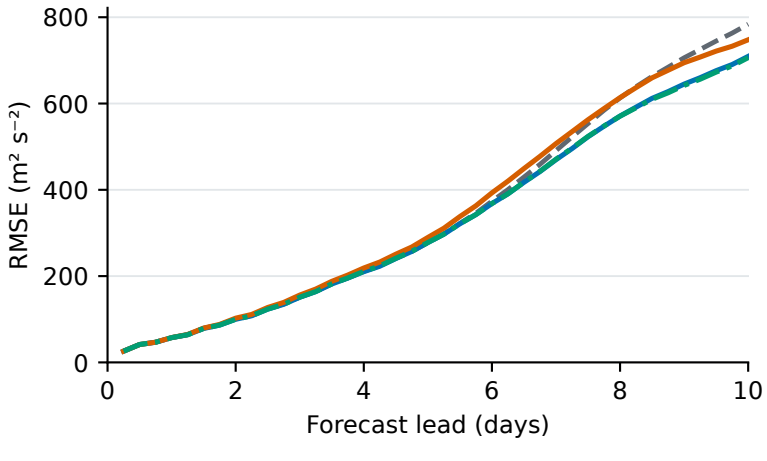
Zonal wind · all pressure levels



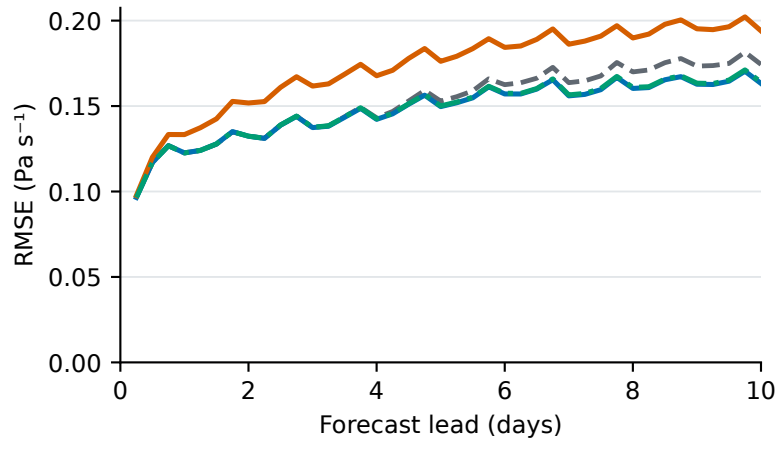
Meridional wind · all levels



Geopotential · all pressure levels



Vertical velocity · all levels



How to read these panels

Lower RMSE is better.
Each panel has its own physical units.

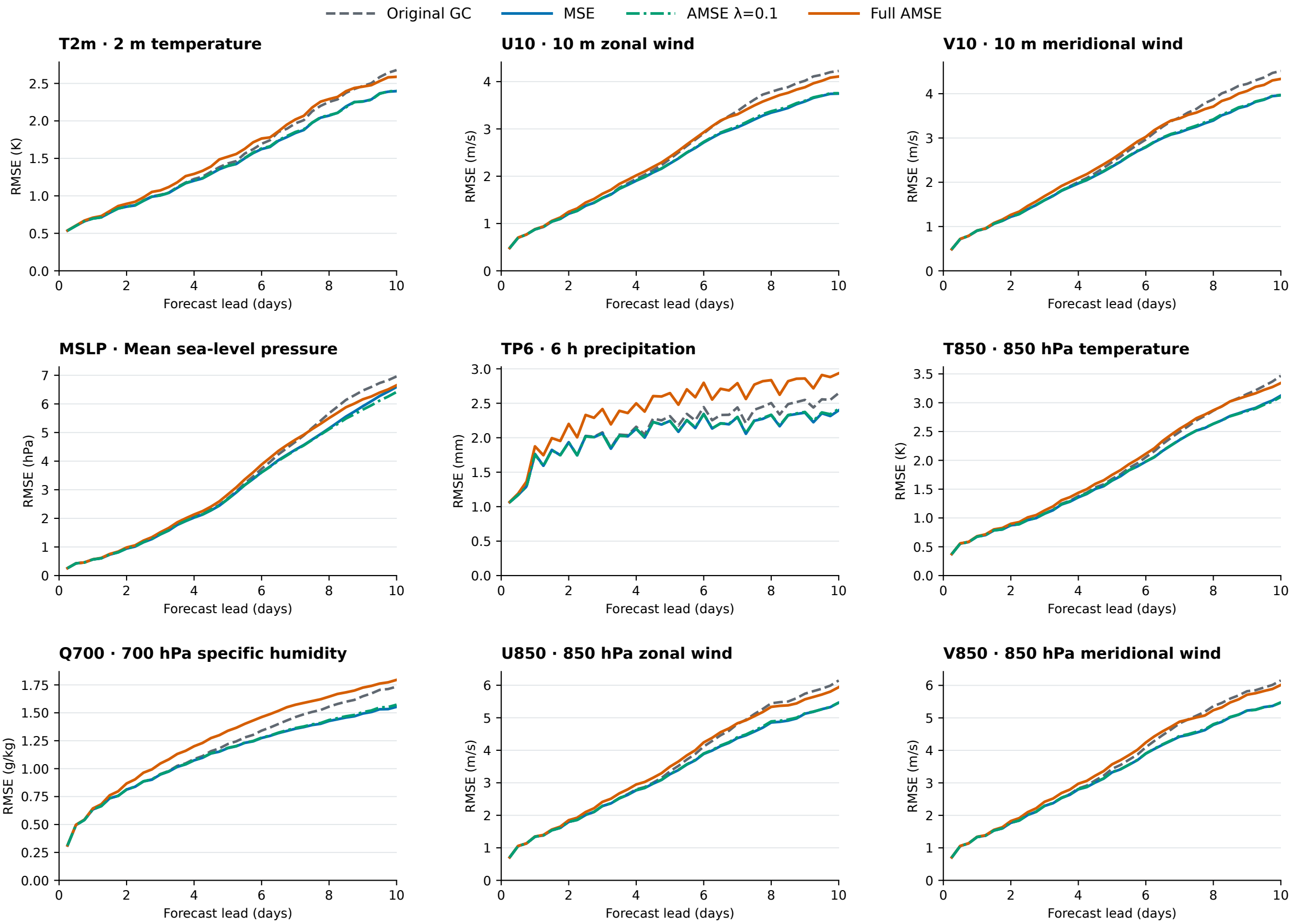
Upper-air groups pool squared error equally across all 13 pressure levels before taking the square root.

Global latitude-weighted errors pooled over the saved starts. These physical-variable RMSEs differ from the exact weighted GraphCast loss.

One common Original GC reference is shown; repeated baseline evaluations have small differences. No uncertainty intervals are available.

Global RMSE for the spectral diagnostic fields

8 matched 2023 starts · 6-hourly leads through day 10 · residual models use SWA checkpoints

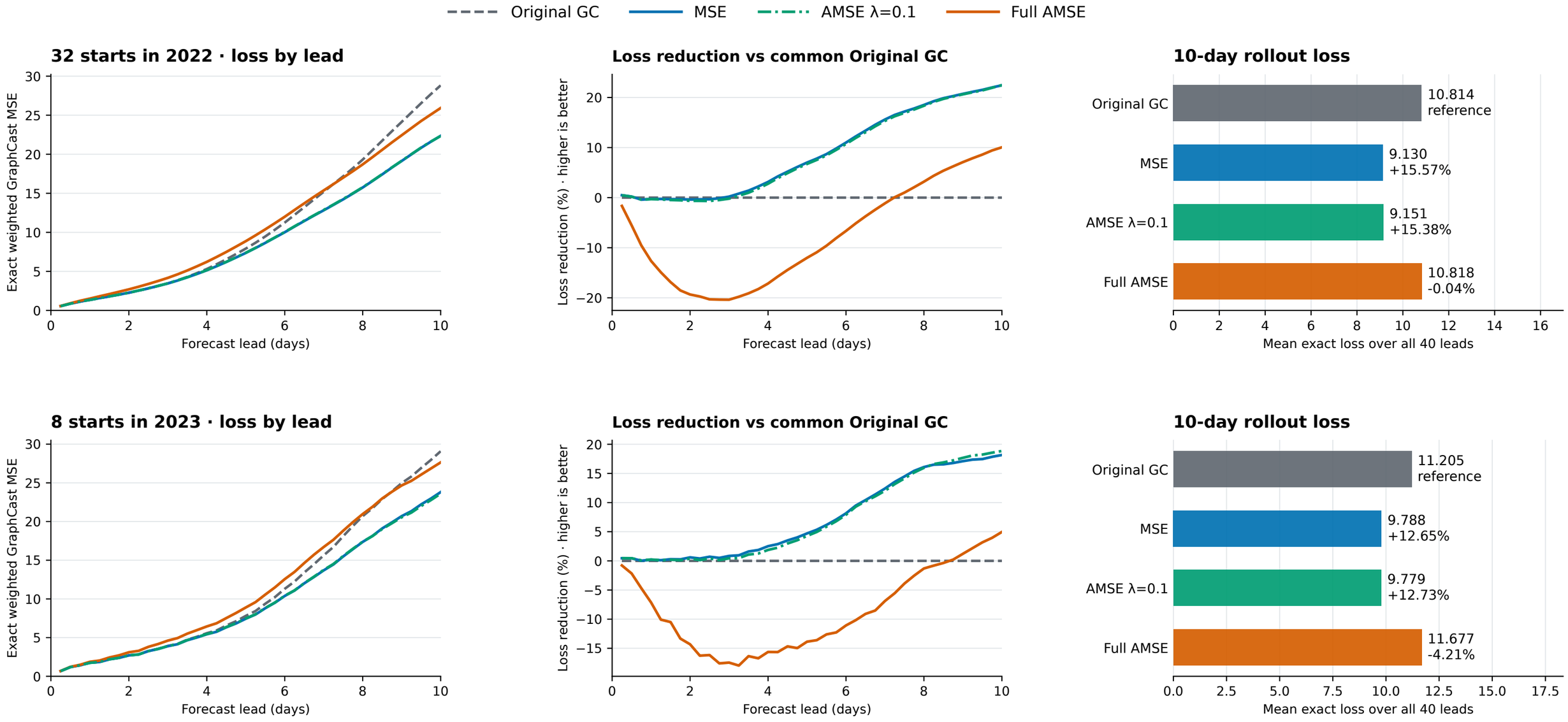


Global latitude-weighted physical RMSE at the specified level; the companion spectral curves use only 30–60° latitude in each hemisphere.

Global wind-speed RMSE (WS10 / WS850) was not saved; it cannot be reconstructed from component RMSEs. No uncertainty intervals.

Exact GraphCast forecast loss

Original variable weights, difference-standard-deviation normalization, pressure-level weights and latitude weights



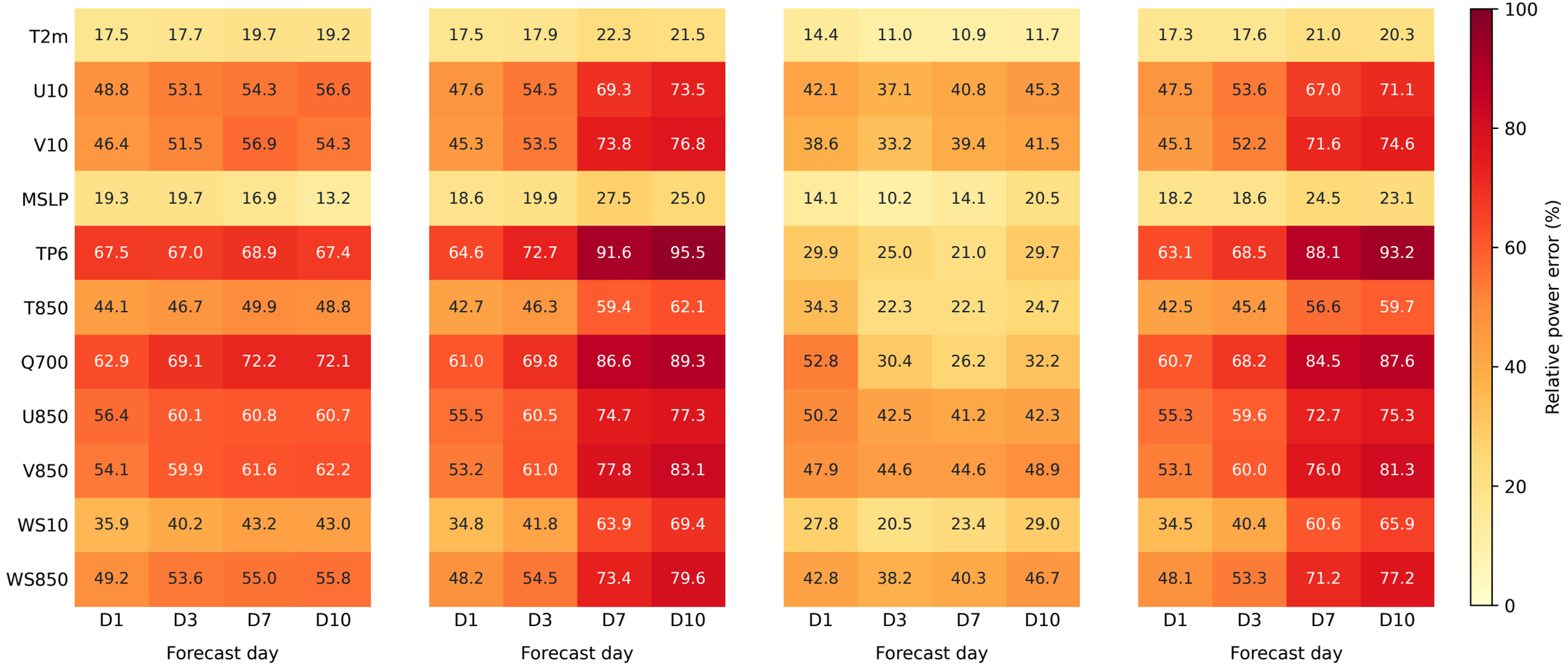
All plotted reductions use the same displayed Original GC within each dataset. They are ratios of losses; they are not RMSE reductions.

Rollout reduction is computed after pooling all 40 leads. Own-paired baseline results and baseline repeatability differences are retained in the data audit.

Spectral power error across all fields

Mean relative power error (%) · Lower is better

Each cell weights wavenumbers k = 1–180 equally. A common color scale is used for all models.



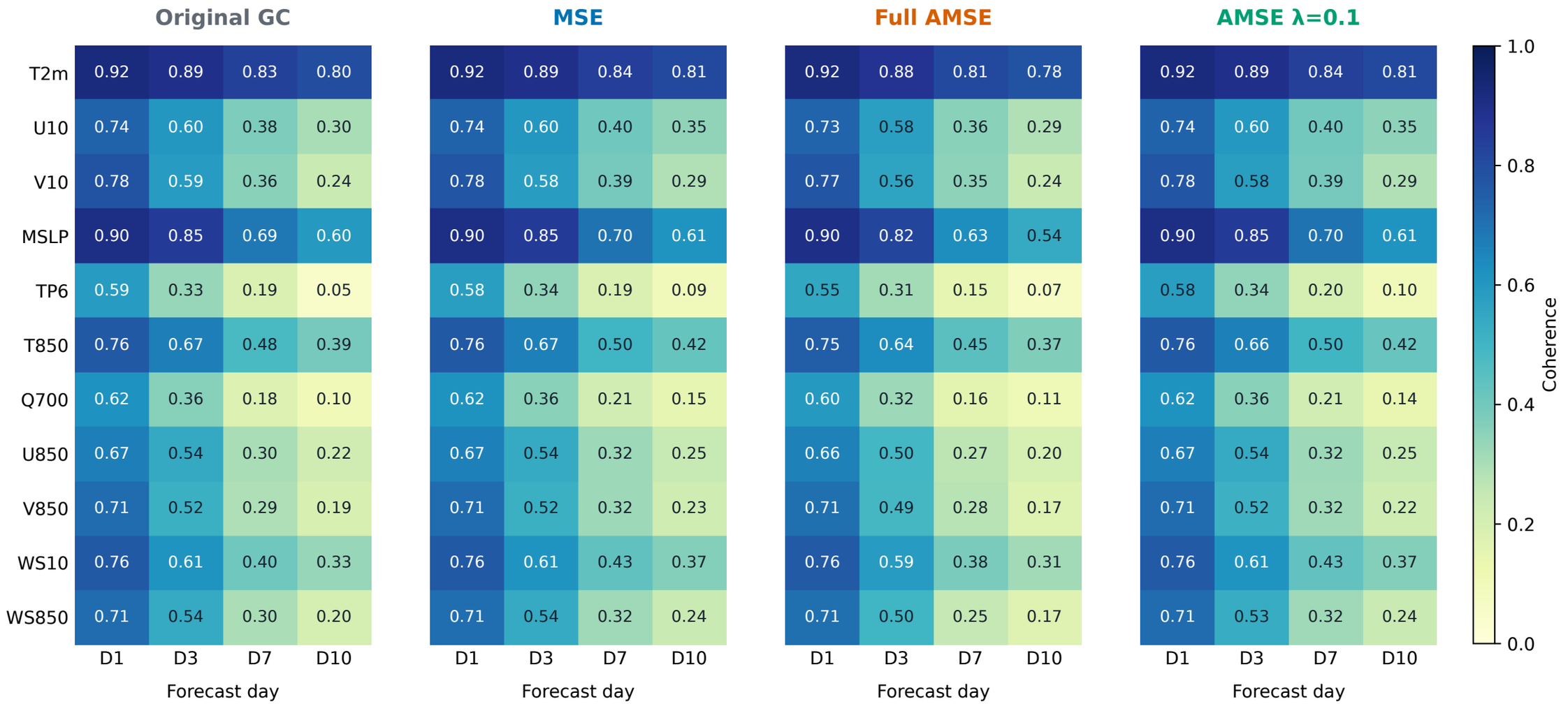
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Field names and physical units are given on the individual diagnostic pages. No uncertainty intervals.

Forecast–truth coherence across all fields

Mean real spectral coherence · Higher is better

Each cell weights wavenumbers k = 1–180 equally. A common color scale is used for all models.



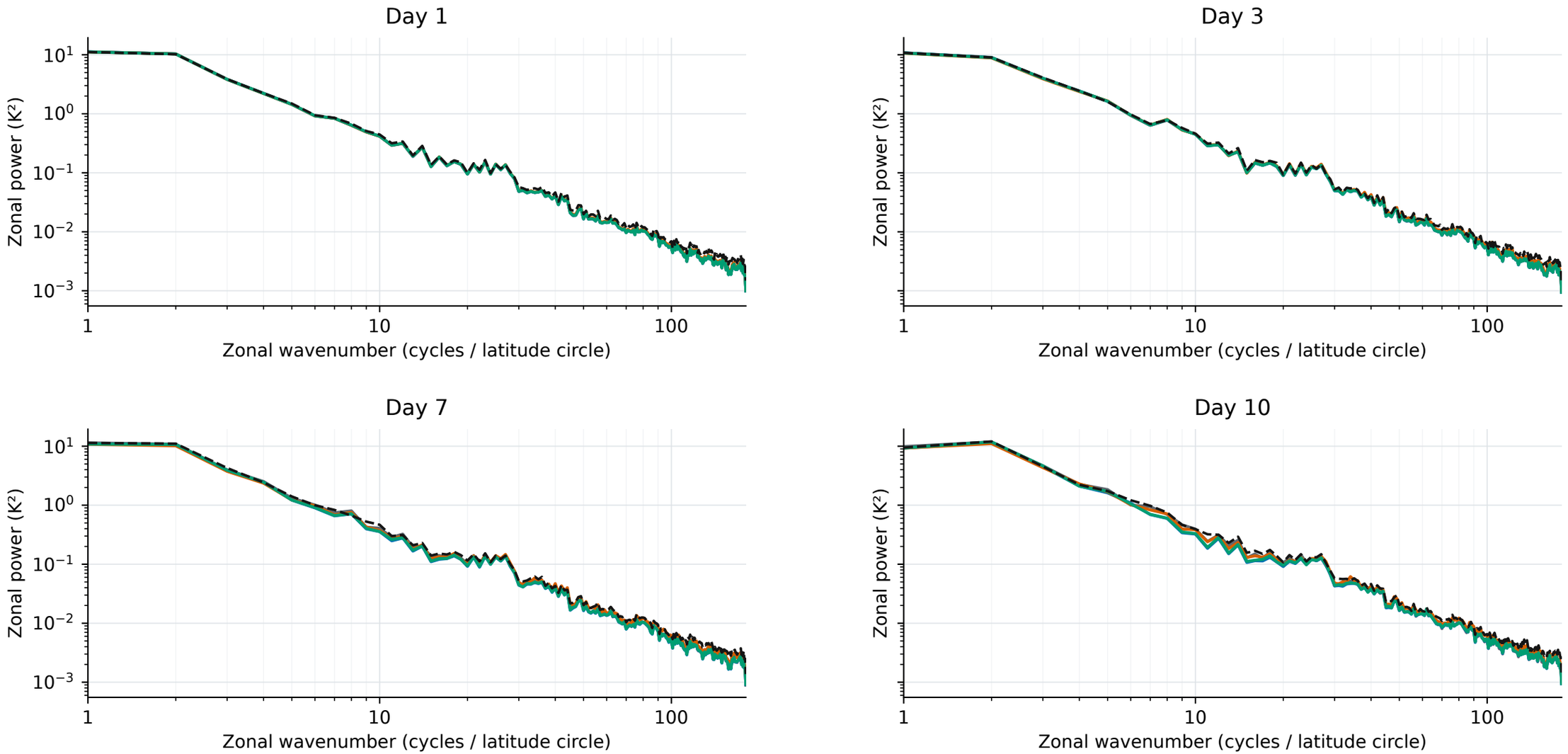
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Field names and physical units are given on the individual diagnostic pages. No uncertainty intervals.

2 m temperature (T2m; K)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



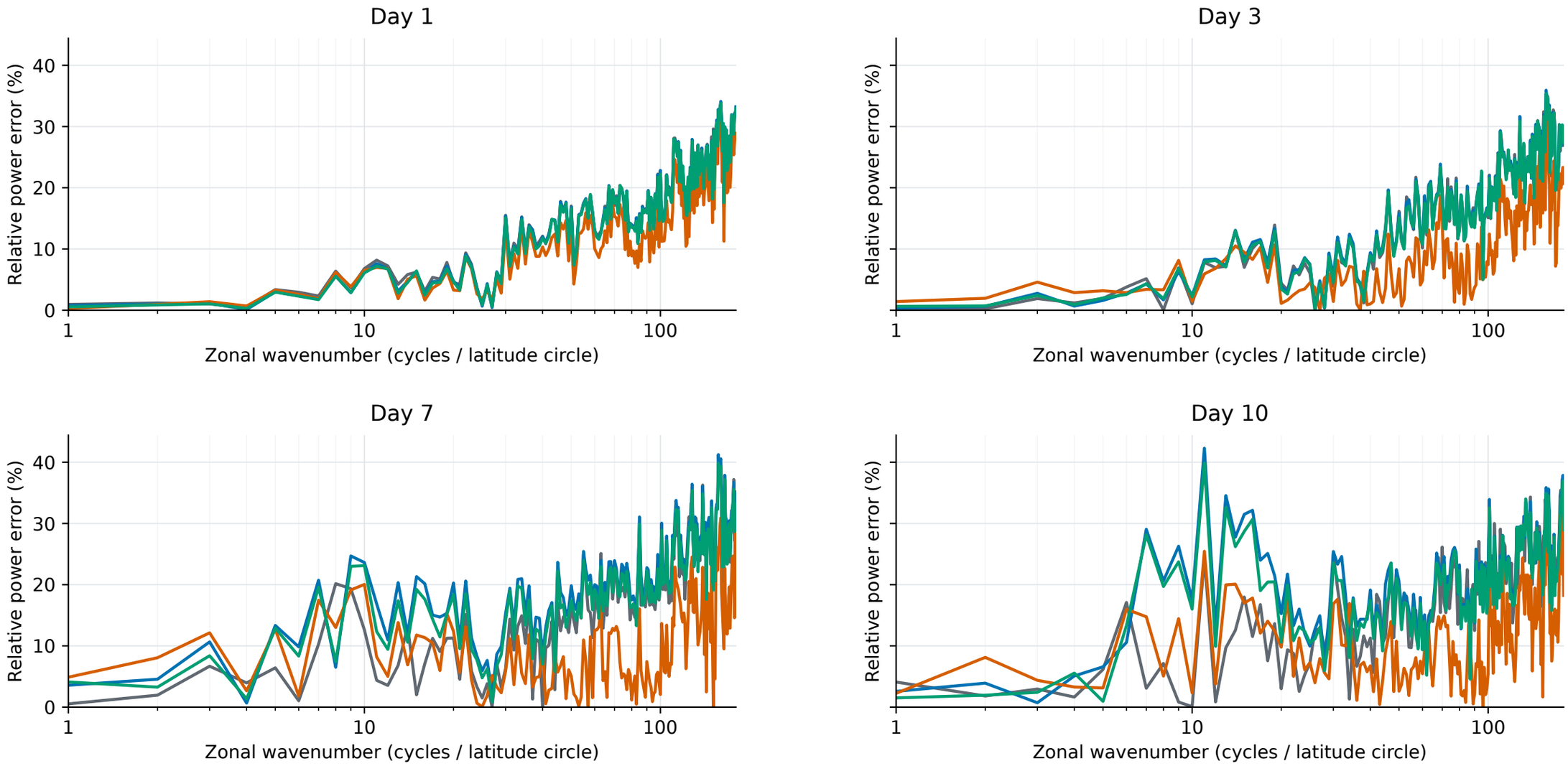
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

2 m temperature (T2m; K)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



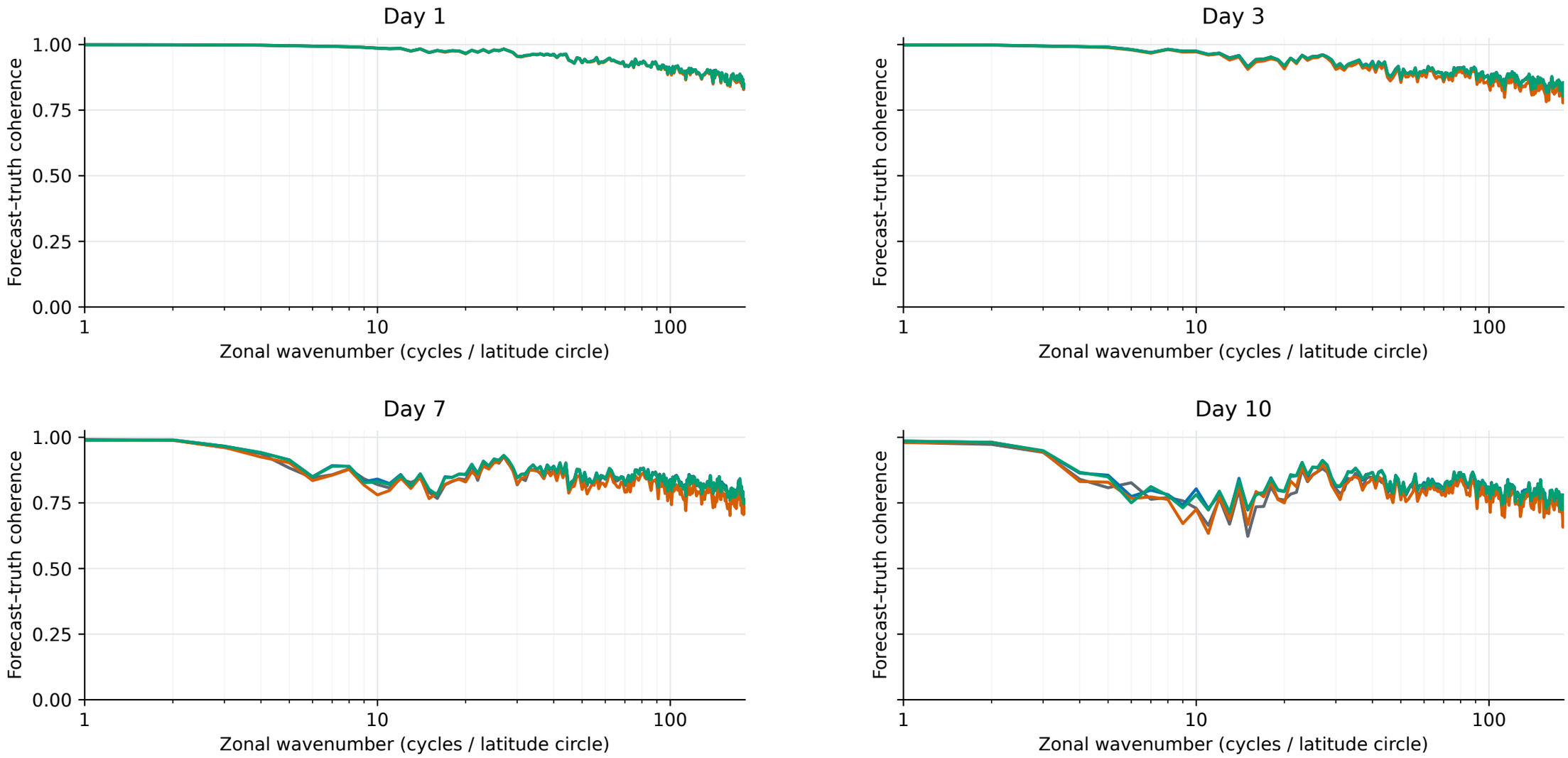
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

2 m temperature (T2m; K)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



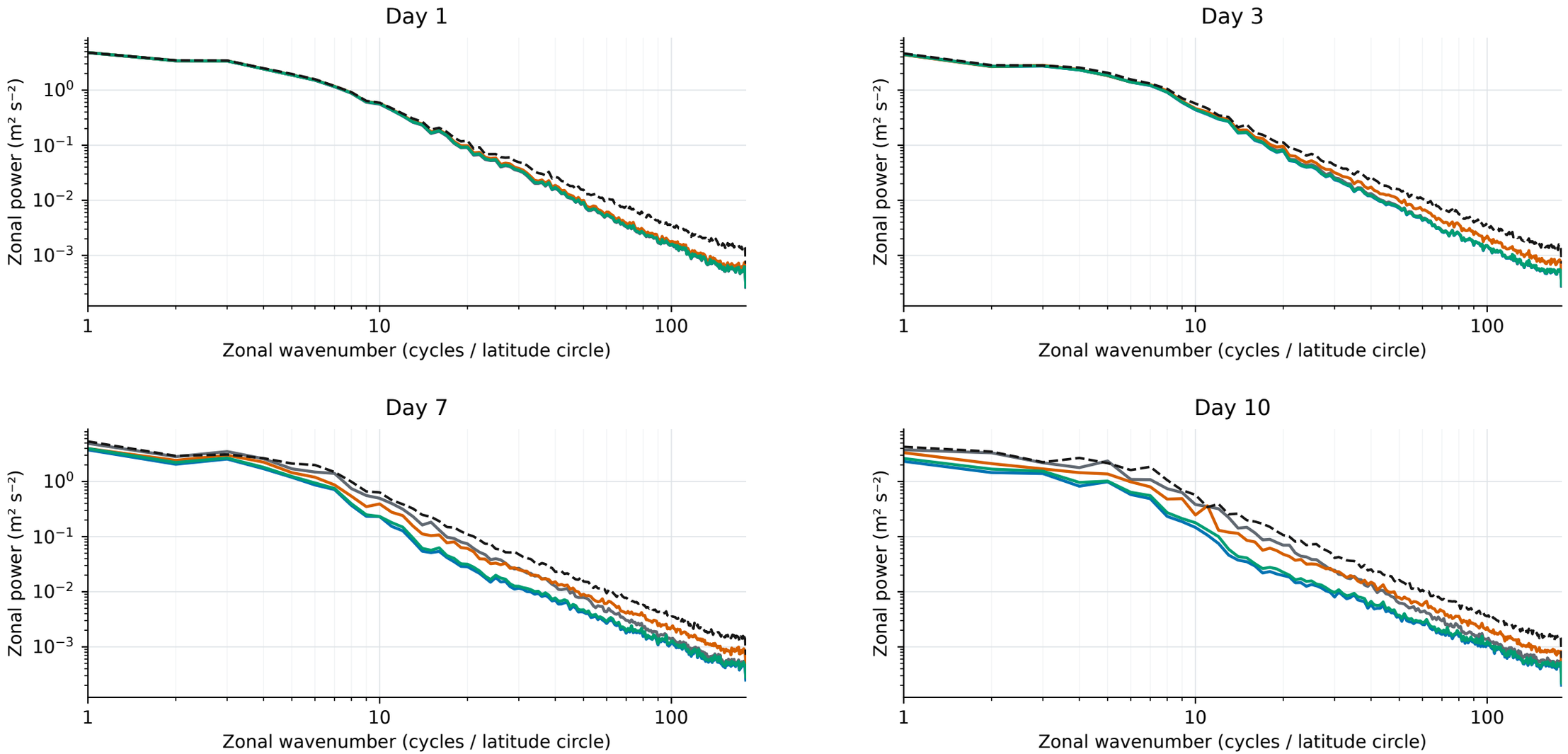
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

10 m zonal wind (U10; m s⁻¹)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



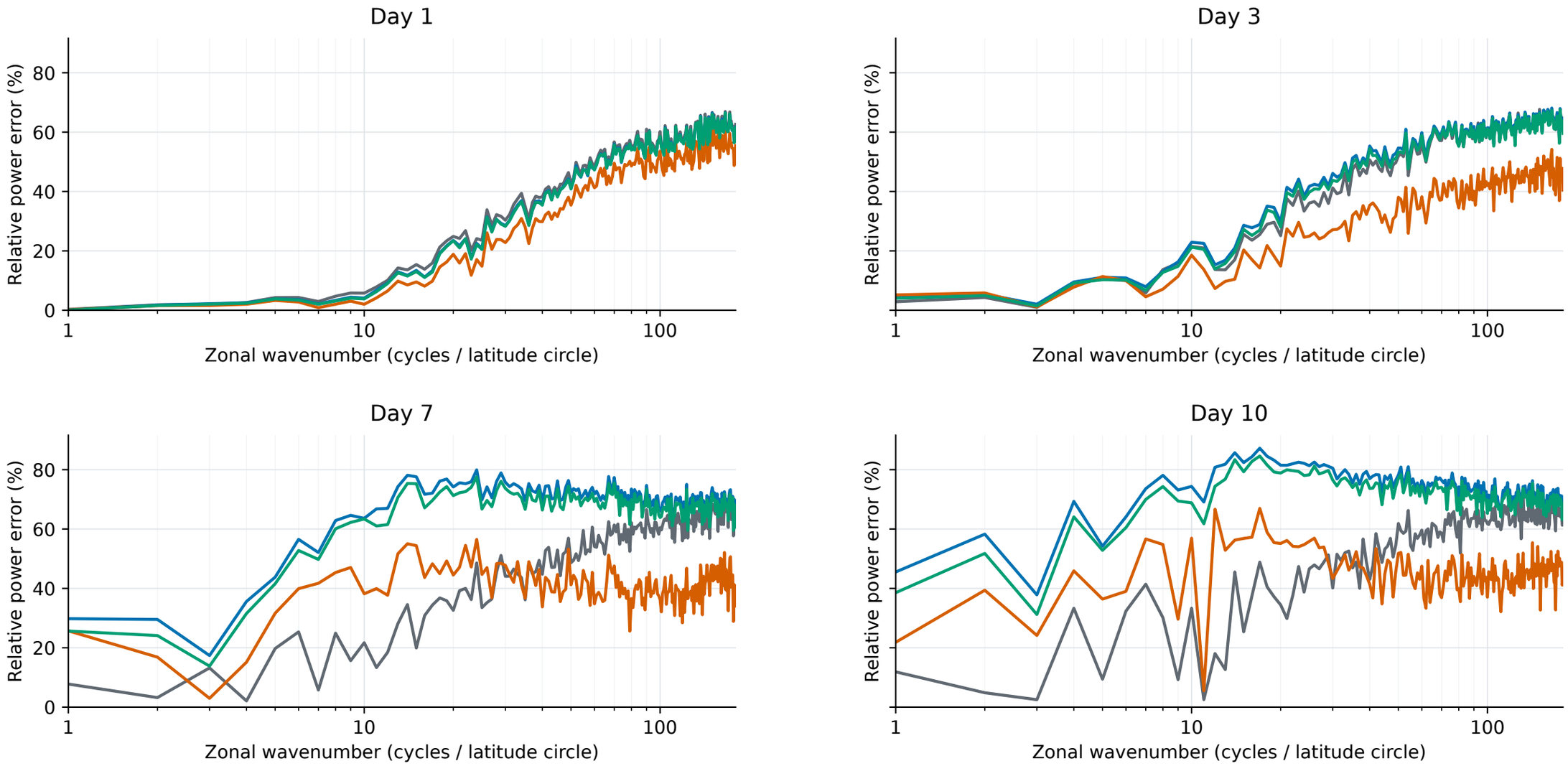
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

10 m zonal wind (U10; m s⁻¹)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



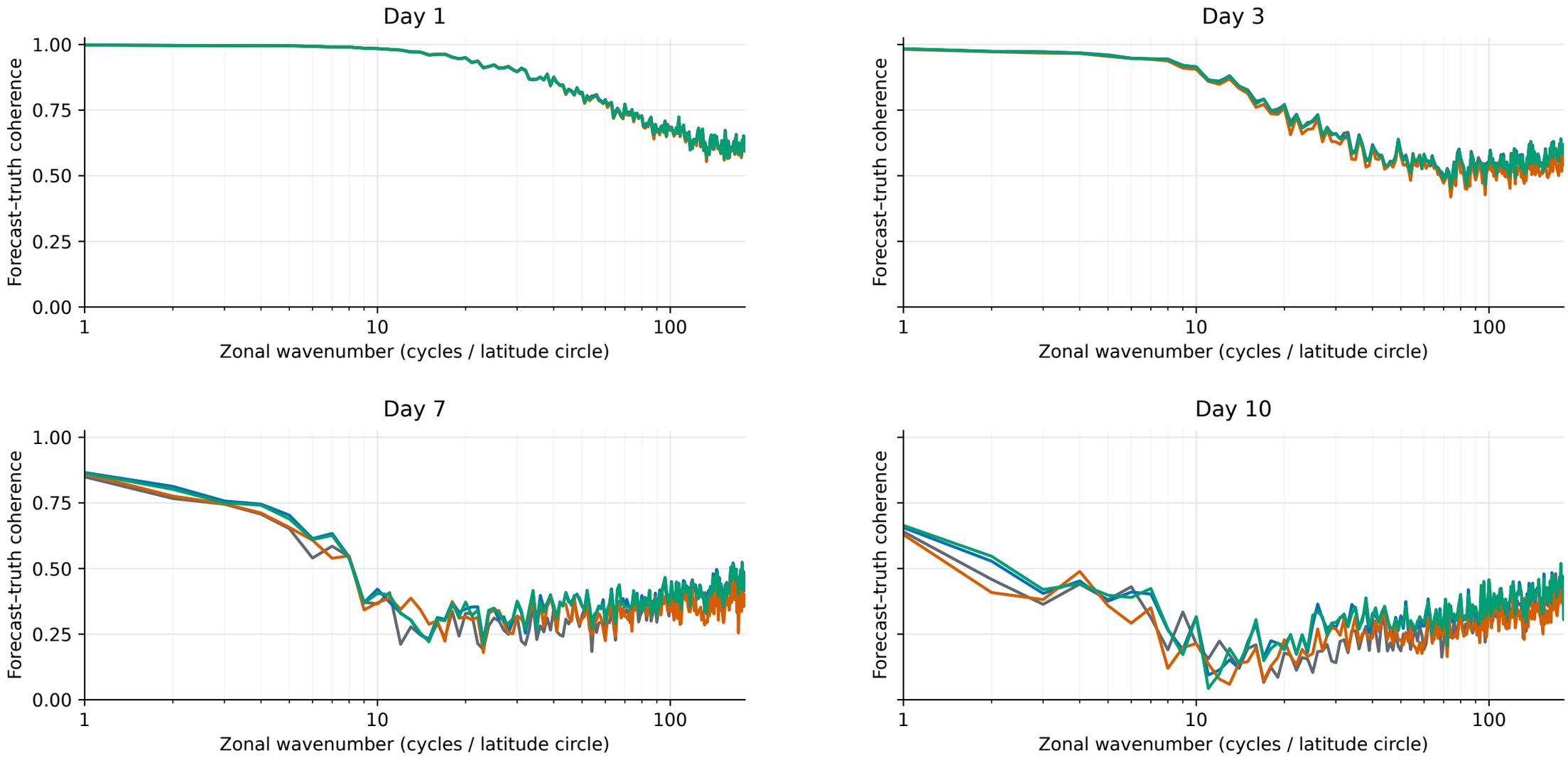
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

10 m zonal wind (U10; m s⁻¹)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



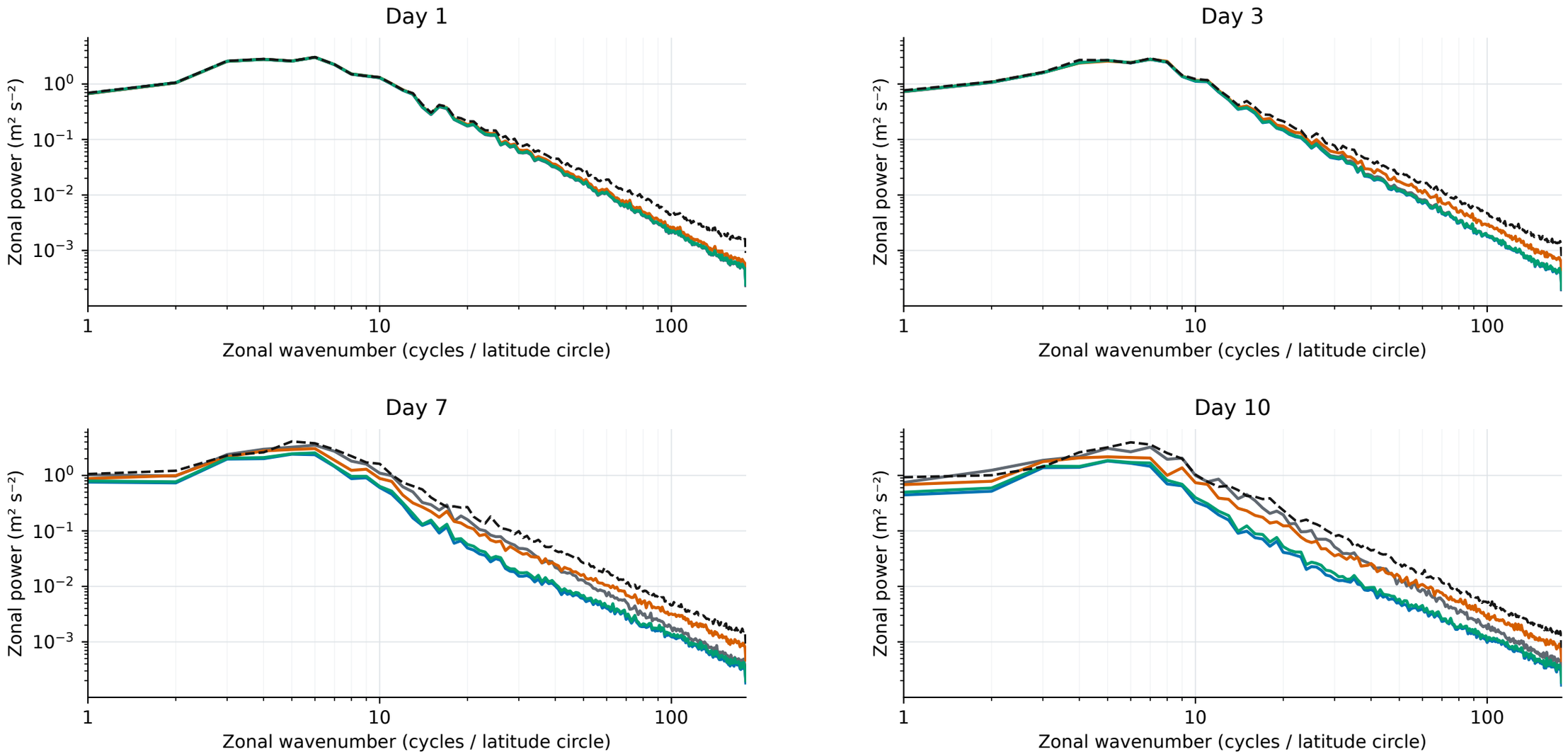
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

10 m meridional wind (V10; m s⁻¹)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



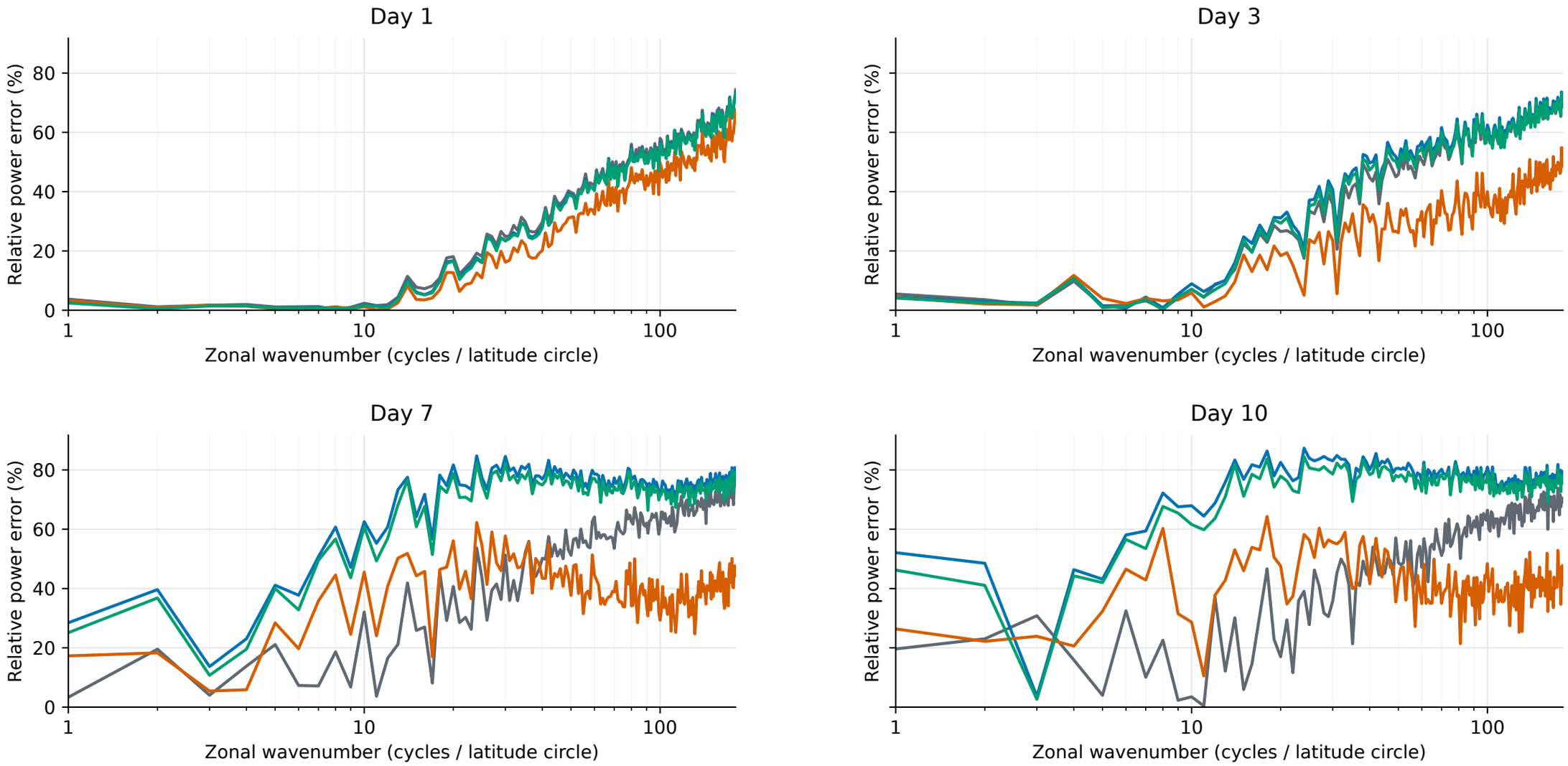
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

10 m meridional wind (V10; m s⁻¹)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



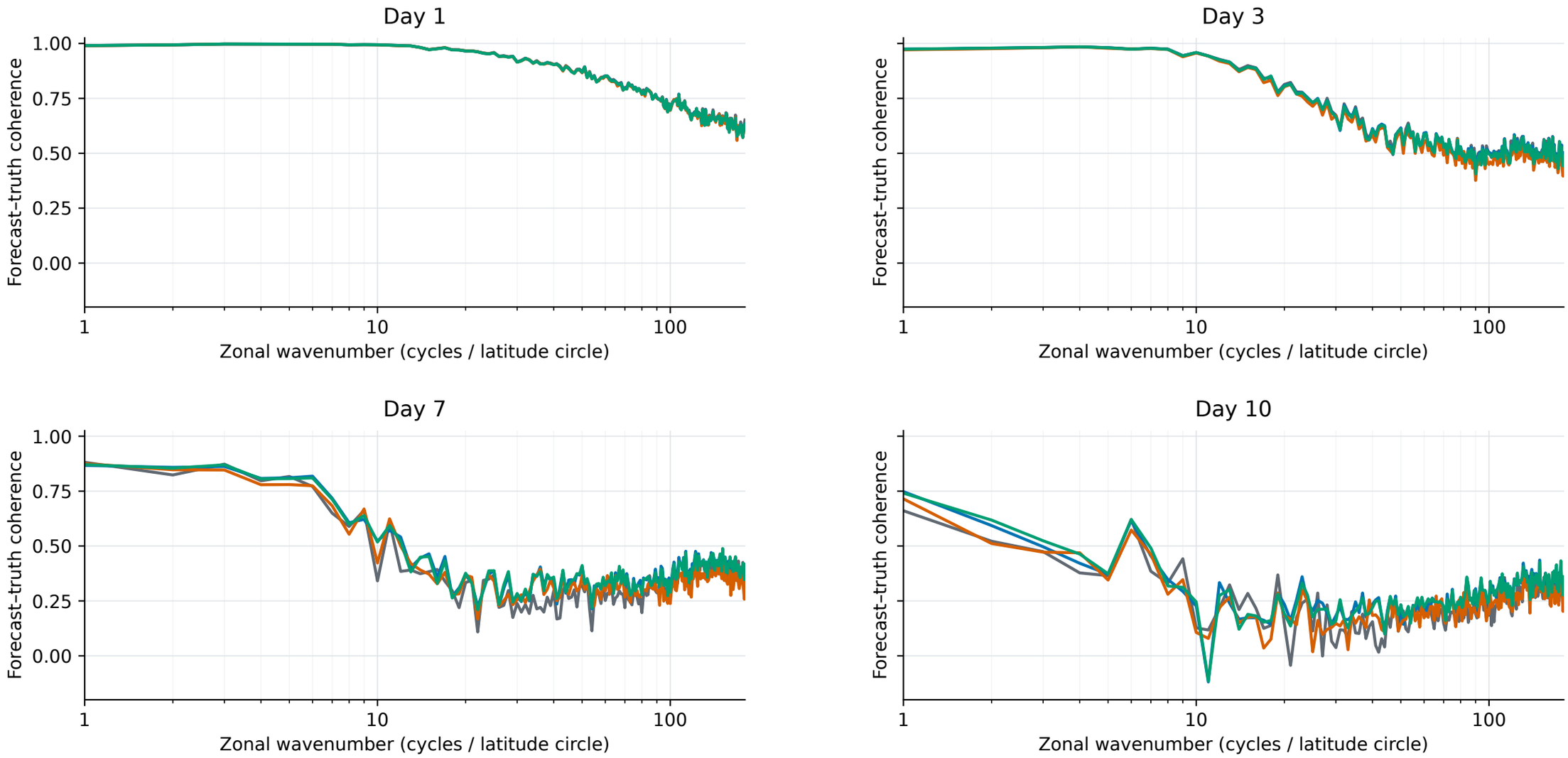
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

10 m meridional wind (V10; m s⁻¹)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



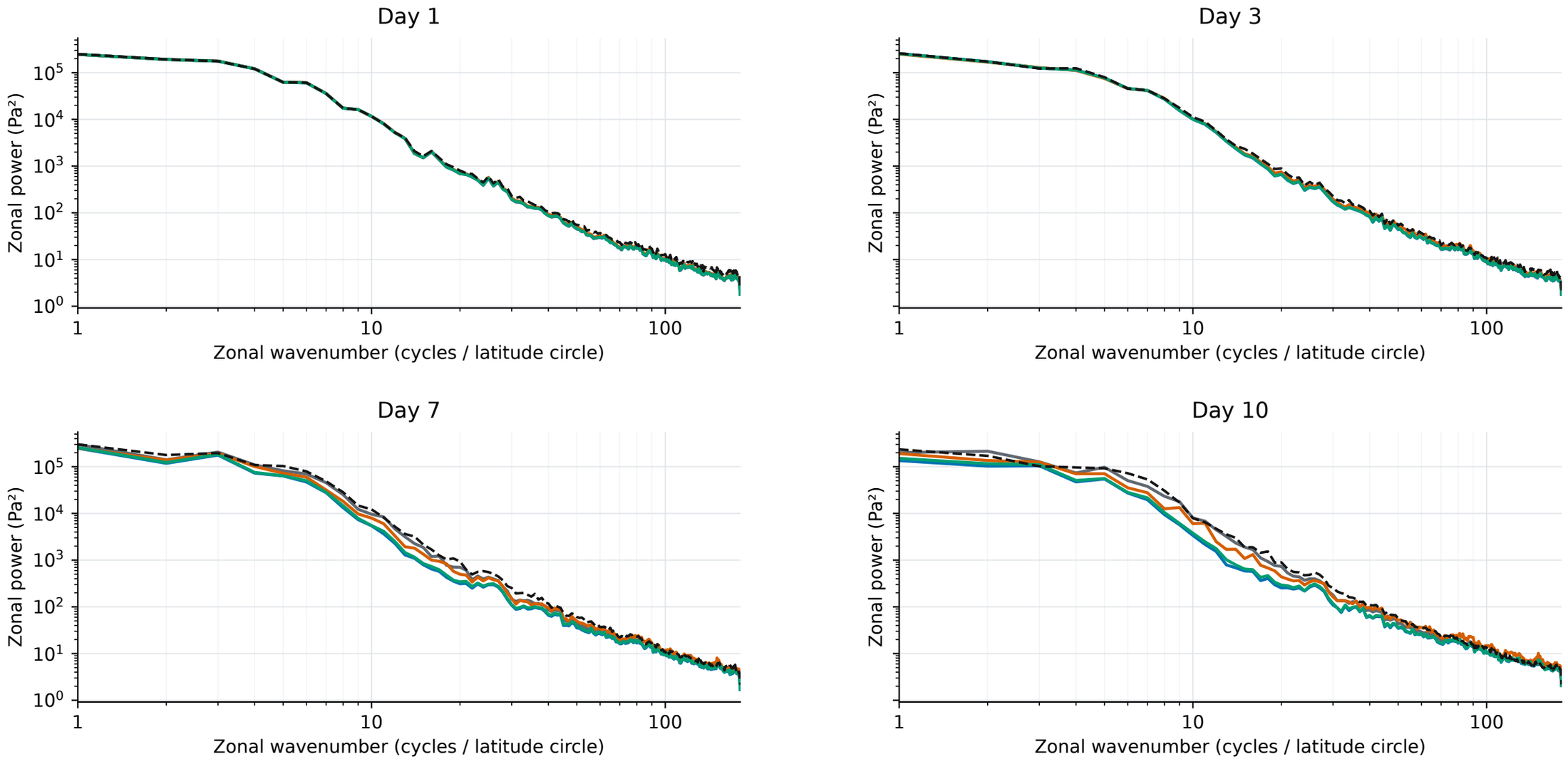
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Mean sea-level pressure (MSLP; Pa)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



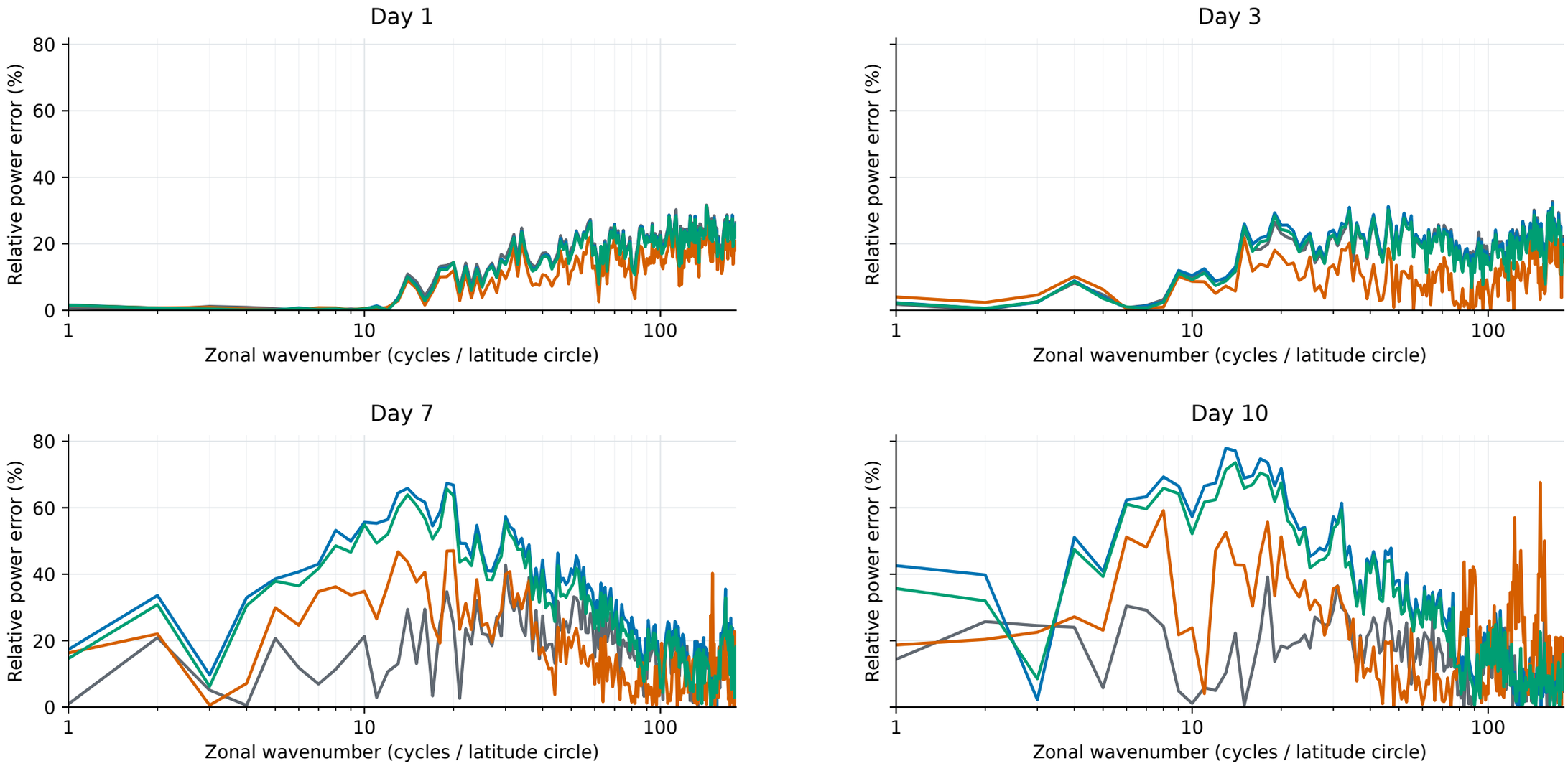
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Mean sea-level pressure (MSLP; Pa)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



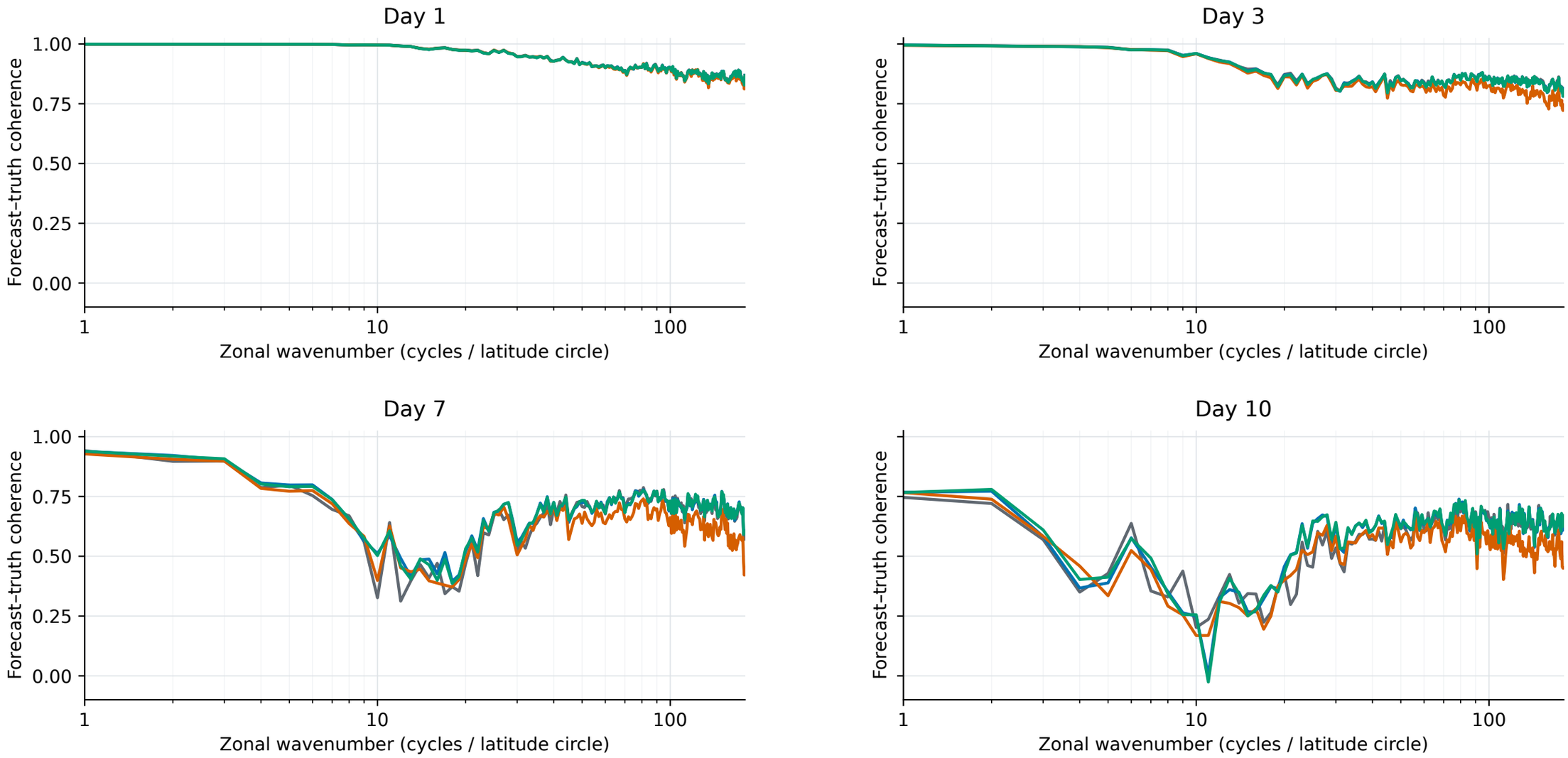
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Mean sea-level pressure (MSLP; Pa)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



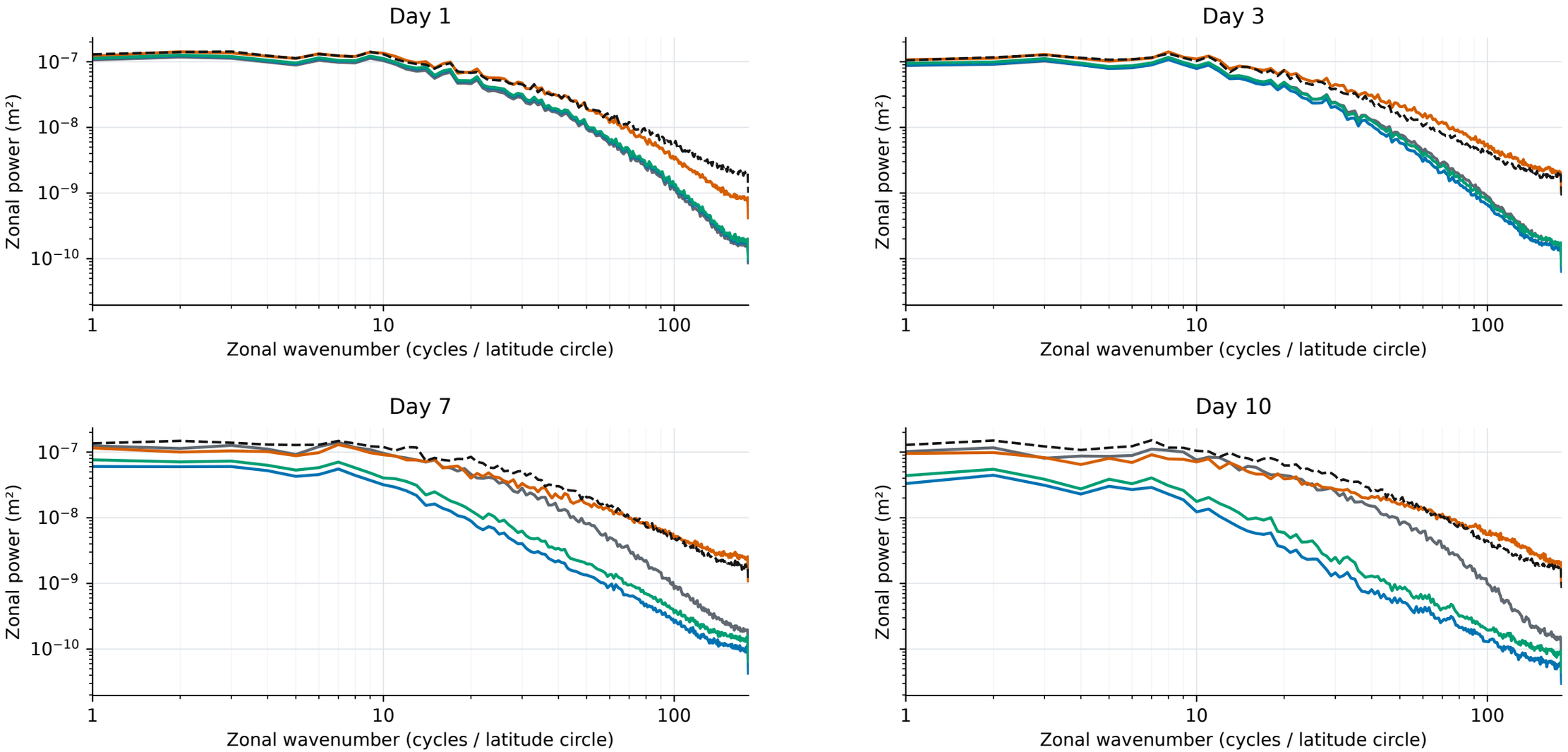
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

6-hour accumulated precipitation (TP6; m)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



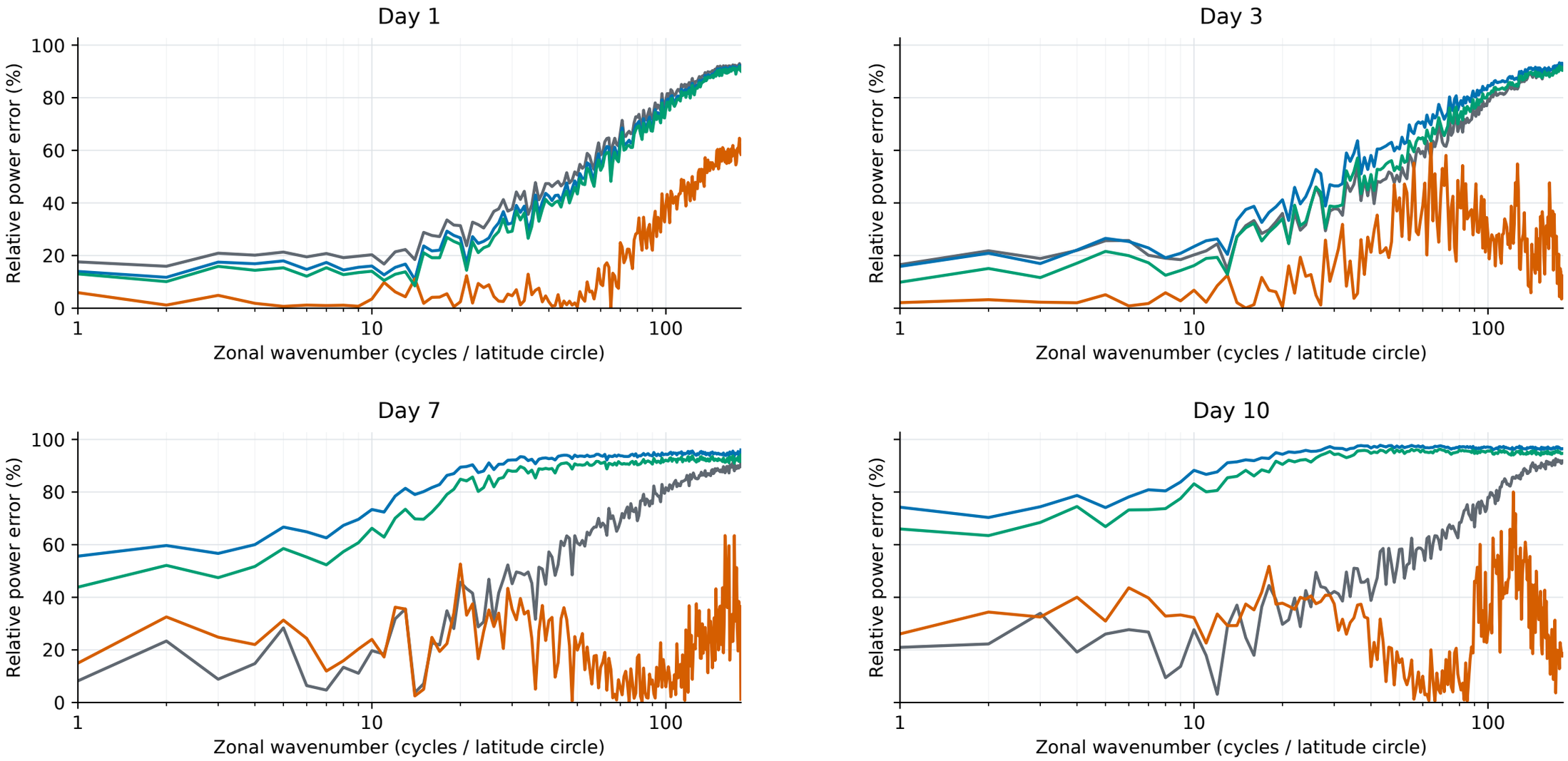
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

6-hour accumulated precipitation (TP6; m)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



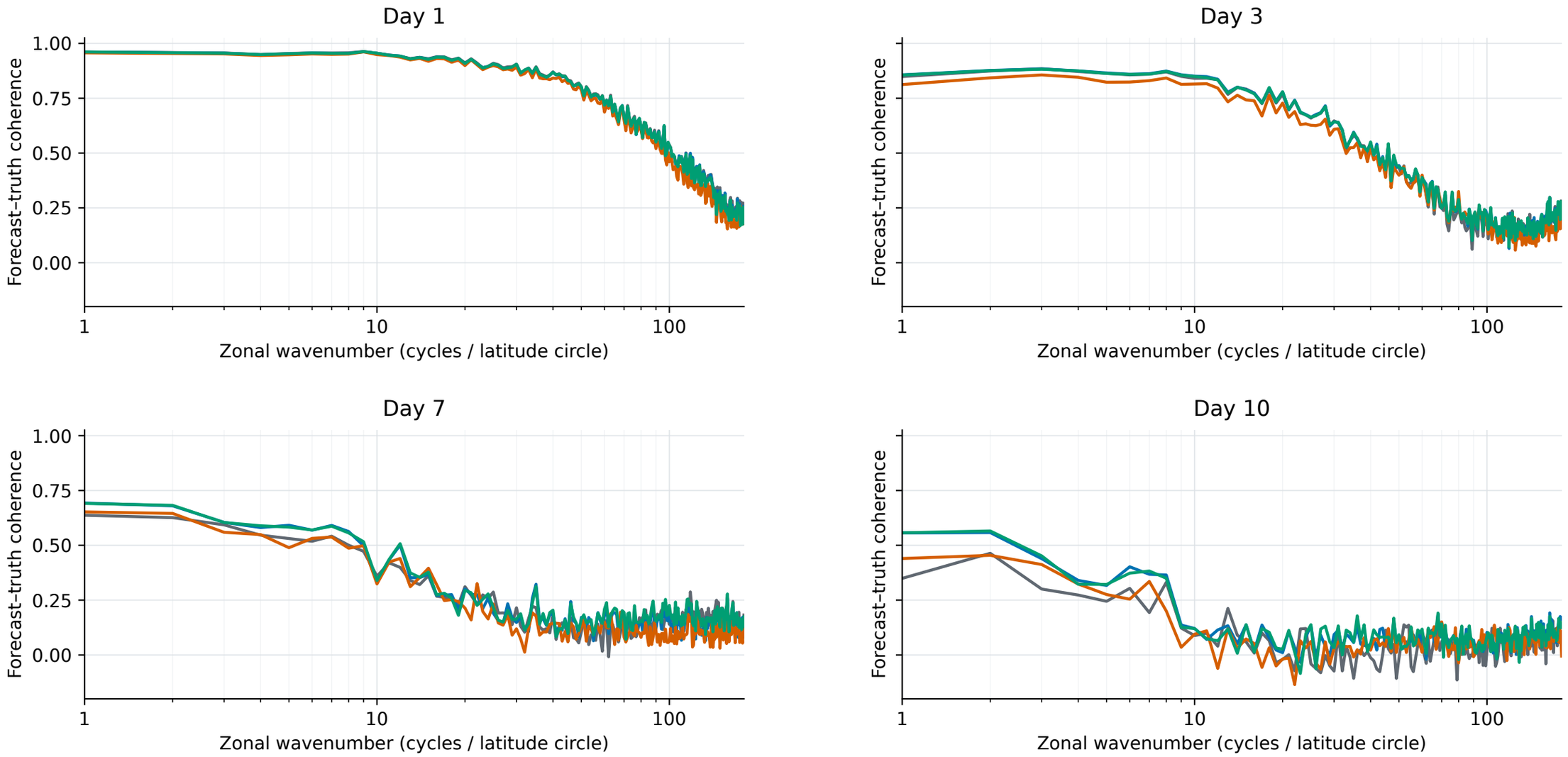
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

6-hour accumulated precipitation (TP6; m)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



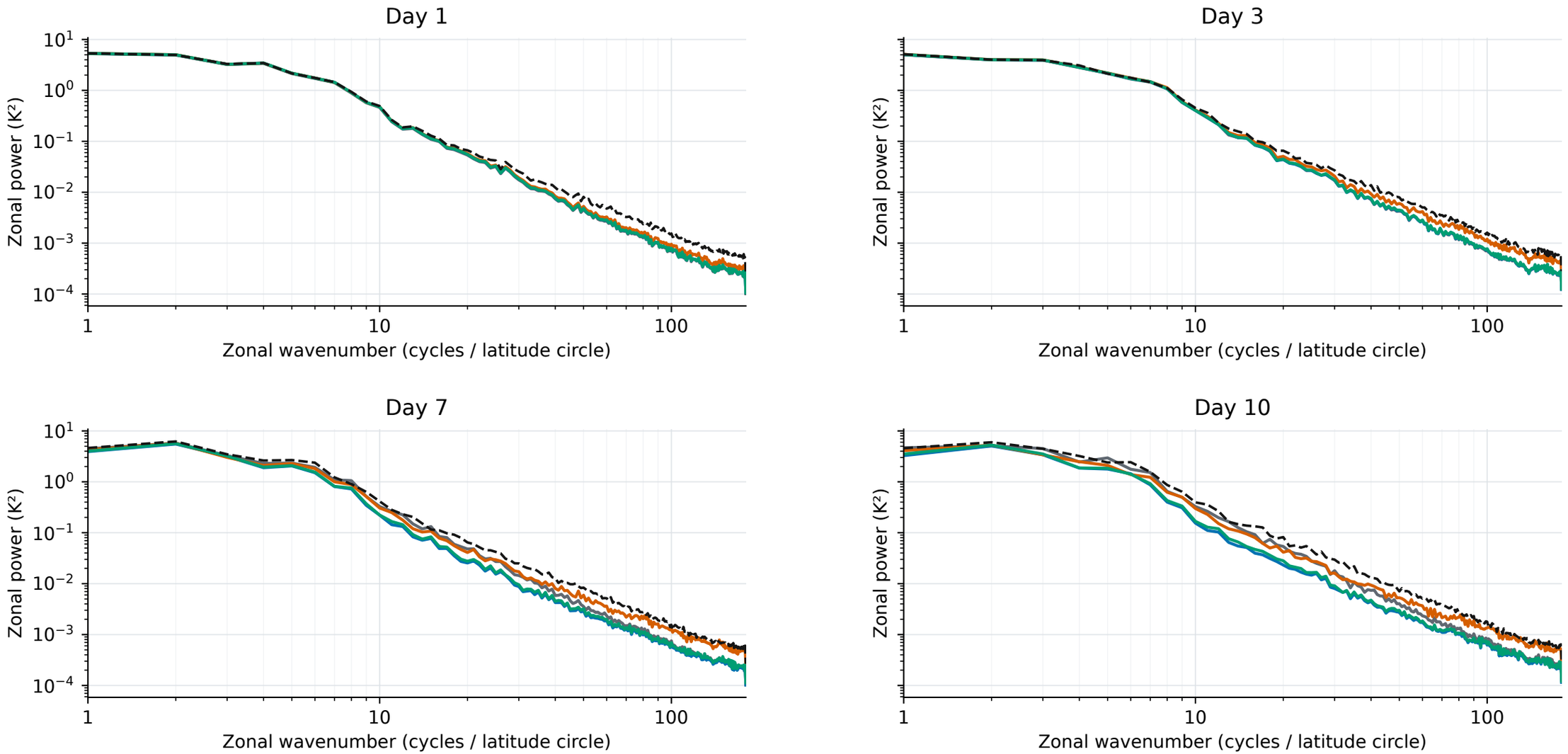
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Temperature at 850 hPa (T850; K)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



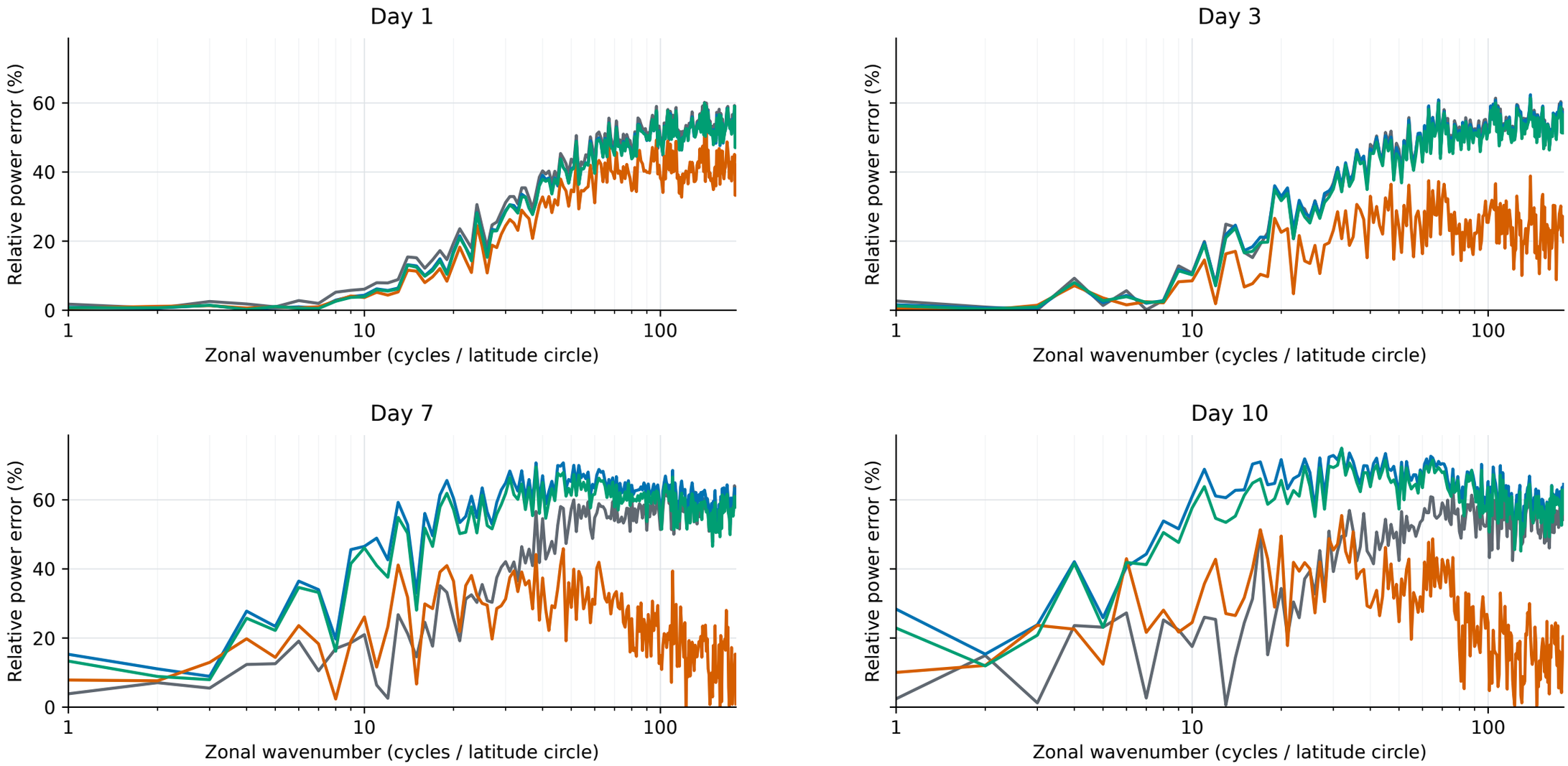
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Temperature at 850 hPa (T850; K)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



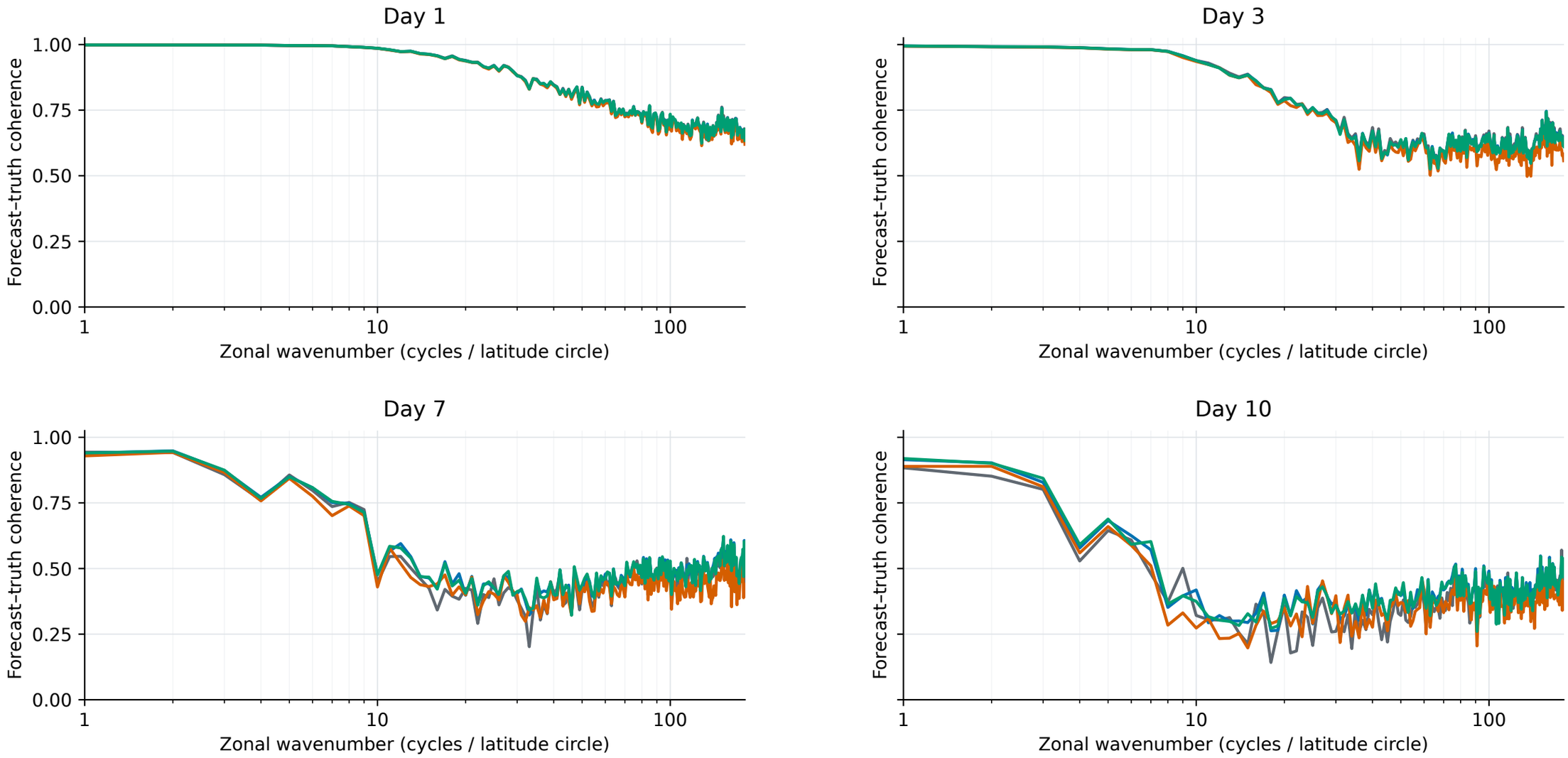
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Temperature at 850 hPa (T850; K)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



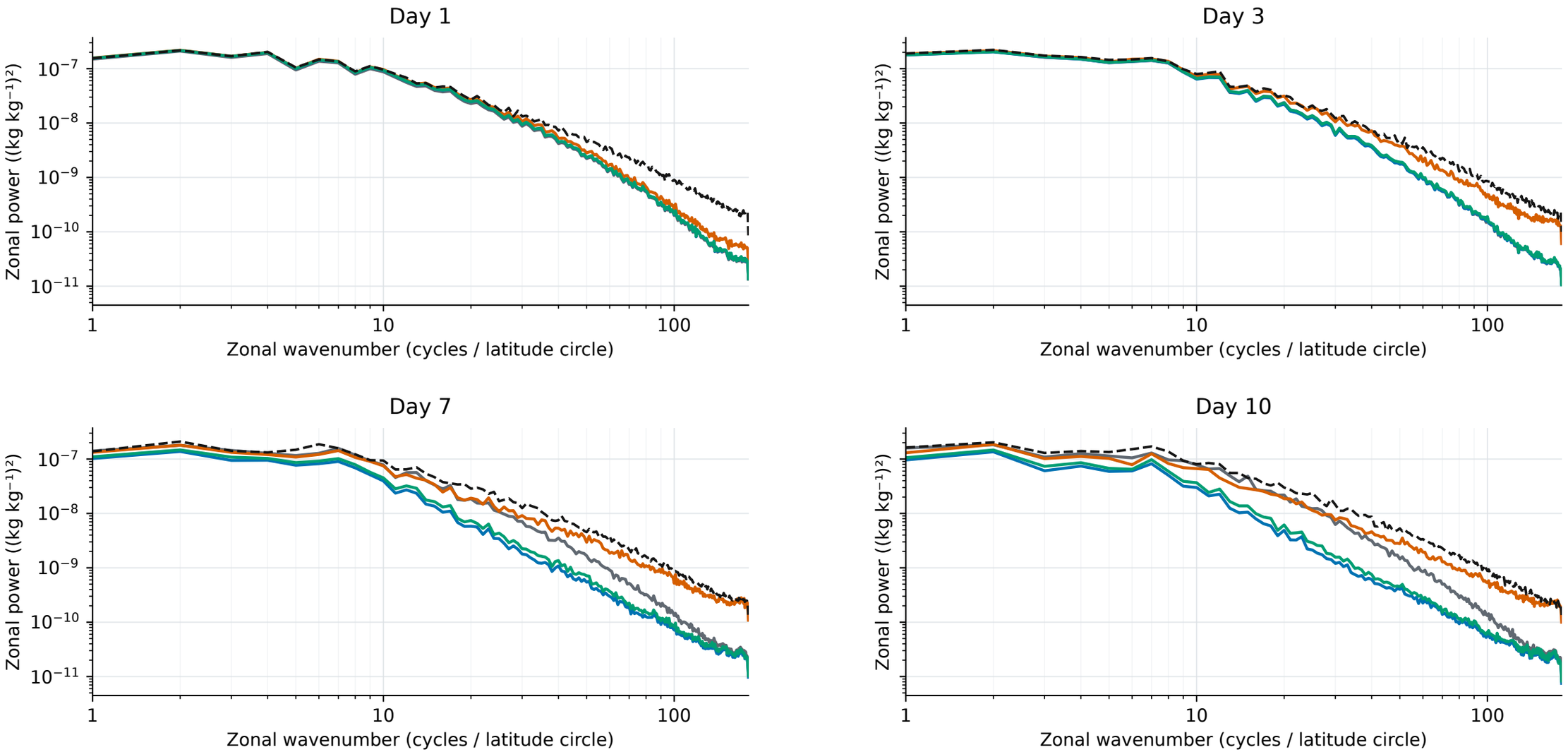
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Specific humidity at 700 hPa (Q700; kg kg⁻¹)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



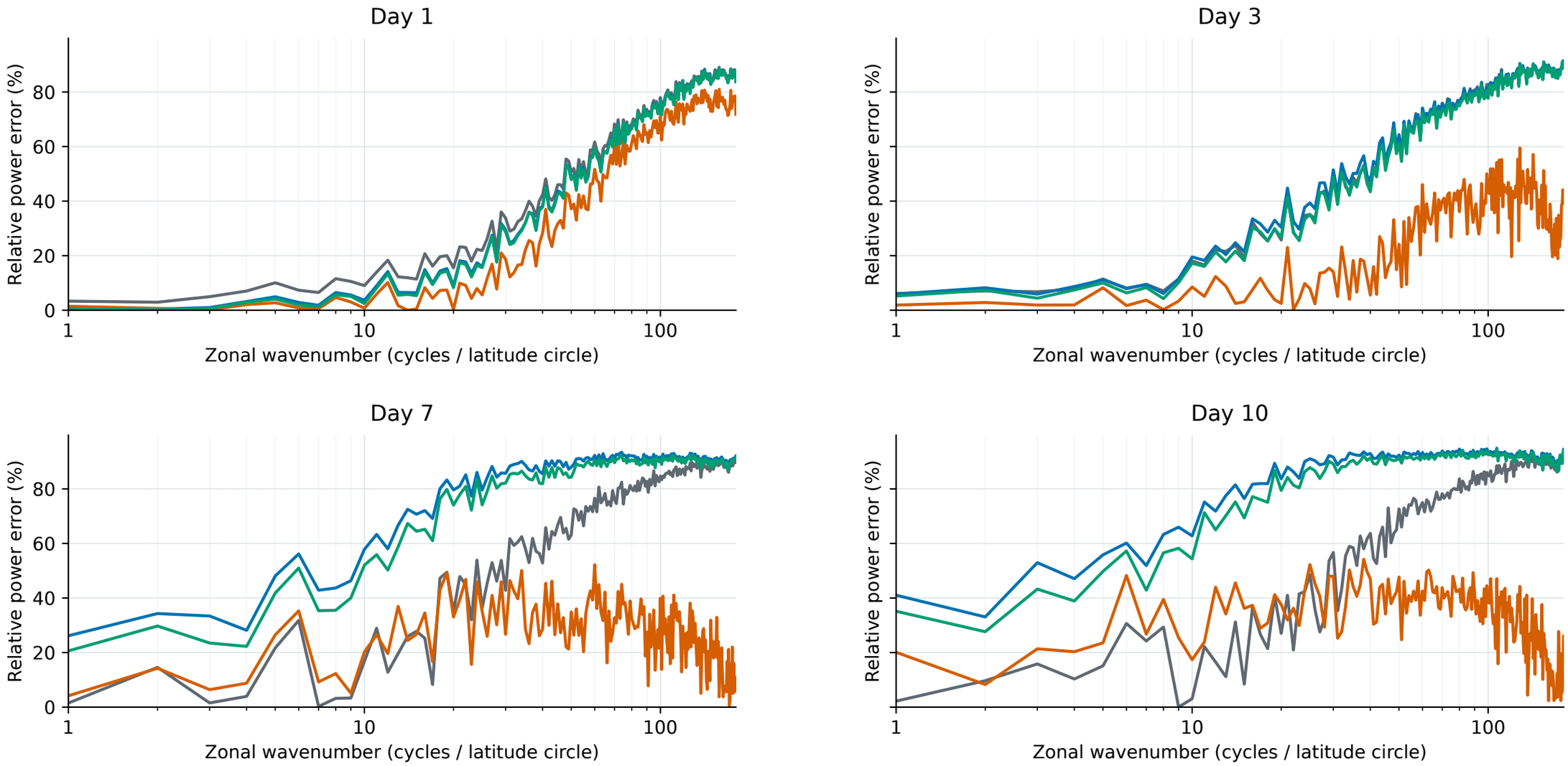
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Specific humidity at 700 hPa (Q700; kg kg⁻¹)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



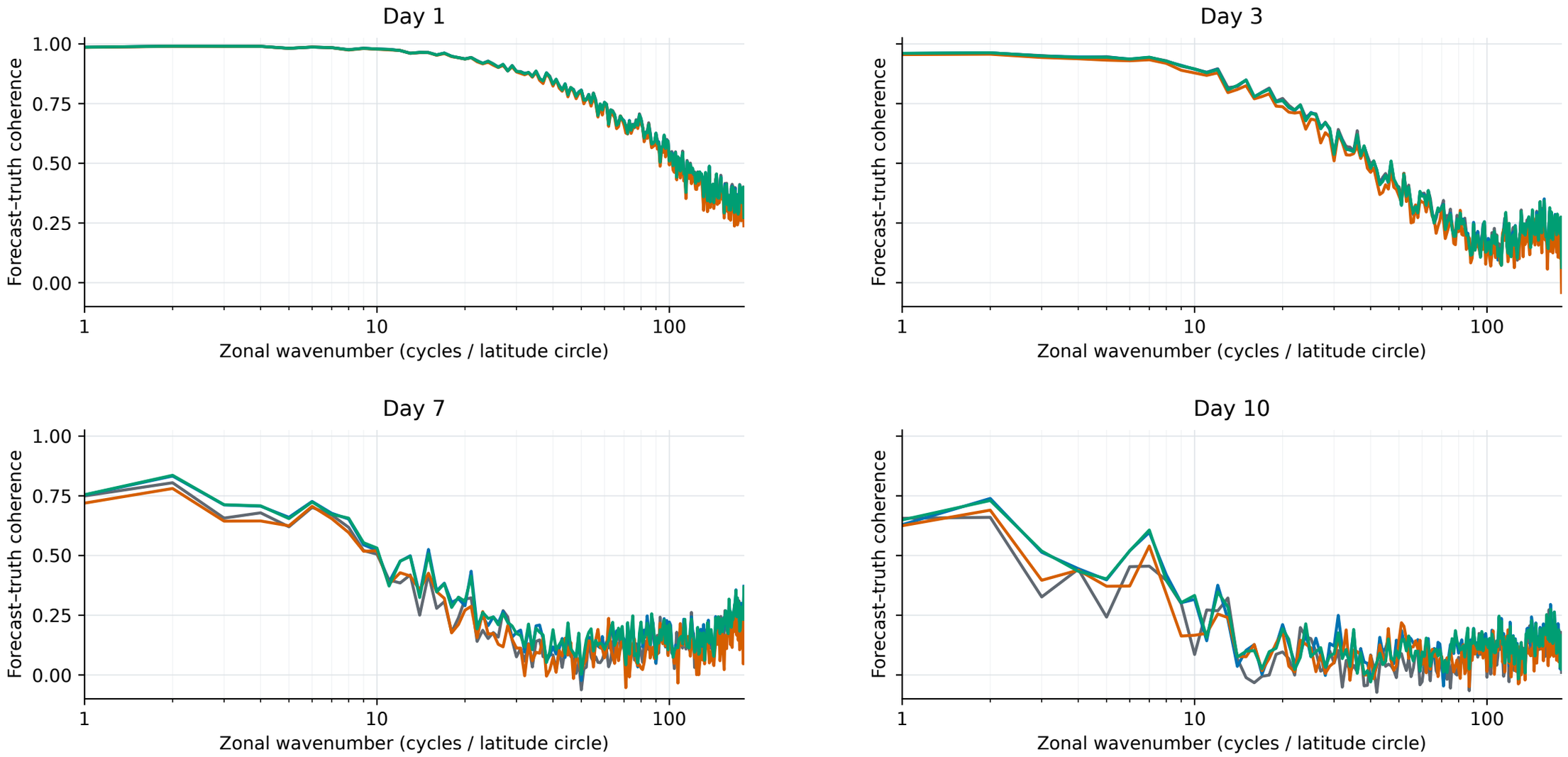
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Specific humidity at 700 hPa (Q700; kg kg⁻¹)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



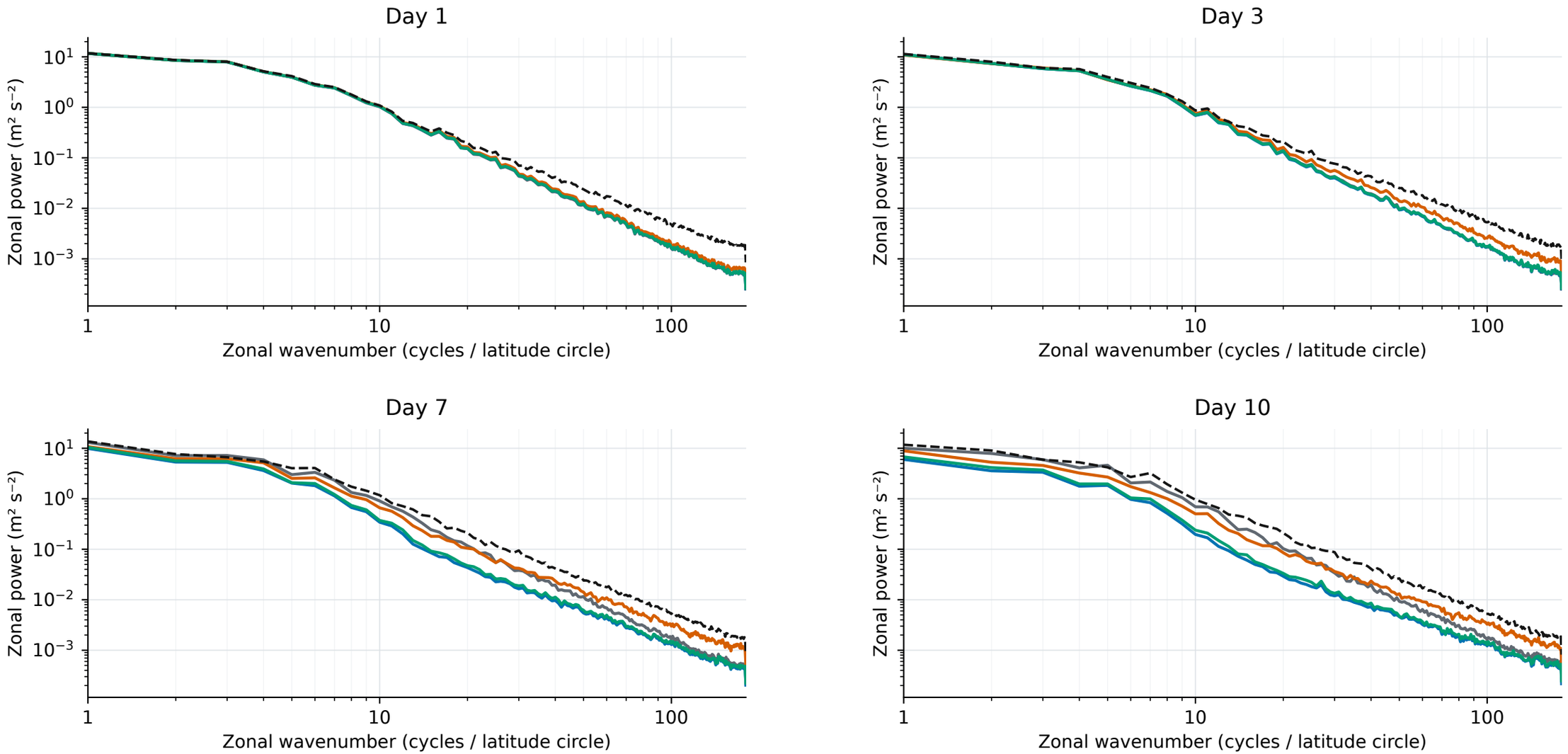
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Zonal wind at 850 hPa (U850; m s⁻¹)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



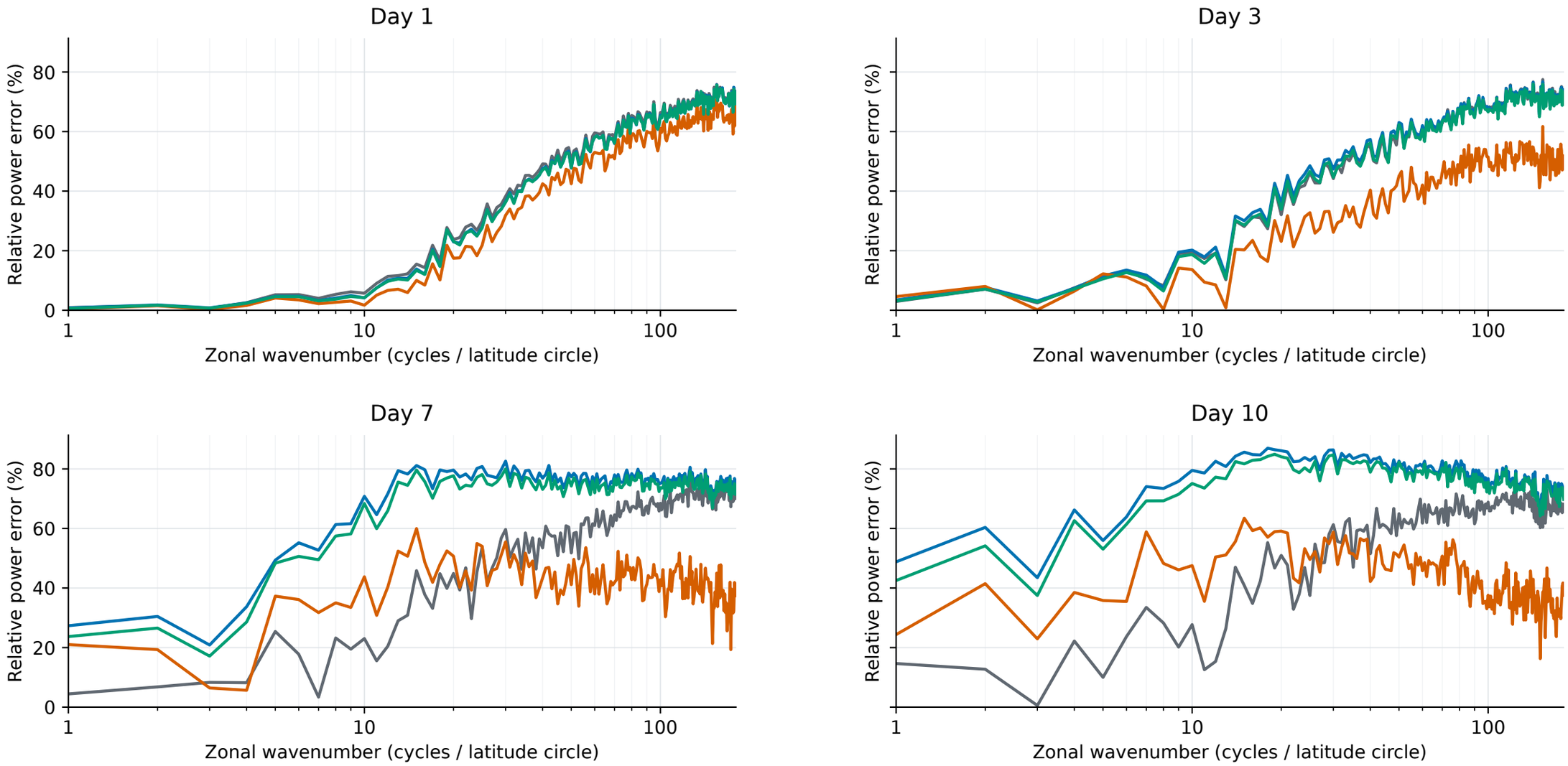
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Zonal wind at 850 hPa (U850; m s⁻¹)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



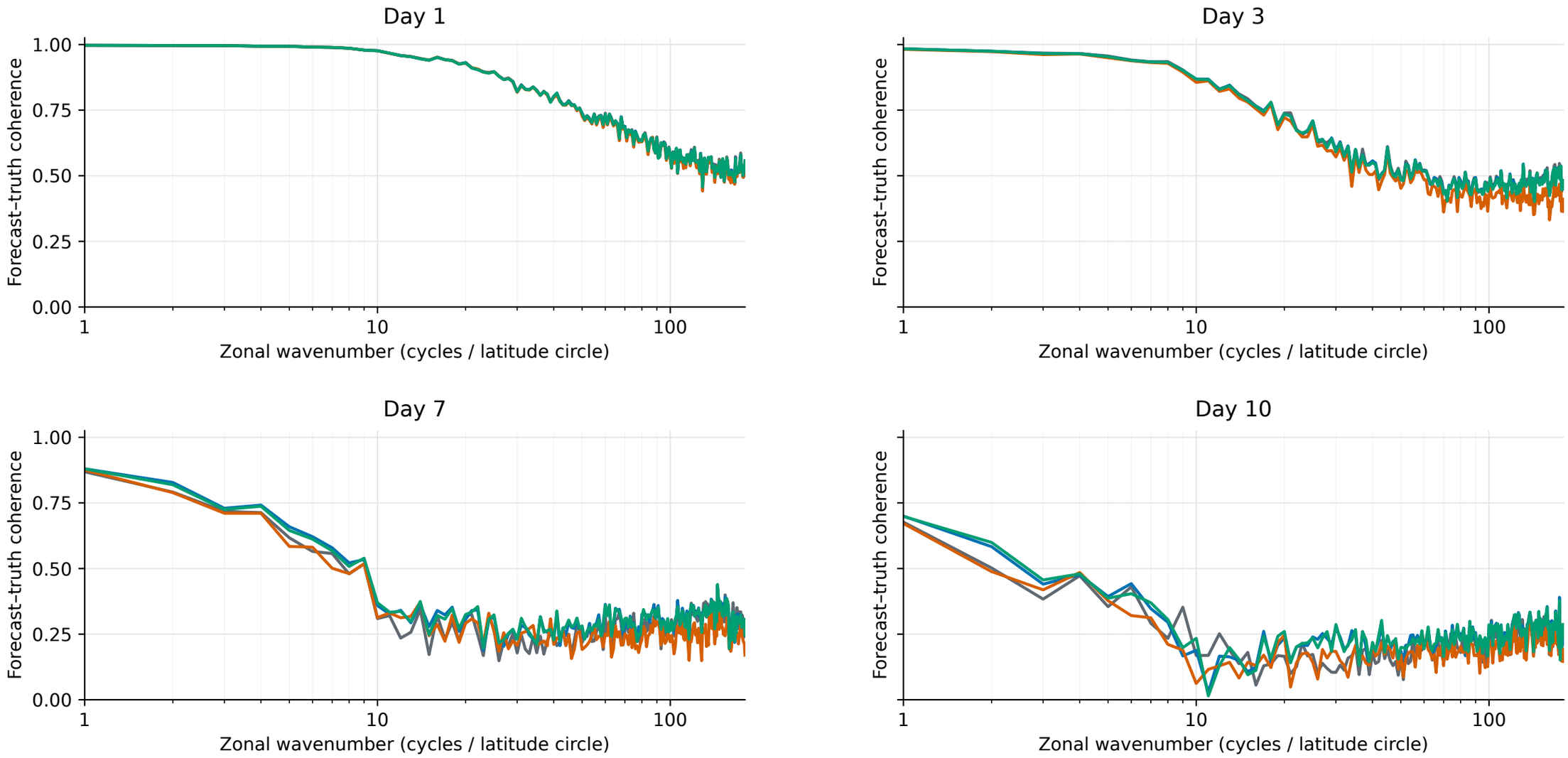
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Zonal wind at 850 hPa (U850; m s⁻¹)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



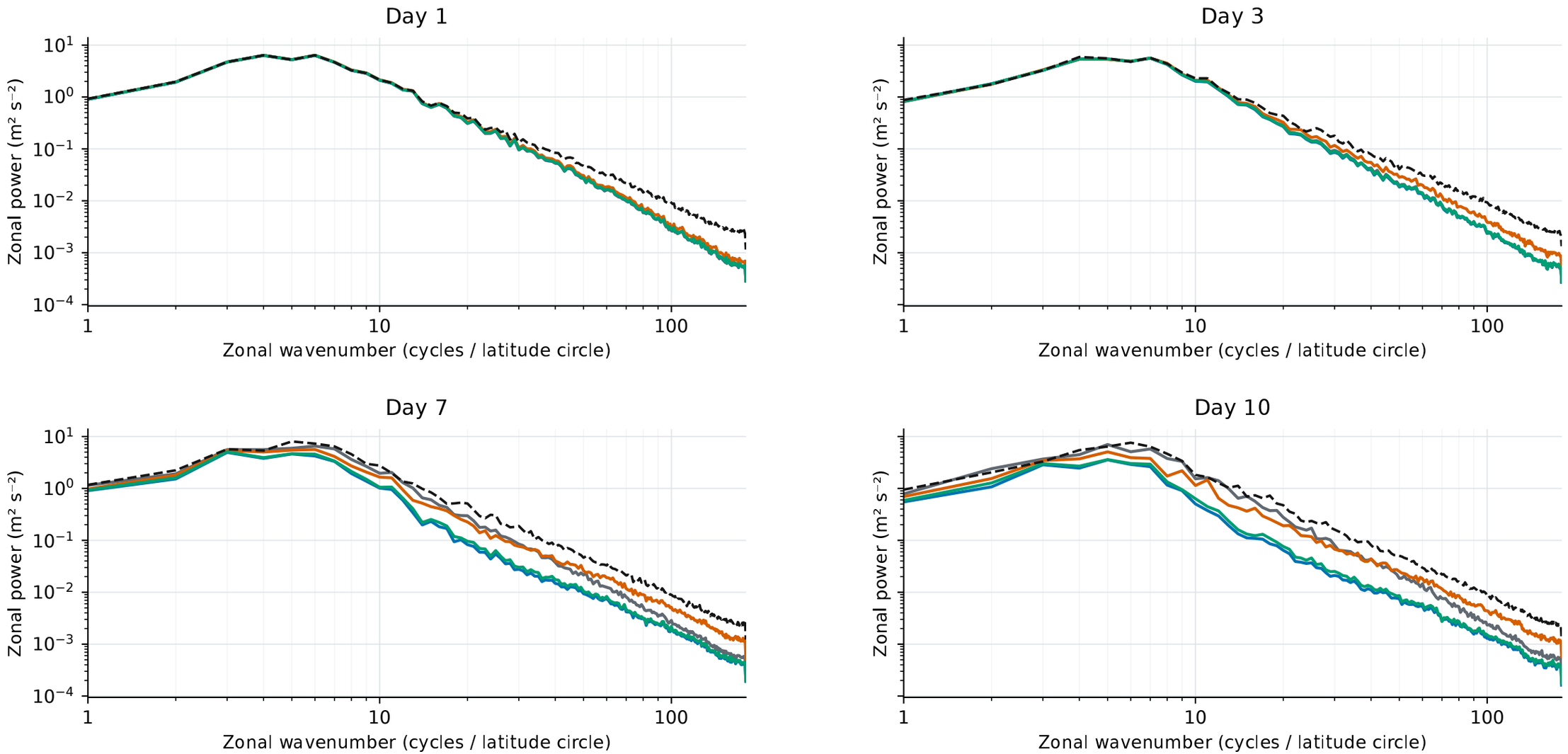
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Meridional wind at 850 hPa (V850; m s⁻¹)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



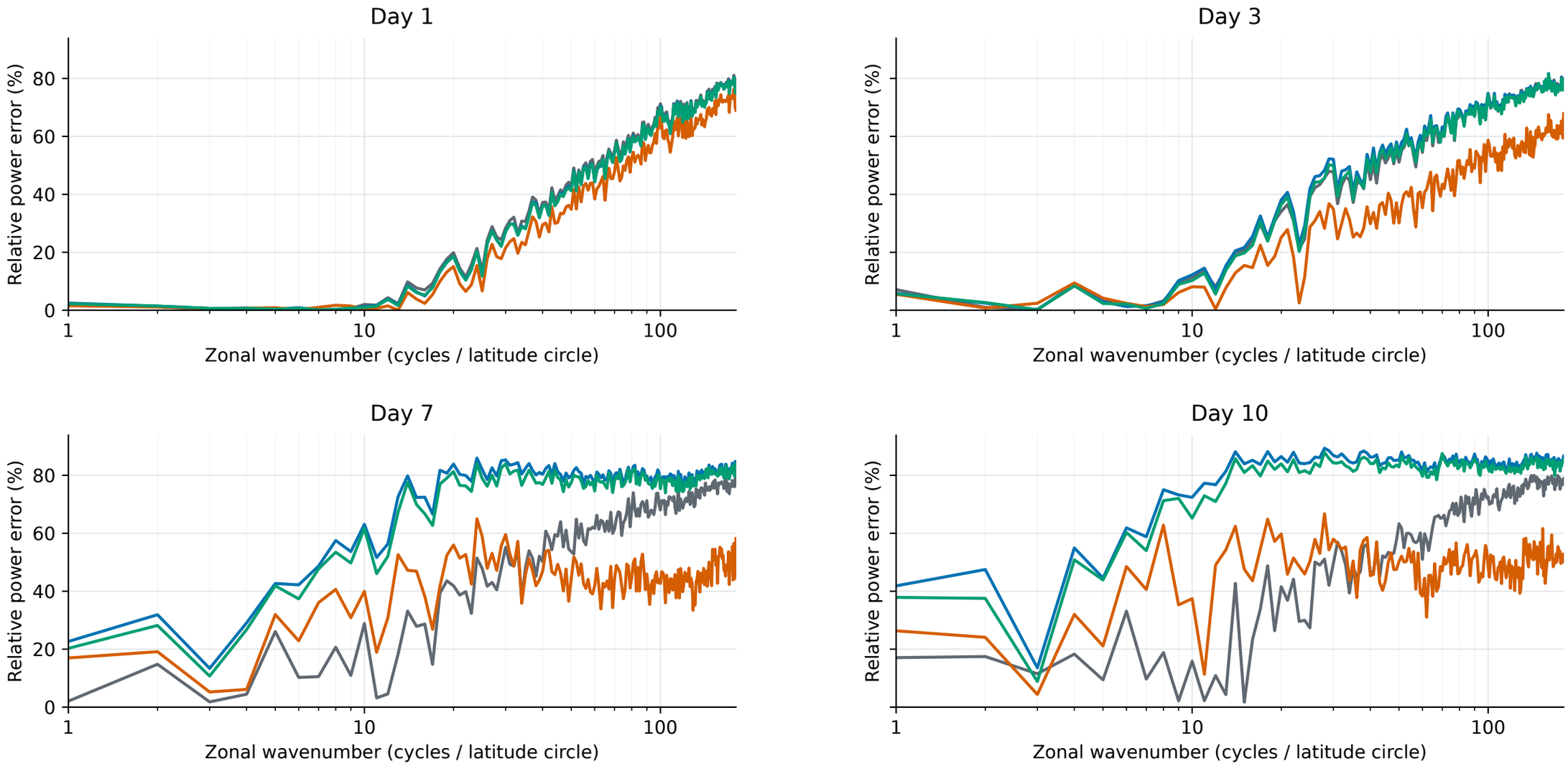
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

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Meridional wind at 850 hPa (V850; m s⁻¹)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



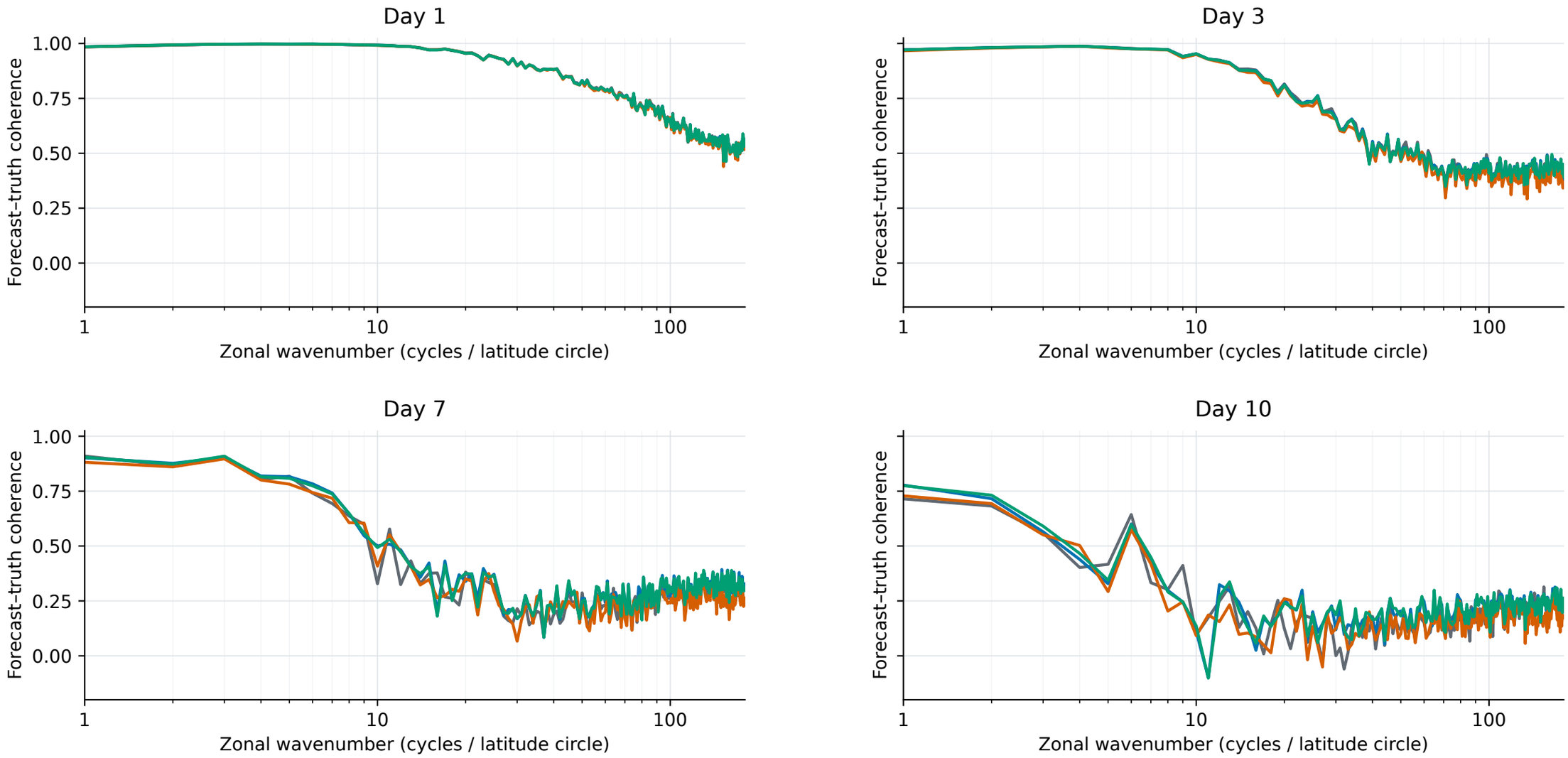
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Meridional wind at 850 hPa (V850; m s⁻¹)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



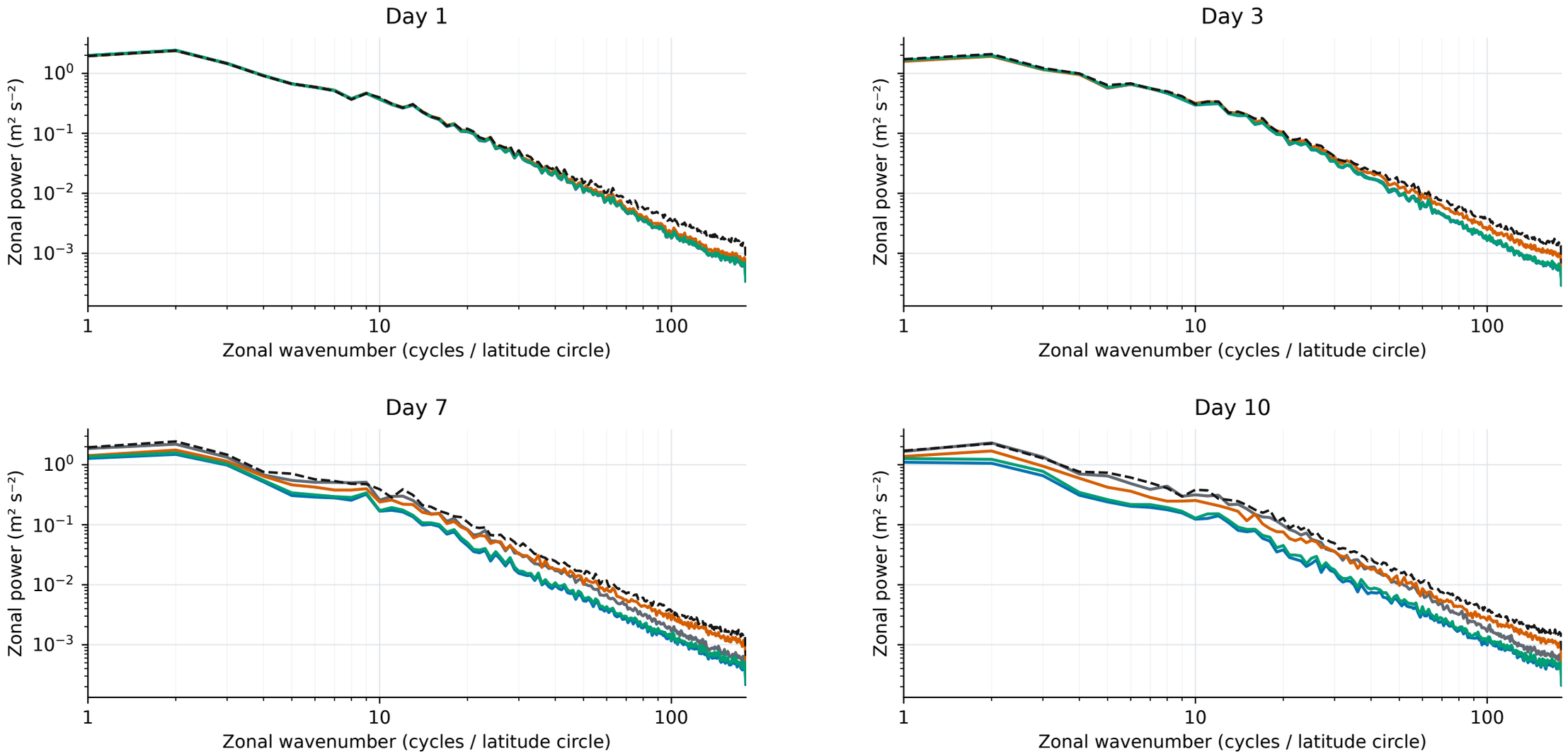
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

10 m wind speed (WS10; m s⁻¹)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



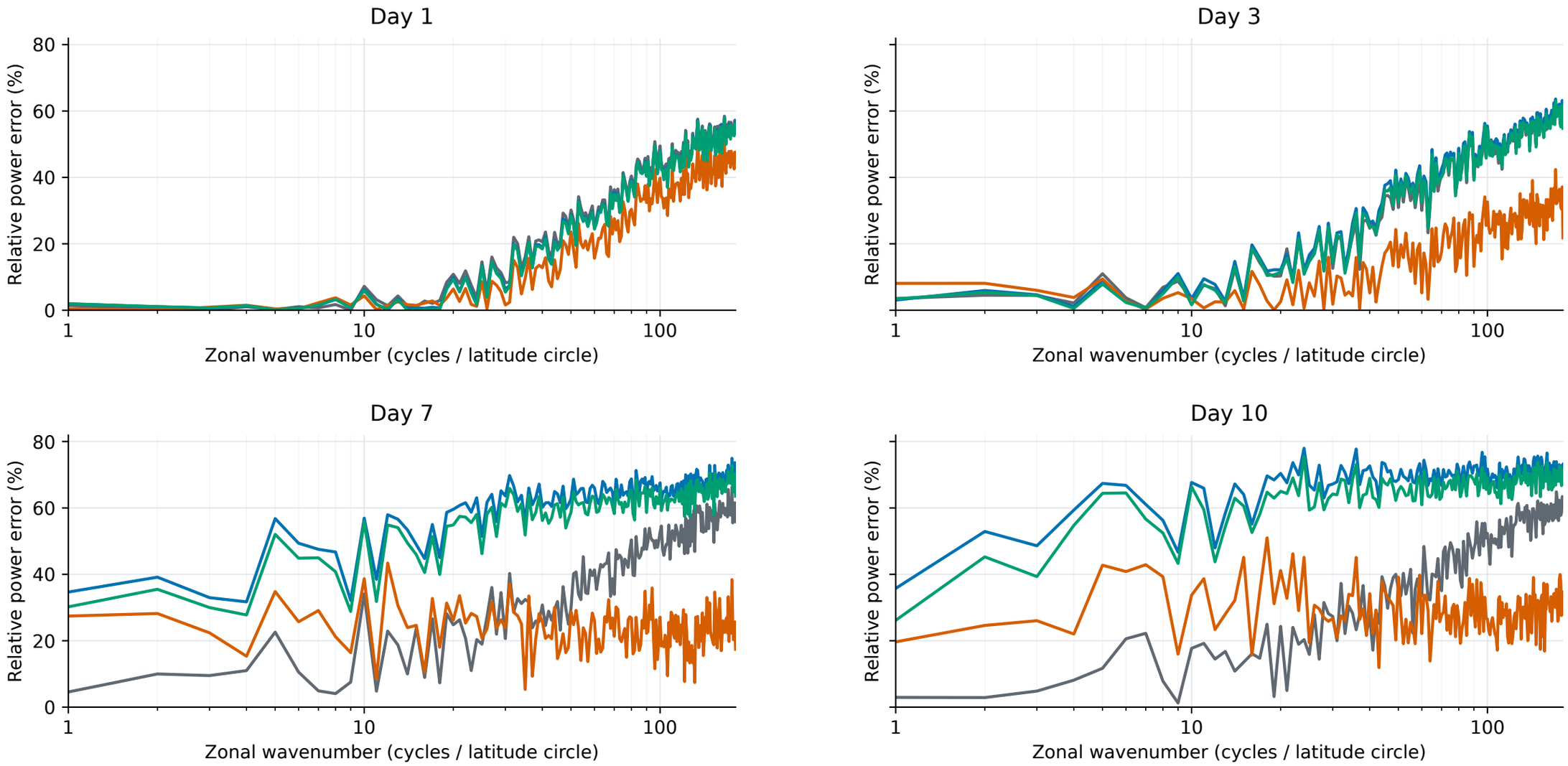
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

10 m wind speed (WS10; m s⁻¹)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



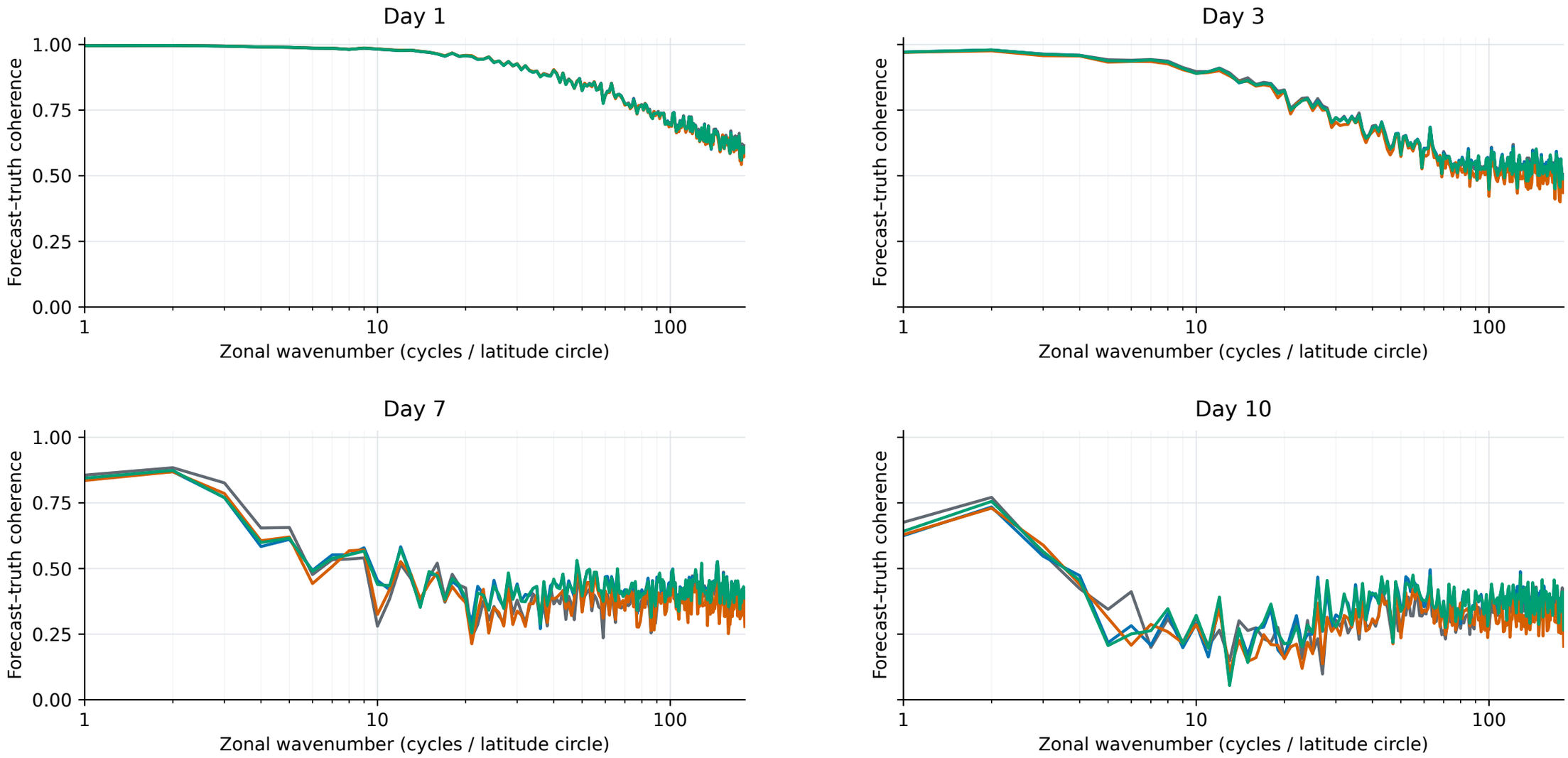
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

10 m wind speed (WS10; m s⁻¹)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



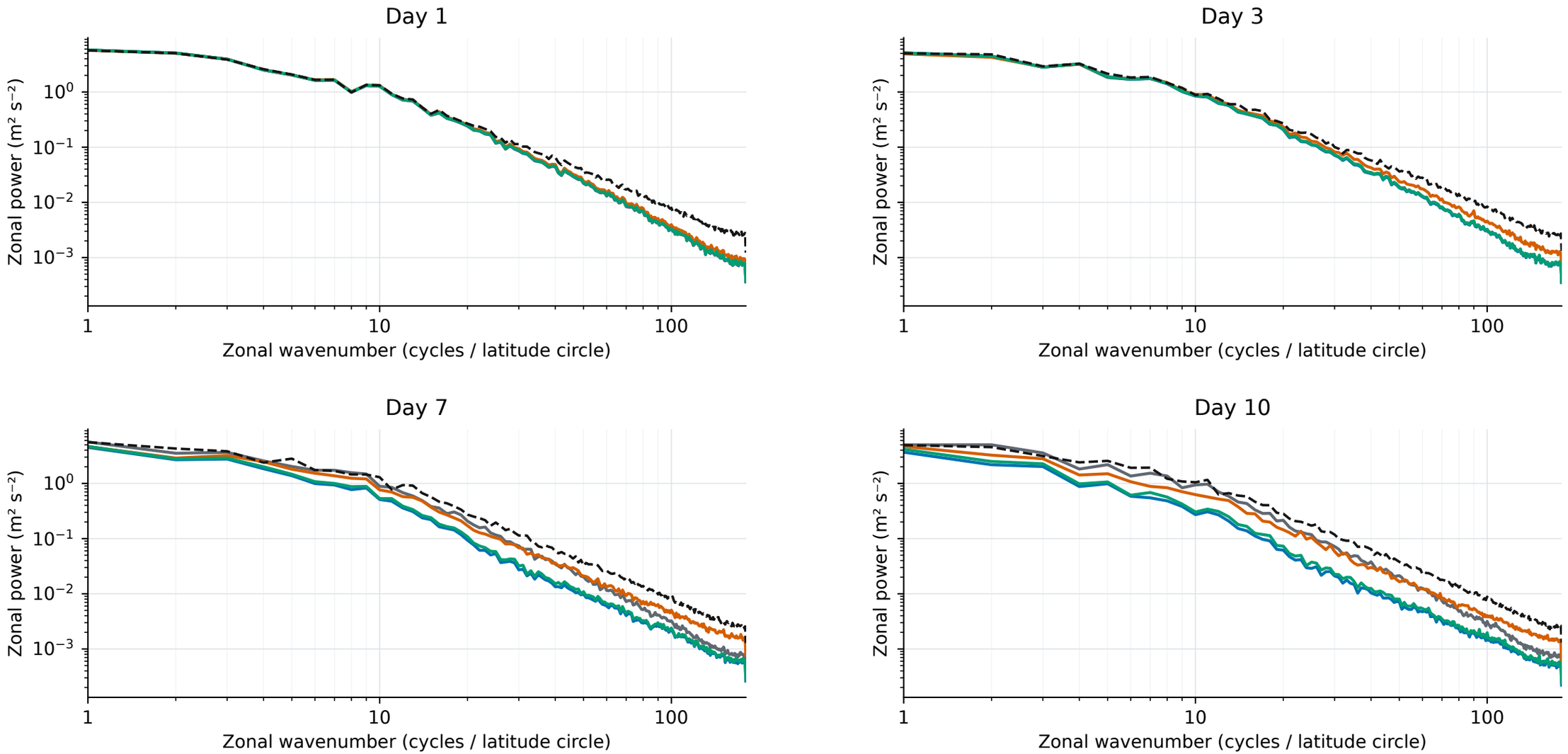
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Wind speed at 850 hPa (WS850; m s⁻¹)

Spectral power

Squared Fourier amplitudes, averaged over latitude and starts



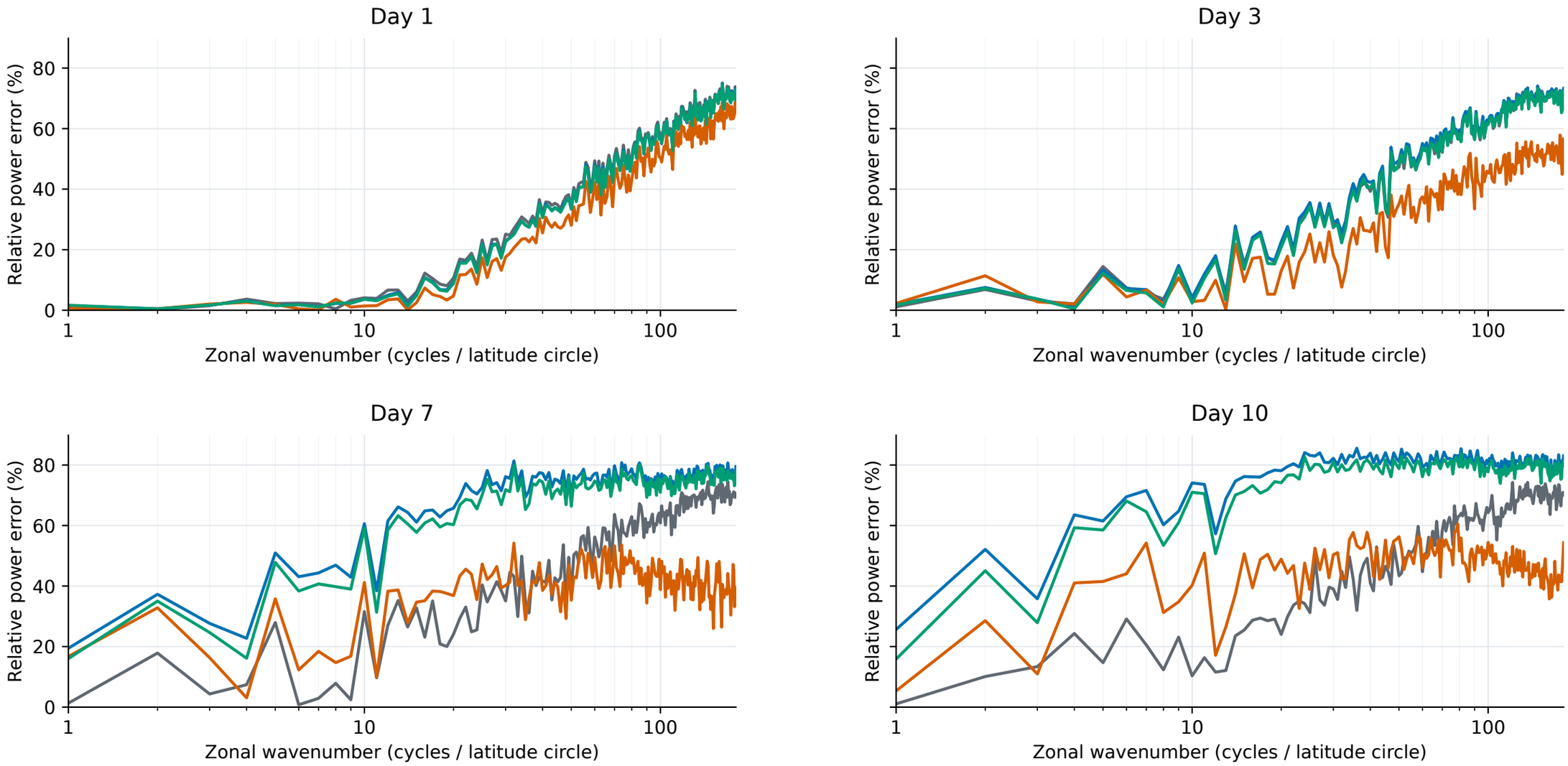
1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Wind speed at 850 hPa (WS850; m s⁻¹)

Relative spectral power error

100 × |forecast power − ERA5 power| / ERA5 power · Lower is better



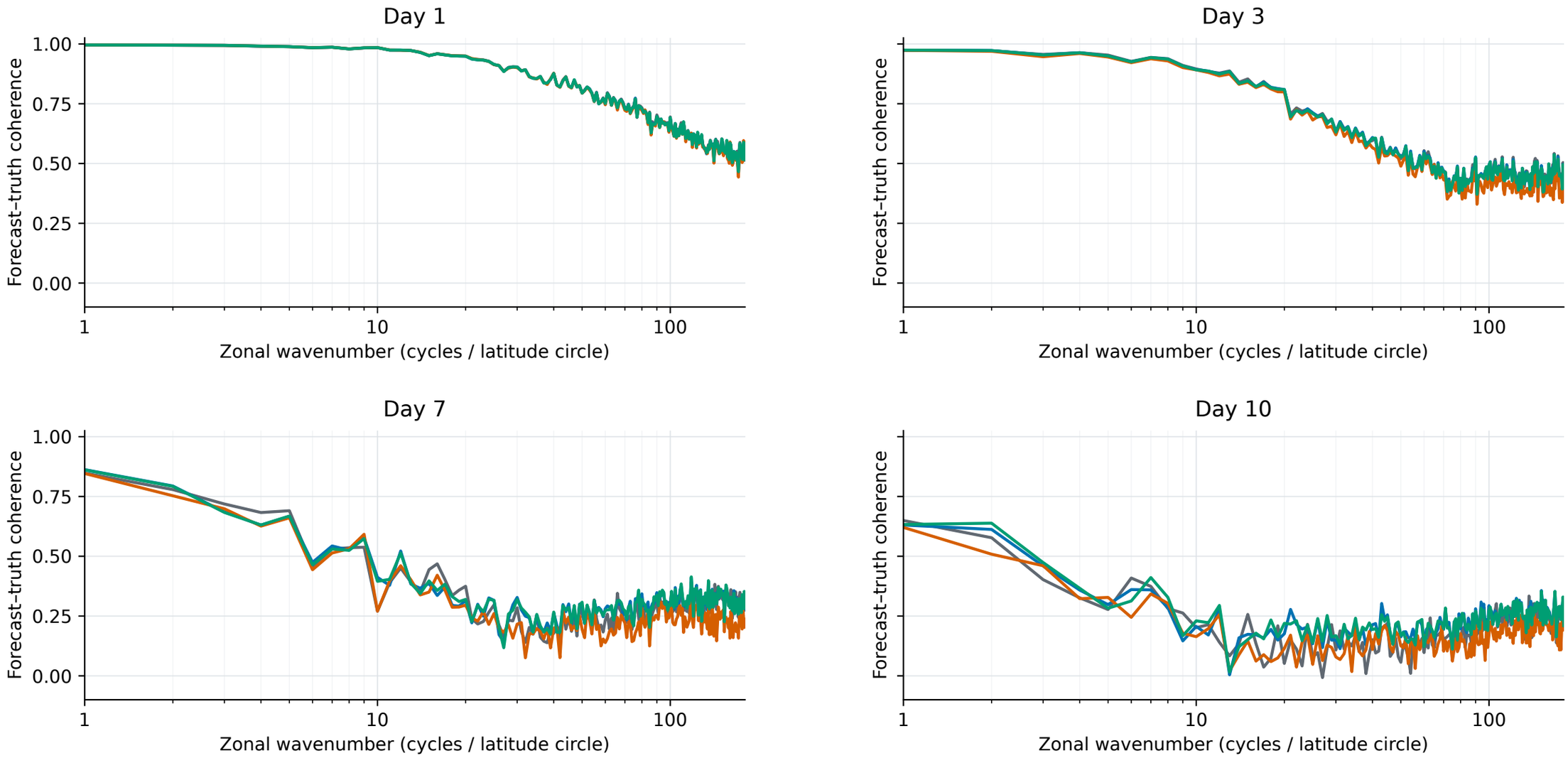
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Wind speed at 850 hPa (WS850; m s⁻¹)

Forecast–truth spectral coherence

Real cospectrum / sqrt(forecast power × ERA5 power) · Higher is better



1° grid · 8 fixed 2023 starts · MSE/AMSE SWA checkpoints · 30–60° latitude in both hemispheres

Raw powers pooled before ratios · Zonal mean removed · k = 1–180 · No smoothing

Reference repeatability and interpretation limits

The figures contain exactly four forecast models

Year	Model	Exact GC rollout loss	vs displayed GC	vs own paired GC
2022	Original GC	10.814070	+0.000%	+0.000%
2022	MSE	9.130295	+15.570%	+15.570%
2022	AMSE λ=0.1	9.150518	+15.383%	+15.404%
2022	Full AMSE	10.818009	-0.036%	-0.017%
2023	Original GC	11.205208	+0.000%	+0.000%
2023	MSE	9.787777	+12.650%	+12.650%
2023	AMSE λ=0.1	9.779006	+12.728%	+13.055%
2023	Full AMSE	11.677263	-4.213%	-4.685%

Repeated original-GC forecasts exhibit numerical drift. All primary plots use one explicitly identified Original GC run, without averaging. The rightmost column retains each model's paired reference to show the practical effect of that choice. The complete audit also records the extra MSE control and per-lead baseline differences.

The spectral sample contains only eight previously inspected starts. The 2022 set contains 32 historical development starts. No confidence intervals, ensemble metrics or unseen test-set claims are made. Pressure-level, latitude-domain and weighting differences must be respected when comparing spectral and global-error charts.

Reproducibility and artifact guide

All source evaluations and checkpoints are preserved

Primary checkpoint policy: SWA of updates 500, 1000, 1500 and 2000. Inference uses 40 six-hour leads, cold initial recurrent state and full corrected feedback. Source manifests identify the original GraphCast checkpoint, data, initialization dates and metric settings.

The report_data.json and spectra.npz files contain the validated four-model inputs. CSV files provide all plotted spectral summaries, all 40 leads of global RMSE and every one of the 83 output channels. Source and checkpoint identities are recorded by SHA256. Standalone PNG and vector PDF files are provided for each figure.

Data and audit directory:

plots/analyze_models/data/resolution_eval/v24_amse_feature_report_20261004

Figure directory:

plots/analyze_models/images/resolution_eval/v24_amse_feature_report_20261004

Generators:

scripts/analyze_models/amse_feature_report_data.py

scripts/analyze_models/plot_v24_amse_feature_report.py

scripts/analyze_models/plot_v24_amse_feature_errors.py

scripts/analyze_models/report_v24_amse_features.py

Rebuild in that order after sourcing scripts/graphcast_env.sh. The process reads saved evaluations, validates them, generates plots and assembles this report. It does not submit training or evaluation jobs.